

Establishment and Adjustment of Calibration Intervals

**Recommended Practice - 1
(RP-1) 2010**



NCSL INTERNATIONAL
Serving the World of Measurement

Recommended Practice-1: Establishment and Adjustment of Calibration Intervals

Recommended Practice (RP-1)

Prepared by:
NCSL International Calibration Intervals Committee

Copyright © 2019 by NCSL International All rights reserved

First edition—May 1979
Second edition—November 1989
Third edition—January 1996
Fourth edition—April 2010
Editorial—April 2010-A (2013)

ISBN 978-1-58464-062-2

Publisher
NCSL International
5766 Central Ave., Suite 150
Boulder, CO 80301
Phone: (303) 440-3339
Fax: (303) 440-3384
Email: info@ncsli.org

Vision:

Be the world's recognized source for measurement science expertise and information

Mission:

To provide the best opportunities for the world's measurement science experts and practitioners

- to network and exchange information;
- to promote measurement education and skill development; and
- to develop means for organizations to resolve measurement challenges



Foreword

This Recommended Practice has been prepared by NCSL International to promote uniformity and the quality in the establishment and adjustment of calibration intervals for measuring and test equipment. To be of real value, this document should not be static, but should be subject to periodic review. NCSL International welcomes comments *and* criticism, which should be addressed to NCSL International.

This Recommended Practice was initiated by the Calibration Interval Committee, coordinated by the cognizant Vice President and approved for publication by the Board of Directors on 31 April 2010.

Acknowledgments

The NCSL International Calibration Interval Committee consists of member delegates and others within the metrology community with expertise in development and/or management of calibration intervals. Committee members represented a variety of organizations, large and small, engaged in the management of instrumentation covering all major measurement technology disciplines. Committee members that have contributed to the 2010 revision of this Recommended Practice are:

Member or Contributor

Mr. Mark J. Kuster (Committee Chair)
Mr. Del Caldwell
Dr. Howard Castrup
Mr. Greg Cenker
Mr. Dave Deaver
Dr. Dennis Dubro
Dr. Steve Dwyer
Mr. William Hinton
Ms. Ding Huang
Dr. Dennis Jackson
Mr. Mitchell Johnson
Mr. Leif King
Dr. Charles A. Motzko
Mr. Richard Ogg
Mr. Derek Porter
Mr. Donald Wyatt

Organization

B&W Pantex, U.S. DOE NNSA Pantex Plant
Calibration Coordination Group
Integrated Sciences Group
Southern California Edison
Fluke Corporation
Pacific Gas & Electric Company
U.S. Naval Surface Warfare Center
Florida Power & Light – Seabrook Station
U.S. Naval Air Station, Patuxent River
U.S. Naval Surface Warfare Center
Donaldson Company
B&W Y-12, U.S. DOE NNSA ORMC
C. A. Motzko & Associates
Agilent Technologies
Boeing Commercial Equipment
Diversified Data Systems

Editorial acknowledgment is due many non-Committee NCSL International members, the NCSL International Board of Directors, and other interested parties who provided valuable comments and suggestions.

Table of Contents

FOREWORD	I
ACKNOWLEDGMENTS.....	II
TABLE OF CONTENTS.....	III
FIGURES.....	V
TABLES.....	VI
CHAPTER 1 GENERAL.....	1
1.1 PURPOSE.....	1
1.2 SCOPE.....	1
1.3 THE GOAL OF INTERVAL ANALYSIS	1
1.4 THE NEED FOR PERIODIC CALIBRATION.....	2
1.5 OPTIMAL INTERVALS.....	3
1.6 DIVERSITY OF METHODS.....	3
1.7 TOPIC ORGANIZATION.....	4
CHAPTER 2 MANAGEMENT BACKGROUND	5
2.1 INTRODUCTION.....	5
2.2 THE NEED FOR INTERVAL ANALYSIS.....	5
2.3 MEASUREMENT RELIABILITY TARGETS	5
2.4 CALIBRATION INTERVAL OBJECTIVES.....	6
2.5 OPTIMAL INTERVALS.....	8
2.6 CALIBRATION INTERVAL-ANALYSIS METHODS	8
2.7 INTERVAL ADJUSTMENT APPROACHES	13
2.8 DATA REQUIREMENTS.....	17
2.9 SYSTEM EVALUATION.....	17
CHAPTER 3 INTERVAL-ANALYSIS PROGRAM ELEMENTS	18
3.1 INTRODUCTION.....	18
3.2 DATA COLLECTION AND STORAGE.....	18
3.3 DATA ANALYSIS	19
3.4 GUARDBAND USE.....	19
3.5 LIMIT TYPES	21
3.6 MEASUREMENT RELIABILITY MODELING AND PROJECTION	22
3.7 ENGINEERING REVIEW.....	22
3.8 LOGISTICS ANALYSIS	22
3.9 IMPOSED REQUIREMENTS	22
3.10 COSTS/BENEFITS ASSESSMENT	24
3.11 PERSONNEL REQUIREMENTS.....	26
3.12 TRAINING AND COMMUNICATIONS	27
CHAPTER 4 INTERVAL-ANALYSIS METHOD SELECTION	28
4.1 INTRODUCTION.....	28
4.2 SELECTION CRITERIA	28
4.3 GENERAL INTERVAL METHOD	29
4.4 BORROWED INTERVALS METHOD	31
4.5 ENGINEERING ANALYSIS METHOD	33
4.6 REACTIVE METHODS.....	35
4.7 MAXIMUM LIKELIHOOD ESTIMATION (MLE) METHODS	39
4.8 METHOD SELECTION DECISION TREES	42
CHAPTER 5 TECHNICAL BACKGROUND	45
5.1 INTRODUCTION.....	45
5.2 UNCERTAINTY GROWTH	45
5.3 MEASUREMENT RELIABILITY	45
5.4 PREDICTIVE METHODS	46
5.5 SETTING MEASUREMENT RELIABILITY TARGETS	58

5.6	INTERVAL CANDIDATE SELECTION	62
5.7	IDENTIFYING OUTLIERS	63
5.8	ENGINEERING ANALYSIS	78
5.9	REACTIVE METHODS.....	79
5.10	INITIAL INTERVALS	79
5.11	SIMILAR ITEM ASSIGNMENT	79
5.12	INSTRUMENT CLASS ASSIGNMENT	79
5.13	ENGINEERING ANALYSIS	80
5.14	EXTERNAL INTERVALS.....	80
5.15	GENERAL INTERVAL	80
CHAPTER 6	REQUIRED DATA ELEMENTS.....	81
6.1	INTRODUCTION AND BACKGROUND	81
6.2	IDENTIFICATION ELEMENTS	82
6.3	TECHNICAL ELEMENTS.....	84
CHAPTER 7	NO PERIODIC CALIBRATION REQUIRED.....	87
REFERENCES.....		88
APPENDIX A	TERMINOLOGY AND DEFINITIONS	94
APPENDIX B	REACTIVE METHODS	98
B.1	INTRODUCTION	98
B.2	METHOD A1 - SIMPLE RESPONSE METHOD	98
B.3	METHOD A2 - INCREMENTAL RESPONSE METHOD	99
B.4	METHOD A3 - INTERVAL TEST METHOD	103
APPENDIX C	METHOD S1 - CLASSICAL METHOD.....	113
C.1	BACKGROUND	113
C.2	RENEW-ALWAYS VERSION	113
C.3	RENEW-AS-NEEDED VERSION	114
C.4	RENEW-IF-FAILED VERSION	115
C.5	METHOD S1 PROS AND CONS	115
APPENDIX D	METHOD S2 - BINOMIAL METHOD.....	117
D.1	MATHEMATICAL DESCRIPTION.....	117
D.2	MEASUREMENT RELIABILITY	117
D.3	THE OUT-OF-TOLERANCE PROCESS.....	117
D.4	THE OUT-OF-TOLERANCE TIME SERIES	118
D.5	ANALYZING THE TIME SERIES	118
D.6	MEASUREMENT RELIABILITY MODELING	120
D.7	RELIABILITY MODEL SELECTION	125
D.8	MEASUREMENT RELIABILITY MODELS	129
D.9	CALIBRATION INTERVAL DETERMINATION.....	140
D.10	METHOD S2 PROS AND CONS	141
APPENDIX E	METHOD S3 - RENEWAL TIME METHOD.....	142
E.1	DEVELOPING THE LIKELIHOOD FUNCTION	142
E.2	GENERALIZING THE LIKELIHOOD FUNCTION.....	143
E.3	THE TOTAL LIKELIHOOD FUNCTION	144
E.4	GROUPING BY RENEWAL TIME	145
E.5	CONSISTENT INTERVAL CASES	145
E.6	LIMITING RENEWAL CASES	146
E.7	EXAMPLE: SIMPLE EXPONENTIAL MODEL	147
E.8	METHOD S3 PROS AND CONS	148
APPENDIX F	ADJUSTING BORROWED INTERVALS	149
F.1	INTRODUCTION	149
F.2	GENERAL CASE	149
F.3	EXPONENTIAL MODEL CASE	149
APPENDIX G	RENEWAL POLICIES.....	151

G.1	INTRODUCTION	151
G.2	DECISION VARIABLES	151
G.3	RENEWAL POLICY SELECTION.....	155
G.4	ANALYTICAL POLICY SELECTION.....	157
G.5	MAINTAINING CONDITION RECEIVED INFORMATION	157
G.6	SUMMARY	158
APPENDIX H	SYSTEM EVALUATION	159
H.1	INTRODUCTION	159
H.2	DEVELOPING A SAMPLING WINDOW.....	159
H.3	SYSTEM EVALUATION GUIDELINES	160
APPENDIX I	SOLVING FOR CALIBRATION INTERVALS	163
I.1	INTRODUCTION	163
I.2	SPECIAL CASES	163
I.3	GENERAL CASES	163
I.4	INVERSE RELIABILITY FUNCTIONS	164
I.5	ADJUSTMENT INTERVALS.....	165
HISTORY.....		166
1989 REVISION		166
1996 REVISION		166
PERMISSION TO REPRODUCE.....	ERROR! BOOKMARK NOT DEFINED.	
PERMISSION TO TRANSLATE	ERROR! BOOKMARK NOT DEFINED.	
DISCLAIMER.....	ERROR! BOOKMARK NOT DEFINED.	

Figures

FIGURE 1-1. RP-1 READER'S GUIDE	4
FIGURE 2-1. INTERVAL-ANALYSIS TAXONOMY.....	14
FIGURE 3-1. ADJUSTMENT VS. REPORTING LIMITS.	21
FIGURE 4-1. SMALL INVENTORY DECISION TREE.	43
FIGURE 4-2. MEDIUM-SIZE INVENTORY DECISION TREE.....	44
FIGURE 4-3. LARGE INVENTORY DECISION TREE.	44
FIGURE 5-1. MEASUREMENT UNCERTAINTY GROWTH	45
FIGURE 5-2. MEASUREMENT RELIABILITY VS. TIME.	46
FIGURE 5-3. MEASUREMENT UNCERTAINTY GROWTH MECHANISMS.	48
FIGURE 5-4. OBSERVED MEASUREMENT RELIABILITY.....	50
FIGURE B-1. TIME (IN CORRECT INTERVALS) TO ARRIVE AT CORRECT INTERVAL	109
FIGURE B-2. STABILITY AT THE CORRECT INTERVAL.....	109
FIGURE D-1. HYPOTHETICAL OBSERVED TIME SERIES.	120
FIGURE D-2. OUT-OF-TOLERANCE STOCHASTIC PROCESS MODEL.....	121
FIGURE D-3. EXPONENTIAL MEASUREMENT RELIABILITY MODEL	130
FIGURE D-4. WEIBULL MEASUREMENT RELIABILITY MODEL	131
FIGURE D-5. MIXED EXPONENTIAL MEASUREMENT RELIABILITY MODEL	132
FIGURE D-6. RANDOM-WALK MEASUREMENT RELIABILITY MODEL	134
FIGURE D-7. RESTRICTED RANDOM-WALK MEASUREMENT RELIABILITY MODEL	135
FIGURE D-8. MODIFIED GAMMA MEASUREMENT RELIABILITY MODEL	136
FIGURE D-9. MORTALITY DRIFT MEASUREMENT RELIABILITY MODEL	136
FIGURE D-10. WARRANTY MEASUREMENT RELIABILITY MODEL.....	137
FIGURE D-11. DRIFT MEASUREMENT RELIABILITY MODEL.....	138
FIGURE D-12. LOGNORMAL MEASUREMENT RELIABILITY MODEL	139
FIGURE H-1. BINOMIAL CONFIDENCE TEST EXAMPLE	161

Tables

TABLE 4-1. GENERAL INTERVAL.....	31
TABLE 4-2. BORROWED INTERVALS METHOD	33
TABLE 4-3. ENGINEERING ANALYSIS METHOD.....	35
TABLE 4-4. REACTIVE METHODOLOGY.....	39
TABLE 4-5. MLE METHODOLOGY RECOMMENDATIONS.....	42
TABLE 5-1. OBSERVED RELIABILITY TIME SERIES	49
TABLE 5-2. SIMULATED GROUP CALIBRATION RESULTS.....	55
TABLE 5-3. EXAMPLE HOMOGENEITY TEST RESULTS.....	56
TABLE 5-4. EXAMPLE OUTLIER IDENTIFICATION DATA.....	69
TABLE 5-5. SORTED OUTLIER IDENTIFICATION DATA.....	69
TABLE 5-6. TECHNICIAN OUTLIER IDENTIFICATION DATA.....	71
TABLE 5-7. USER OUTLIER IDENTIFICATION DATA	72
TABLE 5-8. FACILITY OUTLIER IDENTIFICATION DATA.....	74
TABLE 5-9. TECHNICIAN LOW OOT RATE DATA	77
TABLE 6-1. IDENTIFICATION ELEMENTS.....	82
TABLE 6-2. TECHNICAL ELEMENTS	84
TABLE B-1. EXAMPLE METHOD A3 INTERVAL ADJUSTMENT CRITERIA.....	107
TABLE B-2. EXAMPLE INTERVAL INCREASE CRITERIA	108
TABLE D-1. TYPICAL OUT-OF-TOLERANCE TIME SERIES.....	119
TABLE H-1. SYSTEM EVALUATION TEST RESULTS:	161

Chapter 1 General

1.1 Purpose

This Recommended Practice (RP) is intended to provide a guide for the establishment and adjustment of calibration intervals for equipment subject to periodic calibration.

1.2 Scope

This RP provides information needed to design, implement and manage calibration interval determination, adjustment and evaluation programs. Both management and technical information are presented in this RP. Several methods of calibration interval analysis and adjustment are presented. The advantages and disadvantages of each method are described, and guidelines are given to assist in selecting the best method for a requiring organization.

The management information provides an overview of interval-analysis concepts and program elements and offers guidelines for selecting an appropriate analysis method.

The technical information is intended primarily for use by technically trained personnel assigned the responsibility of designing and developing a calibration interval-analysis system. Because the subject of calibration interval analysis is not commonly treated in generally available technical publications, much of the methodology is presented herein. Where feasible, this methodology is given in the body of the RP, with advanced mathematical and statistical methods deferred to the Appendices. Statistical or other methods that are not described in detail are referenced.

NOTE: — This RP is *not* a design specification. For the implementation of many of the more sophisticated methodologies described herein, it is not feasible to hand this RP to systems development personnel and expect a functioning system to ensue. Participation by cognizant statistical and engineering personnel is also required.

1.3 The Goal of Interval Analysis

It has been asserted that periodic calibration does not prevent out-of-tolerances from occurring. This point has some validity under certain conditions. Actually, whether the assertion is true or not depends on the nature of the out-of-tolerance process, the adjustment or “renewal” policy of the calibrating facility and so on. All this aside, it can be readily appreciated that, while out-of-tolerances may or may not be prevented by periodic calibration, detection of out-of-tolerances and the amount of time that equipment is *used* in an out-of-tolerance condition can certainly be controlled through periodic calibration. Indeed, it can be shown that, for many equipment models and types, there exists a one-to-one correspondence between the calibration interval of an item and the probability that one or more of its attributes will be used while out-of-tolerance.

From these considerations, the **principal goal** or objective of calibration interval analysis that has evolved from the inception of the discipline is **limiting the usage of out-of-tolerance attributes** to an acceptable level. What determines an acceptable level is discussed throughout this RP under the topic heading of **optimal intervals**.

1.4 The Need for Periodic Calibration

Many diverse calibration interval-analysis and management systems have emerged over the past few decades. This is due in no small part to requirements and recommendations set forth in previous and current national and international standards and guiding documents [45662A, Z540-1, Z540.3, 5300.4, IL07, ISO90, ISO03, ISO05 etc.]. An unambiguous example of these requirements can be found in the U.S. Department of Defense [45662A]. The following statement, taken from the 1 August 1988 issue of this standard describes this requirement:

“[MTE] and measurement standards shall be calibrated at periodic intervals established and maintained to assure acceptable accuracy and reliability, where reliability is defined as the probability that the MTE and measurement standard will remain in-tolerance throughout the established interval. Intervals shall be shortened or may be lengthened, by the contractor when the results of previous calibrations indicate that such action is appropriate to maintain acceptable reliability. The contractor shall establish a recall system for the mandatory recall of MTE and measurement standards to assure timely recalibrations, thereby precluding use of an instrument beyond its calibration due date...”

The current requirements in the quality standard [Z540.3] are no less stringent regarding measurement reliability:

“Measuring and test equipment within the scope of the calibration system shall be calibrated at periodic intervals established and maintained to assure acceptable measurement uncertainty, traceability, and reliability...”

"Calibration intervals shall be reviewed regularly and adjusted when necessary to assure continuous compliance of the specified measuring and test equipment performance requirements."

"The calibration system shall include mandatory recall of measuring and test equipment to assure timely recalibrations and preclude use of an item beyond its calibration due date."

The above requirements stem from the fact that a prime objective is that attributes of products fabricated through a product development process and accepted for use through a product testing process will be fielded in an acceptable condition. If measurement uncertainties in the development and testing processes are excessive, the risk increases that this will not be so. As discussed in Chapter 5, under the topic “Uncertainty Growth,” these uncertainties grow with time elapsed since calibration. Controlling uncertainty growth to levels commensurate with acceptable risk is accomplished through periodic calibration.

In recent years, a growing emphasis on controlling the risk of fielding unacceptable products has been evident in the international marketplace. At present, this emphasis is reflected in international and national guidelines that have been developed for computing and expressing measurement uncertainty [ISO95, NIST94]. See also NC94a. Suppliers that control uncertainty through periodic calibration should be in a more favorable market position than those that do not.

In the past few years another trend that relates to controlling uncertainty through calibration interval analysis has also emerged. Managers of calibrating and testing organizations have begun to realize that minimizing the risk of accepting nonconforming products makes good business sense. Controlling uncertainty through periodic calibration is thus becoming viewed as a viable cost control objective. In meeting this objective, another benefit is realized. Controlling uncertainty not only reduces false-accept risk but also reduces the risk that *in*-tolerance attributes will be perceived as being *out*-of-tolerance. The

benefit of reducing this “false-reject” risk is realized in reduced rework and re-test costs [NA89, HC89, NA94].

1.5 Optimal Intervals

Both producers and consumers agree that high product quality is a worthwhile goal. The quality of a product is often intimately connected to the likelihood that its attributes are within tolerance, i.e., that measurement uncertainty is controlled to an acceptable level. Consequently, minimizing uncertainty is an objective supported by producer and consumer alike.

Likewise, both consumer and producer agree that minimizing costs is a worthwhile goal. Because controlling uncertainty requires investments in test and calibration support, the goal of minimizing costs is often viewed as being at odds with the goal of high product quality.

In brief, the following requirements appear to be in conflict:

- The low false-accept and false-reject requirements for accurate, high quality products and a minimum of unnecessary rework and re-test.
- The requirement for minimizing test/calibration support costs.

Clearly, what is required is a balancing of the benefit of reduced uncertainty against the cost of achieving it. This involves defining what levels of uncertainty are acceptable and establishing calibration intervals that correspond to these levels [NA89, HC89, NA94, MK07, MK08, HC08, SD09]. A corollary to this is that the establishment and adjustment of intervals be done in such a way as to arrive at correct intervals in the shortest possible time and at minimum cost. Calibration intervals that meet all these criteria are referred to as *optimal intervals*. The subject of optimal intervals is discussed in detail in Chapter 2.

1.6 Diversity of Methods

The establishment and adjustment of calibration intervals is often one of the most perplexing and frustrating aspects of managing a test and calibration support infrastructure. The talent pool available to the managing facility is usually devoid of interval-analysis practitioners, and auditors and/or technical representatives from customer organizations are without clear guidelines for the evaluation of interval-analysis methods or systems.

The current best practice for establishing and adjusting calibration intervals is that each calibrating and testing organization select from the methods presented herein the one that best matches the organization’s MTE performance goals, data availability, MTE types, and adjustment policies. Calibration encounters disparate equipment types (electrical, electronic, microwave, physical, dimensional, radiometric, etc.) and each organization establishes its own maximum acceptable uncertainty levels and renewal/adjustment policies, determines what attributes to calibrate to what tolerances, sets cost constraints on interval-analysis expenditures, and establishes calibration and testing procedures. Each of these factors has a direct bearing on which calibration interval-analysis method is optimal for a given organization.

Accordingly, this RP presents several interval-analysis methodologies, together with guidelines for selecting the one best suited to a requiring organization.

1.7 Topic Organization

This RP describes engineering, algorithmic and statistical methods for adjusting calibration intervals. **Appendix A** provides a glossary of relevant terms. The overall management background for calibration interval-analysis is presented in Chapter 2. Interval-analysis program elements are described in Chapter 3, and analysis methodology selection criteria are given in Chapter 4. An overview of technical concepts is presented in Chapter 5. Required data elements are described in Chapter 6, and conditions under which periodic calibration is not required are given in Chapter 7. Mathematical details are, for the most part, presented in the Appendices or are referenced.

It is recognized that different interests are represented in the readership of this RP. The diagram in Figure 1-1 may assist the reader in finding material relative to specific applications or needs.

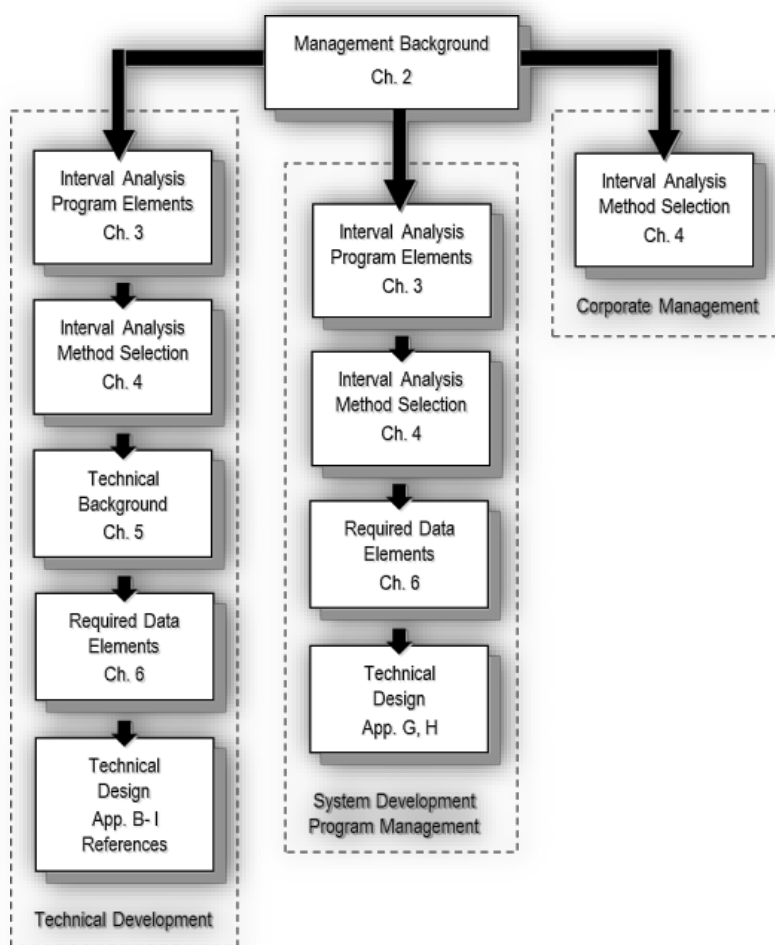


Figure 1-1. RP-1 Reader's Guide

Chapter 2 Management Background

2.1 Introduction

This chapter discusses some of the concepts that are relevant for making decisions regarding the development and/or selection of calibration interval-analysis systems. System program elements are described in more detail in Chapter 3. Specific criteria for selecting an appropriate calibration interval-analysis method are given in Chapter 4.

2.2 The Need for Interval Analysis

MTE (measuring and test equipment) requires calibration to ensure that MTE attributes are performing within appropriate specifications. Because the uncertainties in the values of such attributes tend to grow with time since last calibrated, they require periodic recalibration to maintain end-product quality. For cost-effective operation, intervals between recalibrations should be optimized to achieve a balance between operational support costs and the MTE accuracy required to verify acceptable product quality [NA89, HC89, NA94, MK07, HC08, MK08, SD09].

As the uncertainties in the values of attributes grow with time since calibration, the probability that the attributes of interest will be in-tolerance, known as the measurement reliability, correspondingly diminishes, potentially impacting product quality. Controlling uncertainty growth to an acceptable maximum is therefore equivalent to controlling in-tolerance probability and product quality to an acceptable minimum. This acceptable minimum in-tolerance probability is referred to as the **measurement reliability target**.

2.3 Measurement Reliability Targets

A fundamental quality-control objective is that tests, measurements or other verifications of MTE attributes yield correct accept or reject decisions. Errors in such decisions are directly related to the uncertainties associated with the verification process. One contributor to this uncertainty is the uncertainty in the values of test or calibrating attributes. This uncertainty is a function of the percent of items that are in-tolerance at the time of measurement, i.e., of the **measurement reliability**.

Measurement decision errors can be controlled in part by holding measurement reliabilities of test and calibration systems at acceptable levels. What constitutes an acceptable level is a function of the level of **measurement decision risk** acceptable to management. Measurement decision risks are commonly expressed as the probability of rejecting conforming (in-tolerance) units or accepting nonconforming (out-of-tolerance) units. The first risk is labeled **false-reject risk** and the second is called **false-accept risk**.

What constitutes acceptable risks, then, are the levels of false-reject risk and false-accept risk that are consistent with cost-control requirements (minimize false-reject risk) or quality control objectives (minimize false-accept risk). For example, the quality standard [Z540.3] prescribes false-accept risk requirements and [NC90] includes guidance for the preparation of calibration procedures to meet false-accept risk requirements.

Several sources can be consulted for methods of computing measurement decision risks. A comprehensive list would include references [JF84, HC80, SW84, JL87, AE54, KK84, FG54, NA89, HC89, DD93, DD94, DD95, NA94, HC95a, HC95b, HC95c, JH55, JF95, RK95]. Many more recent references exist also; however, the forthcoming *NCSL International RP-18, Estimation and Evaluation of*

Measurement Decision Risk, is perhaps the most comprehensive compilation on the subject for metrology.

2.4 Calibration Interval Objectives

The immediate objective of calibration interval-analysis systems is the establishment of calibration intervals that ensure that measurement decision risks are under control. In addition to controlling risks, a major objective of any calibration interval-analysis system should be minimizing the analysis cost per interval.

2.4.1 Cost Effectiveness

The objectives of controlling risks and minimizing analysis cost per interval lead to the following criteria for cost-effective calibration interval-analysis systems:

1. Measurement reliability targets correspond to measurement uncertainties commensurate with measurement decision risk control requirements.

Product utility is compromised and operating costs (total support and consequence costs) are increased if incorrect decisions are made during testing. The risk of making these decisions is controlled through holding MTE uncertainties to acceptable levels, although this should be balanced against the costs of attaining those uncertainty levels. This is done by optimizing MTE measurement reliabilities, a topic outside the scope of this RP. These optimum levels are the measurement reliability targets.

2. Calibration intervals lead to observed measurement reliabilities that are in agreement with measurement reliability targets.

For the majority of MTE attributes, measurement reliability decreases with time since calibration. The particular elapsed time since calibration that corresponds to the established measurement reliability target is the desired calibration interval.¹

3. Calibration intervals are determined cost-effectively.

A goal of any calibration interval-analysis system should be that the analysis cost per interval is held to the minimum level needed to meet measurement reliability targets. This can be accomplished if calibration intervals are determined with a minimum of human intervention and manual processing, i.e., if the interval-analysis task is automated. Minimizing human intervention also entails some development and implementation of decision algorithms. Full application of advanced AI methods and tools is not ordinarily required. Simple functions can often be used to approximate human decision processes.

4. Calibration intervals are arrived at in the shortest possible time.

¹ In some applications, periodic MTE recalibrations are not possible (as with MTE on board deep space probes) or are not economically feasible (as with MTE on board orbiting satellites). In these cases, MTE measurement uncertainty is controlled by designing the MTE and ancillary equipment or software to maintain a measurement reliability level that will not fall below the minimum acceptable reliability target for the duration of the mission.

Several methods for determining calibration intervals are currently in use. However, many of them are not capable of meeting criterion 2; i.e., they do not arrive at correct intervals consistently. Certain others are capable of meeting that criterion, but require long periods of time to do so. In most cases, the period required for these methods to arrive at intervals that are consistent with measurement reliability targets exceeds the operational lifetime of the MTE of interest [DJ86a]. Fortunately, there *are* methods that meet criterion 2 and do so in short order. These methods are described in this RP.

5. Analytical results are easily generated and implemented.

In cost-effective systems, analytical results can be easily implemented. The results should be comprehensive, informative and unambiguous. Mechanisms should be in place to couple or transfer the analytical results directly to laboratory or enterprise management software with a minimum of human intervention.

6. System development costs are less than the expected return on investment.

This is often the overriding concern in selecting an interval-analysis methodology. For instance, although certain methods described in this RP can be shown in principle to be decidedly superior to others in terms of meeting objectives 2 to 5 above, the cost of their development and implementation may be higher than their potential benefit. On the other hand, if the cost savings delta between alternative methods exceeds the investment delta, then the magnitude of the investment should not act as a deterrent. This consideration will be discussed in more detail in Chapter 4.

2.4.2 System Responsiveness

To ensure that calibration intervals assigned to equipment reflect current measurement reliability behavior, interval-analysis systems should be responsive to any changes in the makeup of MTE or the policies that govern MTE management and use. This means that systems should be able to respond quickly to new calibration history data generated since the previous analysis. In general, responsiveness is maximized when an initial calibration interval is determined or an existing interval is reevaluated as soon as enough new data have been accumulated to determine an initial interval or change an existing one. (As can be readily seen, the responsiveness feature may sometimes be mediated by the need to minimize calibration interval-analysis costs.)

What constitutes “enough” new data differs from case to case. This question is addressed at appropriate places in this RP.

2.4.3 System Utility and Potential Spin-Offs

The utility of a calibration interval system is evaluated in terms of its effectiveness, ease of use and relevance of analytical results. Included in these results may be a number of “spin-offs,” i.e., by-products of the system. Because of the nature of the data they process and the kinds of analyses they perform, certain calibration interval-analysis systems are more capable of providing spin-offs than other analysis systems by further analyzing the same data used for interval analysis.² Spin-offs known to be of benefit to MTE users and managers of calibration systems include the following:

² The spin-offs discussed in this section are possible consequences of systems that employ Methods S1, S2 or S3, discussed later, on page 23.

One potential spin-off is the identification of MTE with exceptionally high or low uncertainty growth rates (“dogs” or “gems,” respectively). Dogs and gems can be identified by MTE serial number and by manufacturer/model. Identifying serial number dogs helps weed out poor performers (invoking decommissioning, repair, upgrade, or replacement actions) and identifying serial number gems helps in selecting items to be used as check standards. Model number dog and gem identification can also assist in making procurement decisions.

Other potential spin-offs include providing visibility of trends in uncertainty growth rate or calibration interval, identification of users associated with exceptionally high incidences of out-of-tolerance or repair, projection of test and calibration workload changes to be anticipated as a result of calibration interval changes, and identification of calibrating organizations (vendors), calibration procedures, or technicians that generate unusual data patterns.

Calibration interval-analysis systems also offer some unique possibilities as potential test beds for evaluating alternative reliability targets, renewal or adjustment policies, and equipment tolerance limits in terms of their impact on calibration workloads.

Finally, interval-analysis systems provide information needed to estimate reference attribute bias uncertainty, a spin-off that is highly useful in analyzing and reporting uncertainties [HC95a, HC95b, HC95c].

2.5 Optimal Intervals

Calibration intervals that meet reliability targets, are cost-effective, are responsive to changing conditions and are determined in a process that leads to useful spin-offs are considered optimal. Throughout this RP, interval-analysis methods and systems will be evaluated in terms of optimality as stated here.

2.6 Calibration Interval-Analysis Methods

Although this document is a “Recommended Practice,” there is no single interval-analysis method that can be recommended for all calibrating or testing organizations. The method that best suits a given organization is one that is consistent with inventory size, quality objectives, system development and maintenance budgets, available personnel, available automated data processing (ADP) hardware and software, risk management criteria, and potential return on investment.

The various practices that are currently available or are under development can be categorized into five methodological approaches:

- General interval
- Borrowed Intervals
- Engineering Analysis
- Reactive Methods
- Maximum Likelihood Estimation Methods

Each of these approaches is discussed below in general terms.

2.6.1 General Interval Method

Facilities with small homogeneous inventories or little emphasis on controlling measurement reliability sometimes employ a single calibration interval for all MTE. After deciding on the interval to use, this approach is easy to implement and administer. It is, however, the least optimal method with respect to establishing intervals commensurate with measurement-decision risk-control objectives.

The approach is also used, even by organizations with large inventories, to set initial intervals for newly acquired MTE. In this case, a short interval (e.g., two to three months) is the most common choice for a general interval. This is partly because a short interval will accelerate the accumulation of calibration history, thereby tending to spur the determination of an accurate interval. A short interval also provides a sense of well-being from a measurement-assurance standpoint in cases where the appropriate interval is unknown.

The expedient of setting a short interval may, however, lead to exorbitant initial calibration support costs and unnecessary disruptions in equipment use due to frequent recall for calibration. Fortunately, more accurate initial intervals can be obtained by employing certain refinements. These are discussed in the following sections.

2.6.2 Borrowed Intervals Method

Rather than settle on a single common interval, some organizations employ calibration intervals determined by an external organization. If so, it is important that the external organization be similar to the requiring activity with respect to reliability targets, calibration procedures, usage, handling, environment, etc. If there are differences in these areas, modifications may need to be made to the “borrowed” intervals. Borrowed interval modifications may be the result of engineering judgment or may consist of mathematical corrections, as described in Appendix F.

Intervals may also be computed from calibration history data provided externally. For example, the U.S. Department of Defense shares data among the armed services. Large equipment reliability data bases such as [GIDEP] and the Navy’s MIDAS [ML94] may also be consulted. As a word of caution, some foreknowledge is needed of the quality and relevance of data obtained externally to ensure compatibility with the needs of the requiring organization.

2.6.3 Engineering Analysis Method

Engineering considerations may be used to establish and adjust intervals. Typically, engineering analysis means using:

- Similar Item Intervals
- Manufacturer’s Recommended Intervals and Technical Support
- Detailed Component Reliability Analysis

These three considerations are discussed below:

2.6.3.1 Similar Items

Often, MTE is an updated version of an existing product line. It may be the same as its predecessor except for a minor or cosmetic modification. In such cases, the new item should be expected to have performance characteristics similar to its parent model. Often, the parent model will already have an

established calibration history and an assigned calibration interval. If so, the new model can be assigned the recall interval of the parent model.

In like fashion, when no *direct* family relationship can be used, the calibration interval of MTE of similar complexity, similar application, and employing similar design and fabrication technologies may be appropriate. MTE that are **closely related** with respect to these variables are called **similar items**. Equipment that is **broadly related** with respect to these variables composes an **instrument class**. Instrument classes are discussed later.

2.6.3.2 Manufacturer Data / Recommendations

Another source of information is the MTE manufacturer. Manufacturers may provide recommended calibration interval information in their published equipment specifications. These recommendations are sometimes based on analyses of stability at the attribute level. To be valid, they need to accommodate three considerations:

- 1) The attribute tolerance limits;
- 2) A specified period over which the attribute values will be contained within the tolerance limits
- 3) The probability that attributes will be contained within the tolerance limits for the specified period.

Unfortunately, manufacturers are often cognizant of or communicative about only one or, at best, two of these points. Accordingly, some care is appropriate in employing manufacturer interval recommendations. If manufacturer recommended intervals *per se* are in question, supporting data and manufacturer expertise may nevertheless be helpful in setting initial intervals.

For additional information on this subject, see *NCSL International RP-5, Measuring and Test Equipment Specifications*.

2.6.3.3 Design Analysis

Another source of information is the design of the equipment. Cognizant, knowledgeable engineers can often provide valuable information concerning the equipment by identifying, describing and evaluating the calibration critical circuits and components of the equipment in question. An accurate calibration interval prediction may be possible in lieu of calibration history data when the equipment's calibratable measurement attribute aggregate out-of-tolerance rate (OOTR) is determined via circuit analysis and parts performance. The OOTR can be applied, as if it were obtained from field calibration data, to determine an estimate of initial calibration interval.

2.6.4 Reactive Methods

An analysis of calibration results may suggest that an interval change is needed for reasons of risk management or quality control. The simplest analytical methods are those that “react” to calibration results in accordance with a predetermined algorithm. Several algorithms are currently in use or have been proposed for use. They vary from simple “one-liners” to fairly complex statistical procedures. The reactive algorithms described in this RP are the following:

- Method A1 - Simple Response Method
- Method A2 - Incremental Response Method
- Method A3 - Interval Test Method

2.6.4.1 Method A1 - Simple Response Method

With the Simple Response Method, the interval for a given item of MTE is adjusted at each calibration or, at most, after two or three calibrations. Adjustments are either up, if the MTE is found to be in-tolerance, or down, if out-of-tolerance. The magnitude of each adjustment is either a fixed increment or a multiple of the existing interval. A serious drawback of the Simple Response Method is that, since adjustments are made in response to recent calibration results, it is not possible to maintain an item on its “correct” interval.

The Simple Response Method is described in Appendix B. For reasons detailed there and elsewhere in this RP, Method A1 is not recommended but remains documented in this RP to discourage its “reinvention” and maintain awareness of the drawbacks of similar methods.

2.6.4.2 Method A2 - Incremental Response Method

The Incremental Response Method compensates for Method A1’s unending adjustments by progressively shrinking the size of the interval increment at each adjustment. In this way, an item is allowed to approach a final interval asymptotically and remain there, though it does not do so expeditiously. Often, periods as long as five to sixty years are required to reach intervals commensurate with established reliability targets, and considerable flopping around is done in the process.

The Incremental Response Method is described in Appendix B. Like Method A1, Method A2 is not recommended, but remains documented to discourage its use.

2.6.4.3 Method A3 - Interval Test Method

A reactive method that both attains correct intervals in reasonable periods and produces no spasmodic interval fluctuations is the Interval Test Method. In this method, intervals are adjusted only if recent accumulated calibration results are inconsistent with expectations. This consistency is evaluated by statistical testing. The method is described in Appendix B.

2.6.5 Maximum Likelihood Estimation (MLE) Methods

MLE methods are decidedly better than reactive methods at reaching correct intervals. Unfortunately, MLE methods require substantial amounts of data for analysis. Roughly twenty to forty observations (in- or out-of-tolerance events) are needed, depending on the specific method used.

The required number of observations also varies with the homogeneity of the grouping used to accumulate data. For instance, if data are grouped by model number, approximately thirty observations are required. If data are grouped by Instrument Class, about forty observations are needed. If data are accumulated for a single serial number, it is possible to get by with twenty or so observations.

At least three MLE methods are in use or are proposed for implementation. They are:

- Method S1 - Classical Method
- Method S2 - Binomial Method
- Method S3 - Renewal Time Method

2.6.5.1 Method S1 - Classical Method

Method S1 is the simplest and least costly MLE method to implement. It employs classical reliability analysis methods to construct what is called a **likelihood function**. In constructing this function, it is required that the time of occurrence of each out-of-tolerance be known. Unfortunately, this time, referred to as the **failure time**, is almost never known in a calibration context. In this context, we know the in- or

out-of-tolerance status of MTE attributes at the beginning and end of each calibration interval, but not what happens in between.

To circumvent this, the Method S1 estimates failure times. The question is, obviously, how do we estimate a failure time within an interval if all we know is the in- or out-of-tolerance status at the beginning and end of the interval?

The answer is that there is no really good way to make this guess unless the uncertainty growth process follows a particular reliability model, called the **exponential model**. With the exponential model, we can reasonably surmise that each out-of-tolerance occurred halfway between the start and the end of the interval.

With other models, we cannot make a reasonable guess without first knowing the answer. We could use bootstrapping methods to make failure time guesses, but this involves considerable analytical complexity and suffers from the fact that the final answer often depends on what value we use to start the process. So, with the classical method, we are basically stuck with the exponential model.

Unfortunately, given the diversity of current MTE composition and usage, it can be shown that reliance on a single reliability model often leads to suboptimal intervals [HC94].

The upshot of the foregoing is that the Method S1, while more attractive than other MLE methods from the standpoint of simplicity and cost of implementation, may not be cost effective from a total cost perspective.

Method S1 is described in Appendix C.

2.6.5.2 Method S2 - Binomial Method

Unlike Method S1, Method S2 is not restricted to a single reliability model, nor is it hampered by the fact that failure times are unknown. Moreover, Method S2 has been implemented in large-scale automated interval-analysis systems and has performed with impressive success, such as with the Equipment Recall Optimization System (EROS) system [HC78].

With the EROS system, for example, in the first full year of operation, the cost savings due to interval optimization exceeded the entire system development cost by more than forty percent. In addition, system operating costs resulted in a unit cost of twenty-three cents per interval. Reliability targets were reached and a host of spin-offs were generated.

An advantage of Method S2 is that it can easily accommodate virtually any reliability model. This means that Method S2 is suitable for establishing intervals for essentially all types of MTE, both present and future.

The downside of Method S2 is that system development and implementation are expensive and require high-level system analysis and statistical expertise. Method S2 also works best if the “renew always” practice is in effect for attribute adjustment, although “renew-if-failed” and “renew-as-needed” practices can be accommodated as well. Method S2 is described in Appendix D.

2.6.5.3 Method S3 - Renewal Time Method

Method S3 is as robust as Method S2 in its ability to accommodate a variety of reliability models and to analyze unknown failure times. Additionally, Method S3 is more robust than Method S2 with respect to

renewal practice. With Method S3, it does not matter what the renewal practice is, only that calibration history records indicate whether renewals have taken place.

In lieu of this, a specific renewal practice must be assumed. Except for its superior ability to handle renewal alternatives, Method S3 has the same advantages and disadvantages as Method S2. Method S3 is described in Appendix D.

2.6.6 Other Methods

As mentioned elsewhere, the optimal interval adjustment method depends on the organization's requirements. For this reason, a plethora of methods exist in industry, some of which are variants of the methods discussed in this RP. A search of the literature will uncover many proposed methods developed for specific organizations' goals. While many of these other methods may be viable for general use, it is not practical to make a general statement regarding their effectiveness. However, one method under development by the U. S. Navy, which may appear in future editions of this RP, uses intercept reliability models and generalized linear models analysis. See [DJ03b]. Another potential approach is variables data analysis [DJ03a], [HC05].

2.7 Interval Adjustment Approaches

There are four major approaches to calibration interval adjustment illustrated by Figure 2-1. This section discusses each approach in the typical order of consideration when developing an interval-analysis system:

1. Adjustment by serial number
2. Adjustment by model number
3. Adjustment by similar items group
4. Adjustment by instrument class
5. Adjustment by attribute

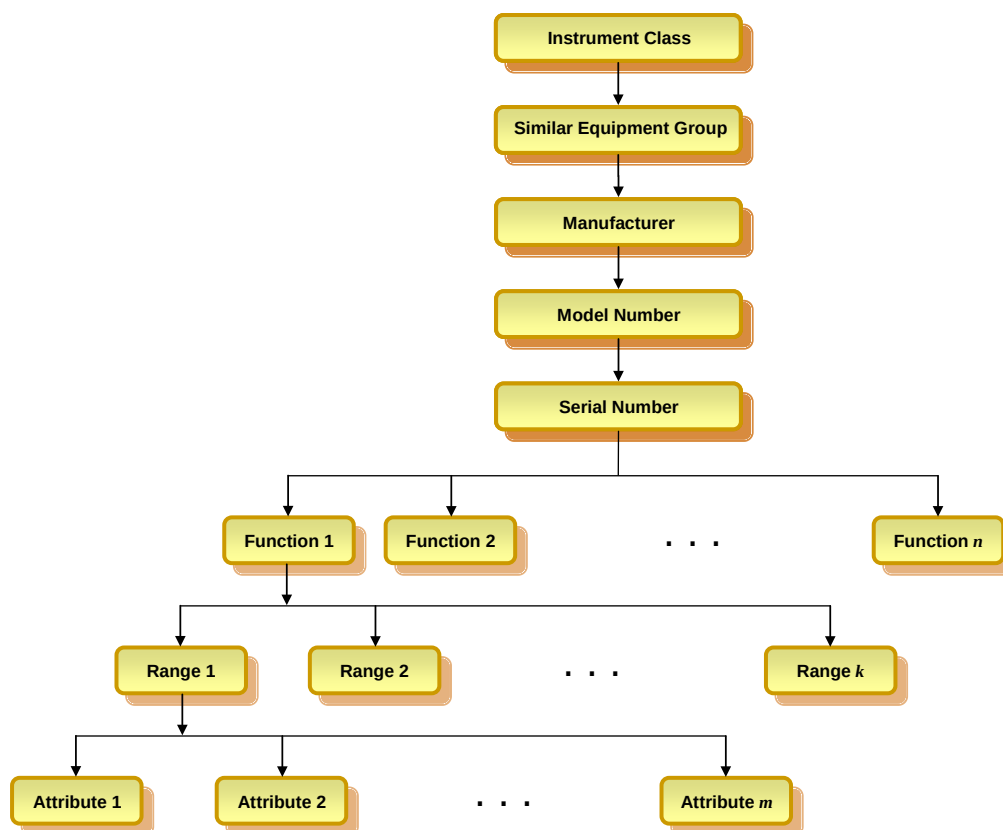


Figure 2-1. Interval-Analysis Taxonomy

2.7.1 Adjustment by Serial Number

Even though serial-numbered items of a given model manufacturer group are similar, they are not necessarily identical. Also, the nature and frequency of the use of individual items and their in-use environmental conditions may vary. Thus, some may perform better and others may perform worse than the average. For this reason, some organizations adjust calibration intervals at the individual serial-number level. The various methods used base such adjustments on the calibration history of each individual item and give simple-to-complicated rules or table look-up procedures. Most of these methods assume that the “correct” calibration interval for an individual instrument is subject to change over its life span, and that, therefore, only data taken from recent calibrations are relevant for establishing its interval.

It has been shown (Ref. *DJ86a*) that, with regard to establishing a “correct” interval for an item, enough relevant data can rarely be accumulated in practice at the single serial number level to achieve this purpose. Even if the restriction of using only recent data could be lifted, it would take several years (often longer than the instrument's useful life) to accumulate sufficient data for an accurate analysis. These considerations argue that calibration intervals cannot, in practice, be rigorously analyzed at the serial-number level.

2.7.2 Adjustment by Model Number

Each serial numbered item of a given model number is typically built to a uniform set of design and component specifications. Moreover, even though design and/or production changes may occur over time, items of the same model number are generally expected to meet a uniform set of published performance

specifications. For these reasons, most serial numbered items of a given model number should be expected to exhibit fairly homogeneous measurement reliability behavior over time, unless demonstrated otherwise.

Grouping by model number often permits the accumulation of sufficient data for statistical analysis and subsequent interval adjustment. Ensuring homogeneous behavior within the group is imperative. For model number grouping, this means that all serial numbers within the group should be subjected to roughly the same usage and are calibrated in accordance with the same procedure to the same accuracy in all attributes.

2.7.2.1 Dog and Gem Identification

The requirements for statistically valid calibration intervals and the need for responsiveness to individual instrument idiosyncrasies can both be addressed by incorporating a means of statistically identifying exceptional equipment or “outliers” within a model number. In such schemes, calibration data are kept by serial number for the given model number. Items with significantly higher and lower out-of-tolerance frequencies than are characteristic of the group may be flagged by serial number. Statistical outliers identified in this way are commonly referred to as “dogs” (high out-of-tolerance rate) and “gems” (low out-of-tolerance rate). The presence of dogs or gems unduly shortens or lengthens the calibration interval for the other items in a model number group. Additionally, removing these outliers from a model number analysis provides greater assurance that the assigned interval is applicable to representative members of the model number group. This practice assumes that outliers will be managed differently from mainstream group members.

2.7.2.2 Dog and Gem Management

Once dogs and gems are identified, considerable latitude is possible regarding their disposition. For example, dogs may require shortened intervals, complete overhaul, removal from service, certification for limited use only, etc. On the other hand, gems may qualify for lengthened intervals or designation as critical support items or higher level standards.

2.7.3 Adjustment by Similar Items Group

A grouping of manufacturer/models that are expected to exhibit similar uncertainty growth mechanisms is called a *similar items group* or *similar equipment group*. Such a group may consist of model numbers that are related by manufacturer and fabrication, such as A and B versions of a model number or stand-alone and rack-mounted versions. The group may include items from different manufacturers, provided they are “equivalent” with respect to function, complexity, fabrication, tolerances and other such factors. A good criterion to use when including items in a similar items group is to require that group members be usable as equipment substitutes. Refer to the Chapter 5 topic “Data Consistency” for quantitative homogeneity tests.

Calibration interval-analysis at the similar-items group level is performed in the same way as analysis at the model number level, with data grouped according to similar-items group rather than model number for interval-analysis and by model number rather than serial number for dog-and-gem analysis. As with analysis by instrument class, identifying model number dogs and gems within a similar items group can assist in making equipment procurement decisions.

2.7.4 Adjustment by Instrument Class

An **instrument class** is a homogeneous grouping of equipment model numbers. If sufficient data for calibration interval-analysis are not available at the model number or similar equipment group level,

pooling of calibration histories from model numbers or groups within a class may yield sufficient data for analysis. The results of such an analysis may be applied to model number items within the class. Once a class has been defined, homogeneity tests should be performed whenever possible to verify the validity of the class grouping (see Chapter 5).

Several criteria are used to define a class. These include commonality of function, application, accuracy, inherent stability, complexity, design and technology. Interestingly, one simple class definition scheme that has proved to be effective consists of subgrouping by acquisition cost within standardized noun nomenclature categories. Apparently, some equipment manufacturers have already performed comparative analyses of the aforementioned criteria and have adjusted prices accordingly.

Calibration interval-analysis at the class level is performed in the same way as analysis at the model number level, with data grouped according to class rather than model number for interval-analysis and by model number or similar items group rather than serial number for dog-and-gem analysis. An interesting consequence of model number dog-and-gem analysis is that flagging model number dogs and gems can provide information for making equipment procurement decisions.

2.7.5 Adjustment by Attribute

Although periodic calibration recall schedules are implemented at the serial number or individual MTE level, uncertainty growth, described on page 2, occurs at the attribute level. For this reason, it makes sense to perform calibration interval-analysis at the attribute level, rather than at the serial-number level. Once data are analyzed and intervals assigned by attribute, algorithms can be employed to develop an item's recall interval from its attribute calibration intervals. Note that the attribute data can be grouped by serial number, model number or at any other level in Figure 2-1, depending on the amount of data available

In the past, calibration history data were not widely available at the attribute level. At best, these data were available at the serial-number level. For this reason, the interval-analysis methods discussed in this RP are usually applied to in- or out-of-tolerance units, rather than to in- or out-of-tolerance attributes. However, there is no reason why these methods cannot be extended to apply to observations recorded by attribute.

At present, calibration history data are becoming more readily available at the attribute level. This is because calibration in general increasingly depends on automated calibration systems in which data collection by attribute is feasible. In addition, in cases where calibrations remain essentially manual, many procedures have calibrating technicians enter measured values by keyboard or other means.

The subject of attribute calibration intervals is a current research topic. Analysis methodologies will be reported in future updates to this RP.

2.7.6 Stratified Calibration

In addition to being superior in terms of uncertainty growth analysis, analyzing and assigning intervals by attribute has another advantage. With attribute interval assignment, stratified calibration becomes feasible.

With stratified calibration, only the shortest interval attribute(s) is (are) calibrated at every MTE resubmission. The next shortest interval attribute is calibrated at every other resubmission, the third shortest at every third resubmission and so on. Such a calibration schedule is similar to maintenance schedules, which have been proven effective for both commercial and military applications.

2.8 Data Requirements

The data collection requirements vary for each interval-analysis method and the desired spin-offs. Ideally then, the choice of interval-analysis systems and calibration laboratory data management systems should be coordinated. If however, as is generally the case, one is selecting an interval-analysis system when the data management system is already in place, or vice versa, the data requirements may impact the choice of systems, restrict the choice of interval-analysis methods, or require modifications to the data management system. For further information, refer to the Chapter 3 topic “Data Collection and Storage,” the Chapter 4 “Data Availability Requirement” topics under each method, and Chapter 6 “Interval-analysis Data Elements.”

2.9 System Evaluation

Just as periodic calibration is necessary to verify the accuracy of MTE, periodic evaluation of a calibration interval-analysis system is necessary to verify its effectiveness. Such evaluations are possible only if predetermined criteria of performance have been established. One such criterion involves comparing observed (recorded) measurement reliabilities against measurement reliability targets.

Agreement between observed measurement reliability and a designated reliability target can be evaluated by comparing the actual percent in-tolerance at calibration (observed measurement reliability) to the designated end-of-period (EOP) reliability target for a random sample of serial numbered items that are representative of the inventory. If the observed measurement reliabilities for the sampled items differ appreciably from the EOP reliability target, the interval-analysis system is in question.

A guideline for evaluating whether measurement reliabilities differ appreciably from target reliabilities is provided in Appendix H. NCSL International included an evaluation tool that performs this evaluation with previous editions of this RP. A current and regularly updated version is now available as freeware on the internet [IE08].

Chapter 3 Interval-Analysis Program Elements

3.1 Introduction

Implementing a calibration interval-analysis capability within an organization can have an impact on facilities, equipment, procedures and personnel. To assist in evaluating this impact, several of the more predominant program elements related to calibration interval-analysis system design, development and maintenance are described below. These elements include:

- Data collection and storage
- Data Analysis
- Guardband Use
- Measurement reliability modeling and projection
- Engineering review
- Logistics analysis
- Imposed Requirements
- Cost /benefits assessment
- Personnel requirements
- Training and communications

3.2 Data Collection and Storage

Calibration history data are required to infer the time dependence of MTE uncertainty growth processes. These data need to be complete, homogeneous, comprehensive and accurate. Required data elements are discussed in Chapter 6.

3.2.1 Completeness

Data are complete when no calibration service actions are missing. Completeness is assured by recording and storing all calibration results.

3.2.2 Homogeneity

If calibration history data are used to infer uncertainty growth processes for a given instrument or equipment type, the data need to be homogeneous with respect to the type. Data are homogeneous when all calibrations on an equipment grouping (e.g., manufacturer/model) are performed to the same tolerances by use of the same procedure.

3.2.3 Comprehensiveness

Data are comprehensive when both “condition received” (received for calibration) and “condition released” (deployed following calibration) are unambiguously specified for each calibration. Depending on the extent to which an interval-analysis system is used to optimize calibration intervals and to realize spin-offs (see below), data comprehensiveness may require that other data elements are also captured. These data elements include date calibrated, date released, serial or other individual ID number, model number and standardized noun nomenclature. Additionally, for detection of facility and technician outliers the calibrating facility designation and technician identity should be recorded and stored for each

calibration. Finally, if intervals are to be analyzed by attribute, calibration procedure step number identification is a required data element.

3.2.4 Accuracy

Data are accurate when they reflect the actual perceived condition of equipment as received for calibration and the actual condition of equipment upon release from calibration. Data accuracy depends on calibrating personnel using data formats properly. Designing these formats with provisions for recording all calibration results noted and all service actions taken can enhance data accuracy.

3.3 Data Analysis

The following conditions are necessary to ensure the accuracy and utility of interval adjustments:

- Calibration history data are complete and comprehensive; a good rule is to require data to be maintained by serial number with all calibrations recorded or accounted for.
- Calibration history data are reviewed and analyzed, and calibration intervals (initial or previously adjusted) are adjusted to meet reliability targets.
- Interval adjustments are made in such a way that reliability requirements are not compromised.

Some amplification is needed as to when review and analysis of calibration history data are appropriate. Review is appropriate when any of the following applies:

- Sufficient data to justify a re-analysis have been accumulated.
- Some relevant procedural or policy modification (changes in calibration procedure, reliability target, equipment application or usage, etc.) has been implemented since the previous interval assignment or adjustment.
- Equipment is known to have a pronounced performance trend, and enough time has elapsed for the trend to require an interval change.

For analyses performed in batch mode on accumulated calibration history, quarterly to annual review and analysis should be sufficient for all but “problem” equipment, critical application equipment, etc.

3.4 Guardband Use

The calibration organization’s guardbanding policy should be reviewed and perhaps supplemented when implementing an interval-analysis program. The quality system may already employ guardbands to reduce false-accept risk, or more rarely, to reduce false-reject risk, due to significant measurement uncertainty in either case. Advanced policies may use guardbands to establish a happy medium between false-accept risks and false-reject risks. If the cost of a false-reject risk is prohibitive, for example, it may be desired to set guardbands that reduce false-reject risk at the expense of increasing false-accept risk. If, on the other hand, the cost of false accepts is prohibitive, it may be desired to reduce this risk at the expense of increasing false-reject risk.

For interval-analysis purposes, however, the decision as to whether an attribute's value represents an out-of-tolerance may be improved by setting reporting guardband limits that equalize false-accept and false-reject risks such that observed reliability is not biased. The attribute is then said to be out-of-tolerance if its observed value lies outside its reporting guardband limits. Therefore, the same guardband limits will

not, in general, serve all purposes. The following sections discuss this in more detail. See also Appendix G.

3.4.1 Compensating for Perception Error

Typically, testing and calibration are performed with safeguards that cause false-accept risks to be lower than false-reject risks. This is characteristic, for example, of calibration or test equipment inventories with pre-test in-tolerance probabilities higher than 50 %. The upshot of this is that, due to the imbalance between false-accept and false-reject risks, the perceived or *observed* percent in-tolerance will be lower than the actual or *true* percent in-tolerance. Observed out-of-tolerances have a higher probability than true out-of-tolerances. Ferling first mentioned this in 1984 as the “True vs. Reported” problem.

As will be discussed in the next section, this discrepancy can have serious repercussions in setting test or calibration intervals. Since these intervals are major cost drivers, the True vs. Reported problem should not be taken lightly.

Through the judicious use of guardband limits, the observed percent in-tolerance can be brought in line with the true in-tolerance percentage. With pre-test in-tolerance probabilities higher than 50 %, this usually means setting test guardband limits outside the tolerance limits. This practice may seem to be at odds with using guardband limits to reduce false-accept risk. Clearly, one guardband limit cannot simultaneously accomplish both goals. This issue will be returned to below in the discussion on Guardband Limit Types. See *NCSL International RP-18, Estimation and Evaluation of Measurement Decision Risk*, for the applicable equations used to set guardband limits, or alternatively, to estimate true measurement reliability from observed measurement reliability.

3.4.2 Implications for Interval Analysis

If intervals are analyzed using test or calibration history and high reliability targets are employed, the intervals ensuing from the analysis process can be seriously impacted by observed out-of-tolerances. In other words, with high reliability targets, even only a few observed out-of-tolerances can drastically shorten intervals.

Since this is the case, and because the length of test or calibration intervals is a major cost driver, it is prudent to ensure that perceived out-of-tolerances not be the result of false-reject risk. This is one of the central reasons why striving for reductions in false-accept risk must be made with caution, because reductions in false-accept risk increase false-reject risk. At the very least, attempts to control false-accept risk should be made with cognizance of the return on investment and an understanding of the trade-off in increased false-reject risk and shortened calibration intervals. Therefore, reliability data should not be generated by comparison with those guardband limits chosen to reduce false-accept limits.

3.5 Limit Types

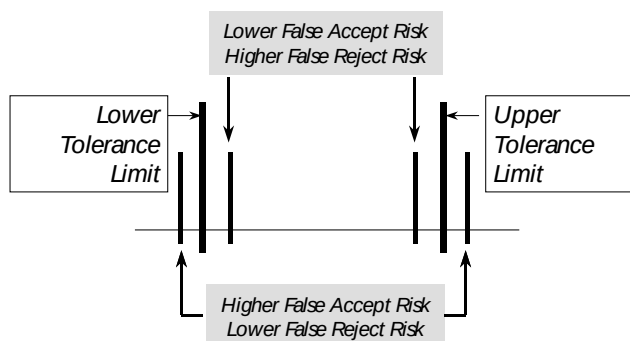


Figure 3-1. Adjustment vs. Reporting Limits. Setting guardband limits inside the tolerance limits reduces false-accept risk but increases false-reject risk. Setting guardband limits outside the tolerance limits has the opposite effect.

To accommodate both the need for low false-accept risks and accurate in-tolerance reporting, two sets of guardband limits must be employed. One, ordinarily set inside the tolerances, would apply to withholding items from use or to triggering attribute adjustment actions. The other, ordinarily set outside the tolerances, would apply to in or out-of-tolerance reporting.

3.5.1 Adjustment Limits

The first set, adjustment limits, are those that are normally thought of when guardbands are discussed. This category includes the guardband limits used to reduce or to control the risk of falsely accepting (releasing) out-of-tolerance items due to measurement uncertainty. As such, adjustment limits are criteria that the *as-left* attribute values must meet before release. Because the observed measurement reliability used to set intervals is an end-of-period metric, the as-left values (beginning-of-period data), and hence the adjustment limits, are ignored. While quality standards vary regarding requirements for statements of conformance with specifications, it should be noted that reporting all as-found values outside the adjustment limits as out-of-tolerance exacerbates the “True vs. Reported” problem and increases the probability that reported failures are false.

NOTE 1: — Adjustment limits are used to flag cases requiring repair, adjustment or rework. Adjustment limits should not be used to determine the end-of-period out-of-tolerance state!

3.5.2 Reporting Limits

Reporting limits are used to compensate for the True vs. Reported problem discussed earlier. An attribute would be *reported* as out-of-tolerance only if its *as-found* value fell outside its reporting limits.

NOTE 2: — Reporting limits are used as pass-fail criteria.

3.5.3 Summary

Separate reporting limits selected to balance false rejects and false accepts provide an unbiased estimate of measurement reliability and should be used where feasible. Failing that, the observed measurement reliability should be derived from the actual tolerance limits in force, which then become the *ipso facto*,

but biased, reporting limits. Measurement reliability should never be estimated with respect to adjustment or guardband limits set strictly to control false accepts.

3.6 Measurement Reliability Modeling and Projection

Uncertainty growth processes are described in terms of mathematical reliability models. Reliability models are used to project measurement reliability as a function of interval, and intervals are computed that are commensurate with reliability targets.

Because attribute drift and other changes are subject to inherently random processes and to random stresses encountered during usage, reliability modeling requires the application of statistical methods. Statistical methods can be used to fit reliability models to uncertainty growth data and to identify exceptional (outlier) circumstances or equipment.

3.7 Engineering Review

Engineering analyses are performed to establish homogeneous MTE groupings (e.g., standardized noun nomenclatures), to provide sanity checks of statistical analysis results, and to develop heuristic interval estimates in cases where calibration data are not sufficient for statistical analysis (e.g., initial intervals).

3.8 Logistics Analysis

Logistics should be considered from an overall cost, risk, and effectiveness standpoint with regard to synchronizing intervals to achievable maintenance schedules or synchronizing intervals for related MTE models, such as mainframes and plug-ins, which are used together.

3.9 Imposed Requirements

3.9.1 Regulated Intervals

Regulated intervals are generally intended to limit false-accept/reject risks of the end products and processes deemed most critical or, in the rare case of a minimum interval, limit support costs for MTE perceived as non-critical. Such constraints have often originated in past environments lacking effective interval-analysis programs and perhaps without observed reliability data on the MTE and specific applications in question. With the benefit of the doubt, a regulated interval may have been based on a borrowed interval or some form of engineering analysis; however, regulated intervals not based on stated risk or reliability specifications are arbitrary. Arbitrary intervals are sub-optimal, and therefore are poor substitutes for modern risk and reliability control methods.

Other imposed requirements will likely be sub-optimal as well. For example, an interval-analysis system using interval data measured only in months will not achieve the results that the same system will achieve by use of interval data measured more precisely, e.g., in days. Even an imposed reliability target may be more costly than determining the optimum reliability target(s) by use of risk analysis if adequate cost and impact data is available to the analyst. The following discussion focuses primarily on the minimum and maximum interval cases but is also applicable to other imposed requirements.

3.9.2 Interpretation

Care is warranted in interpreting regulated intervals, which are sometimes written poorly. A constraint such as “The calibration interval shall be six months.” can be interpreted to mean the interval is

immutable or that the interval shall not *exceed* six months. Other interpretations are possible. If the correct interpretation is less than or equal to six months, the first interpretation could lead to excessive product or process risk. If the intent was indeed six months, no less, no more, then decreasing the interval per the second interpretation might lead to customer dissatisfaction or legal action. Furthermore, interpreting the undefined time (six months) as 183 days might lead to fines and penalties based on another interpretation of 180 days.

3.9.3 Risk Control Impacts

As implied above, regulated intervals can impact risk control. If optimum risk levels are calculated to minimize total costs and the corresponding intervals lie outside the regulated intervals' constraints, then complying with the regulated intervals will shift the risks away from optimum values, thus increasing costs, which is presumably the exact opposite of the regulatory intent. The regulators may consider only one side of the costs (e.g., quality or safety factors), preferring to err on the conservative side, but driving up total cost nonetheless.

3.9.4 Mitigation Options

Obviously, one way to handle regulated intervals is simply to comply with the requirements as written, establishing intervals as close to correct intervals as allowed. This is a convenient path; automated interval-analysis implementations can easily include data fields for the minimum or maximum intervals as well as algorithms to restrict the interval results accordingly. However, the organization(s) will bear increased total cost, either because operational support costs are higher due to shorter-than-correct maximum intervals, or consequence costs associated with reduced product quality are higher due to longer-than-correct minimum intervals.

If it is evident that the regulated interval was motivated more for controlling non-measurement issues such as maintenance or functional reliability rather than measurement reliability, it may be advantageous to establish maintenance intervals that fall within the given constraints and allow the calibration intervals to vary without constraints. This option may require regulatory approval and is clearly less practical if the maintenance procedure invalidates the calibration.

Given that particular MTE are deemed important enough to warrant regulated intervals, it is reasonable to assume an unstated intention that the particular MTE in question meet reliability targets different from those of other MTE. Therefore, another option is to change the MTE reliability targets such that interval-analysis produces intervals within the constraints. Without a risk analysis, there will be a range of reliability targets from which to choose. With risk analysis, the optimum reliability target (and calibration tolerances) subject to the constraints could be determined. See *NCSL International RP-18, Estimation and Evaluation of Measurement Decision Risk*.

If applying separate reliability targets to individual MTE is not appealing, another option is to change the MTE calibration tolerances, assuming the measurement standards are adequate. For example, in the case of a maximum interval constraint that results in reliability greater than the reliability target, the MTE tolerances can be reduced until its reliability at the maximum interval decreases to its reliability target. Effectively, this option simply corrects the stated tolerances to those actually achieved by the MTE at the given interval and reliability target. This strategy may be difficult if the MTE reliability is either too sensitive or insensitive to tolerance changes.

If imposed requirements are redundant, they add no value, and if they contradict effective interval analysis, they are of negative value. That point, along with actual reliability data and interval / risk

analysis results can be presented to policy makers to drive policy changes. Eliminating regulated intervals is the preferred long-term alternative, either altogether in favor of effective interval and risk analysis programs, or at least in favor of prescribed reliability targets. Simply revising the regulated interval to match the analysis result may not be satisfactory; the MTE applications and other factors governing risk and resulting optimum values can change with time, raising the bureaucratic problem of revising written constraints quickly enough to realize net benefits before changing conditions require further revision.

3.9.5 Data Retention

The advent of electronic data storage and digital communications has provided business, consumers, and the public with untold benefits, including access to vast amounts of information and incredible speed in analysis and distribution. Unfortunately, this technological progress comes hand in hand with some disadvantages with regard to such issues as privacy and liability.

The retention of accurately recorded and retrievable calibration data is of upmost importance for calibration interval analysis, not to mention the integrity of the calibration process. Besides this obvious metrological fact, there are additionally many government and corporate directives prescribing the length of time companies must maintain records. Retention periods vary from three to seven years³ and for some industries up to 75 years⁴ or even longer.

Alarming, however, many records-retention directives also specify records destruction at the end of the retention period. Furthermore, legal counsel, without regard to the inherent uncertainty in measurement and mitigation thereof [TM01], often further advocate records destruction policies to minimize potential evidence of liability related to out-of-tolerance MTE attributes and the potential for measurement decision error in accepting product. Calibration databases maintained separately from the official records may or may not be included in such policies, depending on content and case-by-case interpretation. Eliminating or encoding unessential identification fields may be helpful.

While interval-analysis often excludes older data due to significant changes in the calibration process or MTE usage conditions, the lack of data is otherwise a severe handicap, especially to attributes data interval-analysis methods. To be effective, all data relevant to current or future calibration intervals should be retained. The length and depth of the data retention should provide objective evidence of the validity of the calibration interval estimate and support any related calibration failure mode analysis. Failure to retain adequate data will lead to unsupportable intervals and possibly to future liability issues, exactly the opposite of what liability avoidance directives attempt to avoid. While deleting data may have some appeal as a means of limiting liability by destroying “evidence,” the upshot of this supposed protection exposes the organization to greater risk in the end.

3.10 Costs/Benefits Assessment

3.10.1 Operating Costs/Benefits

Obviously, higher frequencies of calibration (shorter intervals) result in higher operational support costs. On the other hand, lengthening intervals corresponds to allowing MTE uncertainties to grow to larger

³ See the Sarbanes-Oxley Act of 2002, often abbreviated as SOX.

⁴ E.g., United States Department of Energy radiological exposure-related records

values. In other words, longer intervals lead to higher probabilities of use of out-of-tolerance MTE for longer periods.

Finding the balance between operational costs and risks associated with the use of out-of-tolerance MTE requires the application of modern technology management methods [NA89, HC89, NA94, DD93, DD94, HC95a, HC95b, HC95c, RK95, MK07, HC08, MK08, SD09, DH09]. These methods enable optimizing calibration frequency through the determination of appropriate measurement reliability targets.

3.10.2 Extended Deployment Considerations

For some applications, MTE cannot be calibrated in accordance with recommended or established calibration schedules after their initial calibration. In these instances, alternatives or supplements to calibration are advisable. In cases where the MTE are highly accurate relative to the tolerances of the attributes of supported items, periodic calibration may not be required. In cases where this condition is not met, a statistical process control supplement involving check standards or other compensatory measures are recommended.

3.10.2.1 High Relative Accuracy

Recent experimentation with new analysis and management tools [NA89, HC89, MK07] has shown that MTE whose testing or calibration accuracies are significantly high relative to the tolerance limits of attributes of the workload items they support seldom require periodic calibration or other process control. The higher the relative accuracy, the less is the need for periodic calibration, other things being equal.

What constitutes a high relative accuracy is determined by case-by-case analyses. Such analyses extrapolate attribute uncertainty growth to extended periods to determine whether maximum expected MTE attribute bias uncertainties increase measurement process uncertainty to such an extent that calibration accuracy becomes inadequate. Whether calibration accuracy is inadequate depends on the specific false-accept and false-reject risk requirements in effect. Moral: Ensure that accuracy remains adequate longer than the required MTE lifetime.

3.10.2.2 Bayesian Methods

Bayesian methods have been developed in recent years to supplement periodic calibration of test and calibration systems [HC84, DJ85, DJ86b, NA94, RC95]. The methods employ role swapping between calibrating or testing systems and units under test or calibration. By role swapping manipulation, recorded MTE under test or calibration measurements can be used to assess the in-tolerance probability of the reference attribute. The process is supplemented by knowledge of time elapsed since calibration of the reference attribute and of the unit under test or calibration. The methods have been extended [HC84, DJ86b, HC91, NA94, HC07] to provide not only an in-tolerance probability for the reference attribute but also an estimate of the attribute's error or bias. NC94a, and *RP-18, Estimation and Evaluation of Measurement Decision Risk*, discuss this topic in detail.

Use of these methods permits on-line statistical analysis of the accuracies of MTE attributes. The methods can be incorporated in ATE, ACE, and product systems by embedding them in measurement controllers. A specification for accomplishing this was provided in 1985 [DJ85] for a prototype manometer calibrator.

3.10.3 Development Costs/Return of Investment

Systems that fail to accurately determine appropriate intervals tend to set intervals that are shorter than necessary. Employing methods such as general interval or engineering analysis, for example, tend to err

on the side of conservatism so that the risk of inadequately supported test systems and products is well within subjective “comfort zones.”

In addition, reactive methods, such as Methods A1 and A2, usually impose a more pronounced interval change to an out-of-tolerance event than to an in-tolerance event. In other words, interval reductions are usually larger or occur more frequently than interval extensions.

In contrast, systems that accurately determine calibration intervals, such as those patterned after Methods S2 or S3, typically cost considerably more to design, develop and implement than heuristic or reactive systems.

The conclusion to be drawn from these considerations is that better systems cost more to put in place but reduce costs during operation. In evaluating return on investment, these opposing costs need to be weighed against each other, with an eye toward minimizing the total [NA89, HC89, NA94].

3.11 Personnel Requirements

Personnel requirements vary with the methodology selected to analyze calibration intervals.

3.11.1 Reactive Systems

3.11.1.1 System Design and Development

Reactive systems (see Chapters 2 and 4) can be designed and developed by personnel without specialized training.

3.11.1.2 System Operation

For reactive systems, the personnel requirements include an understanding of the engineering principles at work in the operation of MTE coupled with an extensive range of experience in using and managing MTE. For reactive systems, operating personnel need to be conversant with procedures for applying interval adjustment algorithms.

3.11.2 Statistical Systems

3.11.2.1 System Design and Development

Highly trained and experienced personnel are required for the design and development of statistical calibration interval-analysis systems. In addition to advanced training in statistics and probability theory, such personnel need to be familiar with MTE uncertainty growth mechanisms in particular and with measurement science and engineering principles in general. Knowledge of calibration facility and associated operations is required, as is familiarity with calibration procedures, calibration formats and calibration history databases. In addition, both scientific and business programming personnel are required for system development.

3.11.2.2 System Operation

No special operational requirements are imposed by statistical systems on engineering or calibration personnel. System operation can be performed by, in most cases, a single individual familiar with system operating procedure. If system changes are needed, system maintenance may require the same skill levels as were required for system development.

3.12 Training and Communications

Training and communications are required to apprise managers, engineers and technicians as to what the interval-analysis system is designed to do and what is required to make its operation successful.

Agreement between system designers and calibrating technicians on terminology, interpretation of data formats and administrative procedures is needed to ensure that system results match real world MTE behavior. In addition, an understanding of the principles of uncertainty growth and an appreciation for how calibration data are used in establishing and adjusting intervals is required to promote data accuracy.

Comprehensive user and system maintenance documentation is also required to ensure successful system operation and longevity. Changes to calibration interval systems should be made by personnel familiar with system theory and operation, and subsequently validated in accordance with applicable requirements. This point cannot be overstressed.

Chapter 4 Interval-Analysis Method Selection

4.1 Introduction

This chapter provides guidelines for selecting the interval-analysis method appropriate for a requiring organization. These guidelines are provided in the form of “ratings” for each of several selection criteria. Tables are provided that summarize these ratings. The selection criteria are designed to promote meeting the calibration interval-analysis objectives stated in Chapter 2:

- Cost effectiveness
- System responsiveness
- System utility

In establishing the ratings, the goal of an interval-analysis system is assumed to be the attainment of the “correct” interval (i.e., one that corresponds to a specified measurement reliability target) in the shortest time at the lowest cost per interval.

All ratings are to be considered relative. For instance, under certain circumstances, the General Interval Method provides the least effective intervals in terms of meeting quality objectives. This method is, accordingly, assigned a rating of “poor” in the “Meets Quality Objectives” category. On the other hand, Method S3 is considered among the best of the available methods in meeting quality objectives. Consequently, this method is rated “excellent” in this category.

The category values and qualifiers for each of the selection criteria are intended to provide rough guidelines only. Flexibility in their application is encouraged. Final selection will depend in large part on the emphasis given by the requiring organization to each of the selection criteria. This is often a matter of corporate preference. Decision tree graphics are presented at the end of this chapter to assist in the selection process.

4.2 Selection Criteria

Several factors are relevant in deciding on the method to use in controlling measurement uncertainty growth. The most often encountered are the following:

1. **Meets Quality Objectives:** The capability to adjust intervals to achieve a specified reliability target, rated qualitatively from “poor” to “excellent.” Establishing a quantitative metric is a current research topic [MK09].
2. **Data Availability Requirement:** The data required for application of the methodology.
3. **Development Budget:** The budget needed for interval-analysis system requirements analysis, design, and development.
4. **Annual Maintenance Budget:** The annual budget needed for system modifications and enhancements during the operational phase of its life cycle.
5. **Annual Operating Budget:** The annual budget needed to operate the interval-analysis system.
6. **Personnel Requirements:**
 - a. **Developer:** Indicates the highest personnel skill level(s) required for system development and maintenance.

- b. User:** Indicates the highest personnel skill level(s) required for system operation.
- 7. Training Requirements:** Indicates the training required to operate and provide data to the system.
- 8. Automated Data Processing (ADP) Requirements:** Refers to the category of processor required for hosting a calibration interval-analysis system or the software involved. “None” applies to cases where calibration interval-analysis would be performed manually. “PC” refers to a desktop processor (“personal computer”). “Server” applies to a processor that can be run in batch mode with the capability for storage and retrieval of large data files.
- 9. System Effectiveness:** Indicates the extent to which reliability objectives are met, renewal policies are accommodated, and the cost per interval is minimized.
- 10. Cost Savings:** The beneficial impact that intervals assigned by the interval-analysis system has on operating costs as compared to random interval assignment. The assigned qualitative ratings range from “none” to “very high”. Research on a quantitative relative cost metric is under way [MK09].

4.3 General Interval Method

1. Meets Quality Objectives

From the standpoint of quality assurance, the following conditions relate to the implementation of a single interval for all MTE in inventory:

- 1) The MTE inventory is small and homogeneous with respect to uncertainty growth.
- 2) Engineering or other knowledge is lacking concerning relative stabilities of MTE models or other groupings.
- 3) The relationship between measurement reliability and measurement decision risk is not understood, so that neglect of out-of-tolerance conditions is unknowingly tolerated.
- 4) The calibration costs due to any overly frequent calibration are less than the cost of interval analysis.
- 5) The MTE inventory is highly stable and all appropriate calibration intervals exceed a maximum allowable interval (in which case, all MTE are calibrated at the maximum interval).
- 6) All MTE in inventory have nominal accuracies that are high relative to products. In such cases, calibration serves to verify this assumption.

2. Data Availability Requirement

No calibration history data are required for implementing a general interval.

3. Development Budget

The development budget for a system employing a general interval is virtually zero.

4. Maintenance Budget

The maintenance budget is zero.

5. Operating Budget

The required operating budget is essentially zero.

6. Personnel Requirements

No specific personnel skills are required for establishing and operating a general interval system.

7. Training Requirements

No special training is required. The only communications requirement is that calibrating technicians know the general interval or that preprinted labels be made available.

8. ADP Requirements

Essentially, no ADP capability is required.

9. System Effectiveness

A general interval system can be effective in terms of controlling measurement decision risks under the following conditions

- 1) The MTE inventory is homogeneous with respect to uncertainty growth.
- 2) A reliability target can be established for the MTE inventory commensurate with acceptable measurement decision risks.
- 3) The general interval has been established by use of Method A3, S1 (if the exponential reliability model is appropriate for all MTE in inventory), S2 or S3.

If these conditions are not met, the general interval will be effective only for MTE whose appropriate intervals are accidentally equal to the general interval. All other MTE will either be over-calibrated or under-calibrated.

10. Cost Savings

In cases where an inventory is small and homogeneous, an interval can in principle be found that is appropriate for all items in inventory. However, in all other cases, the appropriateness of general interval for a given item is the result of a fortuitous accident. This makes interval applicability an entirely random event. For this reason, apart from the homogeneous inventory case, employing a general interval is no better than assigning random intervals, and there is no cost savings to be expected.

Table 4-1. General Interval

Selection Criterion	Value
Meets Quality Objectives	poor
Data Availability Requirement	none
Development Budget	none
Annual Maintenance Budget	none
Annual Operating Budget	none
Personnel Requirements (Developer)	clerical
Personnel Requirements (User)	clerical
Training Requirements	none
Required ADP capability	none
System Effectiveness	poor
Cost Savings	none

4.4 Borrowed Intervals Method

1. Meets Quality Objectives

If the control of measurement decision risk to acceptable levels is a quality objective, then the following conditions need to be met for a borrowed interval:

- 1) The interval is obtained from an organization characterized by at least one of the following:
 - a) The measurement decision risk control objectives are similar to those of the borrowing organization, i.e., both organizations employ the same reliability target for the MTE in question.
 - b) Intervals can be mathematically computed for the borrowing organization's reliability target from the target of the originating organization (if targets are different). This computation usually requires the use of an appropriate reliability model.
- 2) The usage, operating environment, MTE attribute tolerance limits and other variables relevant to uncertainty growth are similar between the originating and the borrowing organizations.
- 3) The borrowed interval has been established by use of Method A2, A3, S1 (if the exponential reliability model is appropriate for the MTE in question), S2 or S3.

2. Data Availability Requirement

No data are required for the borrowed interval approach. If calibration history data are unavailable, and the above conditions are met, a borrowed interval may be as good as can be obtained.

3. Development Budget

The borrowed interval approach requires a nearly zero development budget. The principal development costs are those of locating an originating organization or organizations and verifying that conditions 1 through 3 above are met.

4. Maintenance Budget

The maintenance of a borrowed interval system involves tracking interval changes at the originating organization(s) and implementing the changes at the borrowing organization.

5. Operating Budget

The operating budget for a borrowed interval system is minimal. No computations are involved, except those associated with recomputing intervals if reliability targets differ between the originating and borrowing organizations.

6. Personnel Requirements

If the originating organization's reliability target for its MTE is the same as that of the borrowing organization, then no extraordinary personnel qualifications are required to establish a borrowed interval system. If reliability targets need to be recomputed, knowledge of high school algebra is usually sufficient. For some reliability models, knowledge of calculus may be required.

7. Training Requirements

No special training is required to operate a borrowed interval system. Communications costs are those associated with disseminating borrowed interval information.

8. ADP Requirements

Essentially no ADP capability is required.

9. System Effectiveness

If conditions 1 through 3 above can be met, a borrowed interval system can be as effective as present technology allows. Reliability targets can be achieved and measurement decision risk control objectives can be met.

If conditions 1 through 3 are not met, a borrowed interval may be no better than a general interval, depending on circumstances.

10. Cost Savings

Because of the diversity in calibration procedures, operating environments, equipment usage and so on, an interval that is appropriate for one organization has little likelihood of being appropriate for another. However, little likelihood is not zero likelihood. Accordingly, the cost savings relative to random interval assignment are low but nonzero.

Table 4-2. Borrowed Intervals Method

Selection Criterion	Value
Meets Quality Objectives	poor - fair
Data Availability Requirement	none
Development Budget	none
Annual Maintenance Budget	none
Annual Operating Budget	low
Personnel Requirements (Developer)	low
Personnel Requirements (User)	general ed. - engr.
Training Requirements	none
Required ADP capability	none
System Effectiveness	poor-fair
Cost Savings	very low - low

4.5 Engineering Analysis Method

1. Meets Quality Objectives

For reasons discussed below under System Effectiveness, intervals arrived at through engineering analysis may be only loosely connected to quality objectives. For this reason, if it is desired to meet, rather than exceed or fall short of quality criteria, then engineering interval-analysis is not recommended in general. If, on the other hand, it is desired to exceed quality objectives, regardless of cost, then engineering analysis may be a viable approach.

2. Data Availability Requirement

Data relating to the accuracy and stability of MTE attributes is required for engineering interval analysis. There is no requirement for calibration history data unless such data are used to corroborate or “history match” engineering measurement reliability projections.

3. Development Budget

Assuming that engineering analysis consists of detailed investigations into MTE attribute accuracies and stabilities, the development of an analysis system can run from weeks to years, depending on the variety of MTE in inventory. Much of the cost is involved in setting up attribute information data bases, developing structured analysis guidelines, and setting up a system for interval review and implementation.

4. Maintenance Budget

If designed properly, the maintenance budget for an engineering analysis system should be minimal. System maintenance consists primarily of refining engineering procedures and checklists. Some redesign or optimization of the attribute information database may also be required from time to time.

5. Operating Budget

The operating budget for an engineering analysis system is the highest of any of the interval-analysis methods documented in this RP. This is due to the fact that considerable manual effort is required for each interval. Depending on the stability of the MTE inventory, the annual operating cost may rival the initial development cost. Effort is also required to update the attribute information database.

6. Personnel Requirements

Engineering personnel with considerable experience with MTE behavior over time and with the ability to understand measurement reliability concepts are required for an engineering analysis system. Such personnel should have a strong background in physics, mathematics and “equipment zoology.”

7. Training Requirements

For the Engineering Analysis Method, the training budget is likely to be high. This training manifests itself in:

- Training of engineers in the principles of measurement reliability and uncertainty growth control.
- Training of engineers in following structured analysis procedures.
- Continual updating of engineering expertise and familiarity with MTE technology.

8. ADP Requirements

Little to no ADP capability is required.

9. System Effectiveness

It is exceedingly difficult to convert engineering knowledge into an interval projection that is consistent with a specified reliability target. Often, the best that can be done is to make interval assignments or changes that correspond loosely to changes in measurement reliability.

Engineering analysis may, however, be effective in identifying MTE attributes that require special handling or consideration.

10. Cost Savings

Given the comments under “Meets Quality Objectives” above, it may seem that the engineering analysis method is no better than the general interval method. However, even at its worst, engineering analysis is not expected to be a completely blind exercise. Nevertheless, because of its high personnel, training and operating cost, the return on investment is not likely to greatly exceed that of the general interval method.

Table 4-3. Engineering Analysis Method

Selection Criterion	Value
Meets Quality Objectives	poor
Data Availability Requirement	low
Development Budget	low to moderate
Annual Maintenance Budget	low
Annual Operating Budget	high
Personnel Requirements (Developer)	sr. engr.
Personnel Requirements (User)	sr. engr.
Training Requirements	high
Required ADP capability	none
System Effectiveness	fair
Cost Savings	very low

4.6 Reactive Methods

The three reactive methods discussed in Chapter 3 differ with respect to selection criteria ratings. A summary of these differences is shown in Table 4-4.

1. Meets Quality Objectives

Method A1

Although one can tailor the Method A1 algorithm parameters to a long-term average measurement reliability target, the reliability achieved for any individual instrument is essentially a hit-or-miss affair. Accordingly, Method A1 cannot be said to be effective in meeting quality objectives.

Method A2

With Method A2, the reliability target governs the size of an interval adjustment in effect. However, the method is prone to producing interval adjustments when adjustments are not called for. For this reason, it can be considered fair only with respect to meeting quality objectives.

Method A3

Method A3 adjusts intervals to meet reliability targets and also avoids unnecessary adjustments. It is considered good with respect to meeting quality objectives.

2. Data Availability Requirement

Method A1

Only the results of the current calibration and the current assigned interval are required for interval adjustment.

Method A2

The data required for interval adjustment using Method A2 consist of a tracking index (iteration counter), a variable adjustment parameter, the current assigned interval and the results of the current calibration.

Method A3

The data required for Method A3 consist of the assigned interval and a history of calibration results running from the current calibration back to the calibration following the most recent interval adjustment.

3. Development Budget

Method A1

The development budget for this method is minimal.

Method A2

Method A2 can be applied by calibrating technicians, but works most efficiently if implemented on a PC or server with access to the required data indicated above. The development budget for this method ranges from minimal to low.

Method A3

Method A3 should be implemented on a PC or network server. The required development budget is moderate.

4. Maintenance Budget

Method A1

This method requires virtually no maintenance unless it is desired to change the adjustment algorithm to alter the measurement reliability that results from using it.

Method A2

Method A2 requires little or no maintenance budget.

Method A3

If designed properly, this method is virtually maintenance free.

5. Operating Budget

Method A1

This method typically requires that interval adjustments be computed by calibrating technicians. The operating budget is, accordingly, in the moderate to high range, though automation is possible.

Method A2

If interval adjustments are computed manually, either by calibrating technicians or by support engineers, the required operating budget for this method is high. If the method is implemented on a PC or server, the operating budget is low.

Method A3

Because Method A3 is implemented on a PC or server, the required operating budget is low.

6a. Personnel Requirements (Developer)

Method A1

Most implementations of Method A1 have historically been accomplished by either management or technical personnel with minimal mathematical training. If it is desired to tailor the adjustment

RP-1: Establishment and Adjustment of Calibration Intervals

algorithm's convergence period or ability to maintain an interval, senior statistical personnel may be helpful.

Method A2

If Method A2 utilizes calibrating technicians to compute interval changes, the method can be implemented by general technical personnel. If interval changes are to be automated, development will require journeyman-level systems analysis and engineering personnel.

Method A3

Method A3 implementation requires journeyman-level systems analysts and statisticians.

6b. Personnel Requirements (User)

Method A1

With method A1, personnel are required to multiply the current interval by a decimal fraction. The required skill level is a general high school education or equivalent.

Method A2

If interval adjustments are made manually, a general engineering skill level is required. If adjustments are made automatically, only a minimal clerical skill level is required.

Method A3

The skill level required for operation of Method A3 is general clerical.

7. Training Requirements

Method A1

Training requirements are minimal.

Method A2

Depending on whether interval adjustments are automated or manually computed, the training requirements range from low to moderate.

Method A3

Little or no training is required for Method A3.

8. ADP Requirements

Method A1

No ADP capability is required for Method A1.

Method A2

If Method A2 is automated, an application capable of tracking the initial calibration intervals and adjustment parameters of each instrument is required as a minimum. If Method A2 is implemented manually, the ADP requirement consists of engineering pocket calculators distributed to calibrating technicians.

Method A3

The minimum ADP requirement for Method A3 is a PC.

9. System Effectiveness

Method A1

Method A1, while economical to implement, is somewhat costly to operate. Furthermore, it is not effective in meeting quality objectives. This is because (1) the method requires long periods of time to reach desired reliability goals; and (2) the method achieves reliability goals only “on average.” That is, the average reliability of a population of serial numbered items slowly iterates toward the reliability target, but each item subject to interval adjustment spends very little of its life cycle on an interval commensurate with this target. Method A1's effectiveness must be considered poor.

Method A2

Method A2 can be economical to operate and it may produce intervals that come in line with quality objectives. However, the period required for this to happen is excessive and interval fluctuations are experienced in the process. For these reasons, its effectiveness is considered only poor to fair.

Method A3

Like method A2, method A3 can be operated with minimal expense. Moreover, if the selection of initial intervals is fairly accurate, the method yields the correct intervals in a relatively short period with little or no fluctuation. If initial interval selection is inaccurate, the period required for solution is lengthened and the amount of fluctuation is increased. Even so, the period required for solution and the amount of fluctuation experienced are both considerably lower than for Method A2. The effectiveness of Method A3 is considered in the “fair to good” range.

10. Cost Savings

Method A1

Although Method A1 is inexpensive to implement, its poor system effectiveness makes it little better than a random interval system. For this reason, cost savings are low.

Method A2

Method A2 suffers from the same slow pace that characterizes Method A1. However, with Method A2, because interval increments shrink as interval adjustments progress, each item has a chance of eventually reaching an interval commensurate with its reliability target. Prior to this, however, interval assignment is not significantly better than random assignment. Synthesizing between these two points yields a moderate rating for Method A2's cost savings.

Method A3

Method A3 may be viewed as an approach that begins with an engineering analysis or a borrowed interval and then makes interval adjustments statistically. While system development costs and initial interval costs may be low to moderate, the cost of interval adjustment is almost nonexistent. In addition, Method A3 offers significant improvement over Methods A1 and A2 in finding and retaining correct intervals. For these reasons, the cost savings inherent in Method A3 are considered high.

Table 4-4. Reactive Methodology

Analysis Methodology			
Selection Criterion	A1	A2	A3
Meets Quality Objectives	N/A	poor	good
Data Availability Requirement	current cal	recent cal history	recent cal history
Development Budget	low	minimal -low	low to moderate
Annual Maintenance Budget	low	none	none
Annual Operating Budget	moderate - high	low – high*	low
Personnel Requirements (Developer)	general education	general technical systems analyst*	systems analyst statistician
Personnel Requirements (User)	cal tech	clerical – engr*	clerical
Training Requirements	low	low – moderate*	low
Required ADP capability	none	none – PC*	PC
System Effectiveness	poor	poor – fair	fair to good
Cost Savings	low	moderate	moderate to high

*Depending on whether implementation is manual or automated (see discussion)

4.7 Maximum Likelihood Estimation (MLE) Methods

The three maximum likelihood estimation methods discussed in Chapter 2 differ with respect to selection criteria ratings, as shown in Table 4-5 following the ratings discussion.

1. Meets Quality Objectives

State-of-the-art MLE methods have been shown to be optimal in terms of attaining reliability targets and minimizing operating costs [HC94]. If maintaining quality objectives is a paramount concern, methods S2 and S3 should be considered strong favorites. Method S1, while significantly better than the General Interval, the Borrowed Interval, Method A1 or Method A2, is limited by its exclusive reliance on a single reliability model.

2. Data Availability Requirement

MLE methods require that calibration history be maintained for each serial-numbered MTE item. If intervals are to be determined by attribute, calibration history is needed for each calibrated attribute. MLE methods generally require more calibration history than method A3, and methods S2 and S3 in particular are more effective if the data for a given item or item group contain a variety of assigned interval values.

NOTE: — Use Caution. For systems using MLE methods, data accuracy, continuity and consistency are critical. Considerable care must be taken in the design of data-input documents or other vehicles. It has been found that calibrating technicians' lack of understanding or trust in the purpose and utility of requested information on calibration data forms, or the clarity of instructions regarding the data being collected may promote inaccurate, sloppy or even intentionally erroneous data [HC78].

3. Development Budget

Designing and developing systems that employ state-of-the-art MLE methods can be an expensive proposition. System development costs typically run in the **\$1M to \$2M** range (in 2007 U.S. dollars) for Methods **S2** and **S3** and around **\$100K** for Method **S1**. As such it is generally more feasible to pursue commercially available systems.

Cost/Benefit Considerations:

While development costs are high, **state-of-the-art MLE methods have been known to return the initial investment during the first or second year of operation** [HC94]. In addition, such methods are likely to be more applicable to future MTE designs and to future technology management requirements than less sophisticated methods. This can translate to greater system longevity and lower life cycle maintenance costs.

Another significant factor in budgeting for development and maintenance is the benefit to be derived from calibration interval-analysis spin-offs. Cost savings and cost avoidances made possible by supplemental diagnostic and reporting capabilities need to be included with operational cost factors in weighing system development and maintenance costs against potential benefits.

Obviously, organizations with large inventories of equipment and with large annual calibration workloads will benefit the most from investing in optimal methods. Such organizations also are more likely to be able to afford a development budget sufficient for the implementation of these methods.

4. Maintenance Budget

If properly designed, the annual system maintenance budget is minimal.

5. Operating Budget

Depending on the extent to which system operation is automated, system operation may consist of updating some initial run criteria and clicking a “run” button. In cases where it is felt that extensive manual review of computed intervals or other engineering input is required, operating costs may become high. In most cases, such manual intervention can largely be avoided by good system design.

6a. Personnel Requirements (Developer)

Highly trained and experienced systems, engineering and statistical personnel are required for the design of MLE calibration interval-analysis systems. In addition to having had advanced training in statistics and probability theory, such personnel need to be familiar with MTE uncertainty growth mechanisms in particular and with measurement science and engineering principles in general. Knowledge of calibration facility and associated operations is required, as is familiarity with calibration procedures, calibration formats and calibration history databases.

System development requires both scientific and business programmers.

6a. Personnel Requirements Operator

Once developed and implemented, system operation may range from what is essentially a clerical function to an engineering analysis and evaluation function. The personnel level required depends on the extent to which system operation is automated.

7. Training Requirements

Training is required to apprise managers, engineers and technicians as to what the interval-analysis system is designed to do and what is required to ensure its successful operation. Agreement between system designers and calibrating technicians on terminology, interpretation of data formats and administrative procedures is needed to ensure that system results match real-world MTE behavior. In addition, to promote system accuracy, calibrating technicians should understand the principles of uncertainty growth and appreciate how calibration data are used in establishing and adjusting intervals.

8. Required ADP Capability

MLE methods require considerable data manipulation and number-crunching capability. However, most PCs should be adequate, depending on the system features and on the size of the calibration history database. If the data base is extensive and all possible features are implemented (see Chapter 6), then a database server may also be useful. Of more concern is the required software, which for methods S2 and S3, must have sophisticated statistics and numeric solving capability, and if the analysis process is to be highly automated, a database interface for exchanging calibration data and resultant interval information.

9. System Effectiveness

The use of Methods S2 and S3 leads to interval-analysis systems that are optimal with respect to controlling measurement decision risk to levels commensurate with quality objectives. In addition, if system design is done in such a way as to minimize manual processing, these methods can also lead to a low cost per interval. Method S1's cost per interval is also potentially low, but its effectiveness with regard to controlling measurement decision risk does not compare favorably with the other MLE methods.

10. Cost Savings

If the requiring organization has an annual calibration workload in the neighborhood of several thousand or more calibrations, then the cost savings to be realized from MLE methods are decidedly higher than random interval assignment. This is especially so for Methods S2 and S3. These methods achieve a high-to-very high rating due to their ability to easily accommodate a variety of uncertainty growth mechanisms.

Table 4-5. MLE Methodology Recommendations

Analysis Methodology			
Selection Criterion	S1	S2	S3
Meets Quality Objectives	good	good to excellent	good to excellent
Data Availability Requirement	cal history	cal history action taken	cal history action taken
Development Budget	moderate	high	high
Annual Maintenance Budget	low	low	low
Annual Operating Budget	low	low	low
Personnel Requirements (Developer)	sr. stat. sr. sys. anlys.	sr. stat. sr. sys. analyst	sr. stat. sr. sys. Analyst
Personnel Requirements (User)	cal tech	cal tech	cal tech
Training Requirements	low	moderate	moderate
Required ADP capability	PC	PC	PC
System Effectiveness	good	good to excellent	excellent
Cost Savings	moderate	high to very high	high to very high

4.8 Method Selection Decision Trees

In this section, three guideline decision trees are offered that highlight the factors considered in selecting an interval-analysis method that is optimal for a given requiring organization. The following criteria were used in developing these guidelines:

4.8.1 Calibration Workload:

- Large** - 5,000 or more serial-numbered items, where items can be grouped into model number or other homogeneous family groupings
- Medium** - From around 500 items to 5,000 items
- Small** - Less than around 500 items

4.8.2 Quality Assurance (QA) Emphasis:

- High** - Quality requirements stem from the need to field products with a high probability of conformance with specifications.
- Average** - Quality requirements stem from the need to meet national documentary consensus standards.
- Low** - The interval-analysis system is required to satisfy an essentially cosmetic requirement for periodic recalibration of MTE.

4.8.3 Cost Factor:

- Development** - Includes system design, development and maintenance.
- Operation** - Includes system operation, calibration costs, rework costs and the cost of false accepts.
- Total** - The sum of development and operation costs weighted by QA emphasis.

4.8.4 Data Availability:

Calibration Records - The as-received and as-released condition of MTE are available, along with corresponding resubmission times.

Engineering - Calibration records are not available. The only source of in-house information on MTE stability and accuracy is engineering knowledge and technical experience.

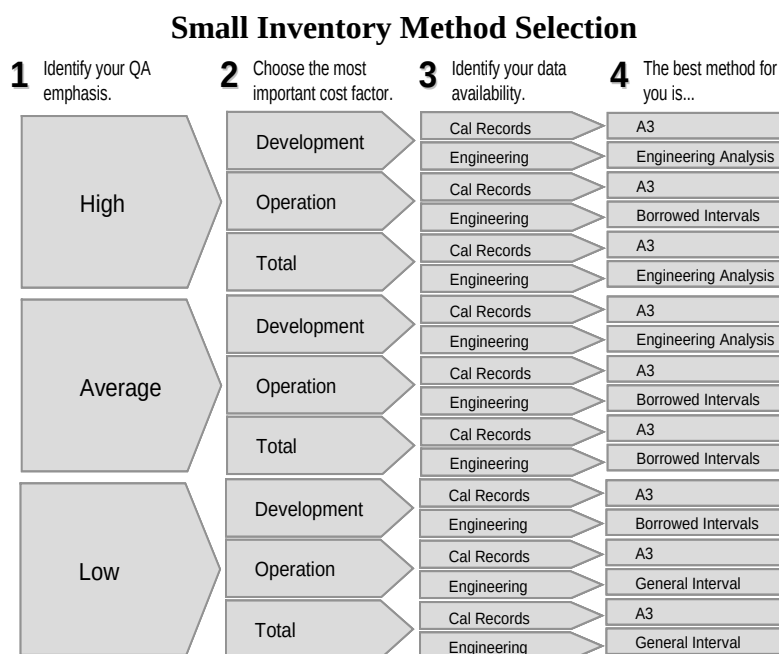


Figure 4-1. Small Inventory Decision Tree. The criteria are summarized for deciding on an appropriate interval-analysis system for requiring organizations with small calibration workloads.

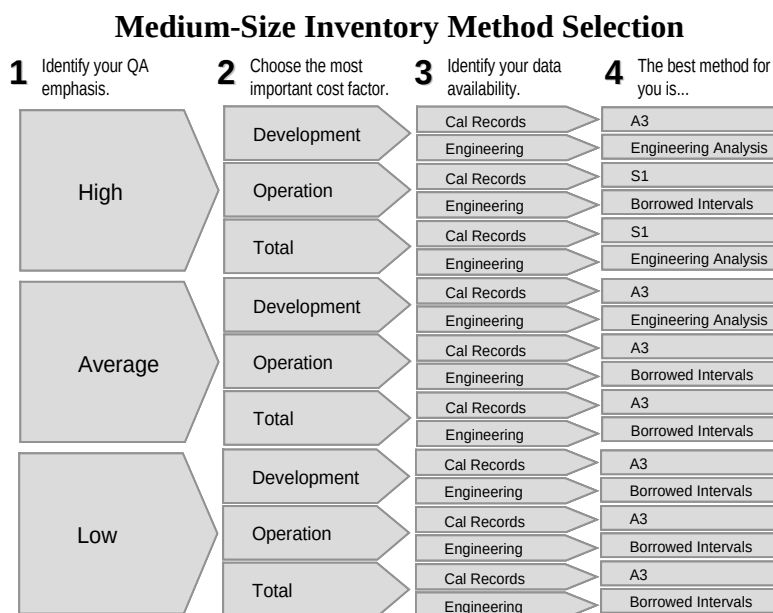


Figure 4-2. Medium-Size Inventory Decision Tree. The criteria are summarized for deciding on an appropriate interval-analysis system for requiring organizations with medium-size calibration workloads.

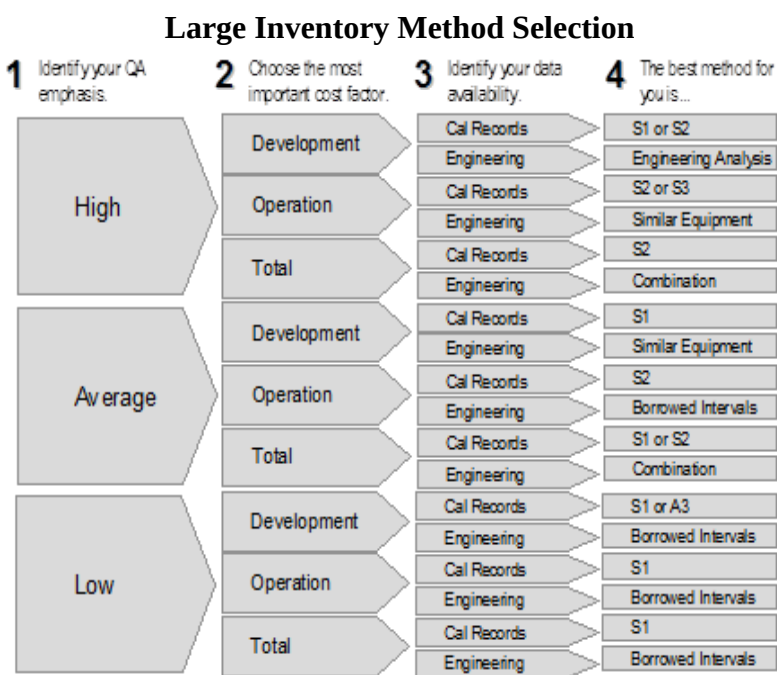


Figure 4-3. Large Inventory Decision Tree. The criteria are summarized for deciding on an appropriate interval-analysis system for requiring organizations with large calibration workloads.

Chapter 5 Technical Background

5.1 Introduction

Technical concepts relevant to the design and development of calibration interval-analysis systems are described in this chapter. Reliability analysis methodologies discussed in this chapter are described in detail in the Appendices.

5.2 Uncertainty Growth

Our knowledge of the values of the measurable attributes of a calibrated item begins to diminish from the time the item is calibrated. This loss of knowledge of the values of attributes over time is called **uncertainty growth**. For many attributes, there is a point where uncertainty growth reaches an unacceptable level, creating a need for recalibration. Determining the period required for an attribute's uncertainty to grow to an unacceptable level is the principal endeavor of calibration interval analysis.

An unacceptable level of uncertainty corresponds to an unacceptable out-of-tolerance probability and a higher expected incidence of out-of-tolerance conditions. For analysis purposes, an out-of-tolerance condition is regarded as a kind of “failure,” similar to a component or other functional failure. However, unlike functional failures that are obvious to equipment users and operators, *out-of-tolerance failures usually go undetected during use*. The detection of such failures occurs during calibration, provided of course that the calibration process uncertainty is sufficiently low.

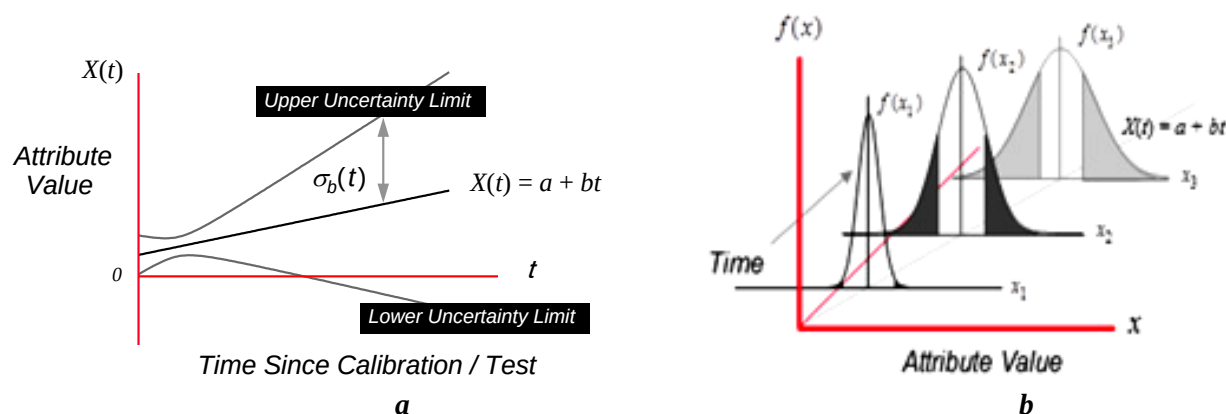


Figure 5-1. Measurement Uncertainty Growth. Uncertainty growth over time for a typical attribute. The curve in *a* shows the growth in uncertainty of the predicted value of an attribute *x*. The sequence in *b* shows corresponding statistical distributions at three different times. The uncertainty growth is reflected in the spreads in the curves. The out-of-tolerance probabilities at the times shown are represented by the shaded areas under the curves (the total area of each curve is equal to unity.). As can be seen, the growth in uncertainty over time corresponds to a growth in out-of-tolerance probability over time.

5.3 Measurement Reliability

Measurement uncertainty is controlled in part by requiring that MTE perform within assigned specifications or *tolerance limits* during use. This is achieved by periodic comparison to higher-level standards or equipment during calibration. Intervals between periodic calibrations are established and

adjusted in such a way as to maintain acceptable levels of confidence that MTE are performing within their specified tolerance limits during use.

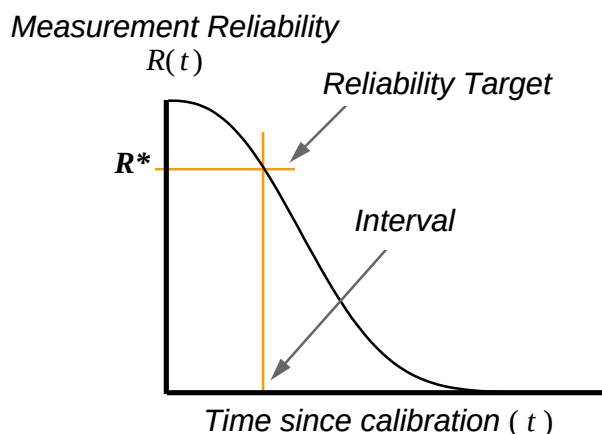


Figure 5-2. Measurement Reliability vs. Time. The statistical picture of uncertainty growth in Figure 5-1b shows that the in-tolerance probability, or measurement reliability, decreases with time since calibration. Plotting this quantity vs. time suggests that measurement reliability can be modeled by a time-varying function. Determining this function is the principal aim of statistical calibration interval-analysis methods.

A useful measure of this level of confidence is **measurement reliability**. Measurement reliability is defined as the **probability that an MTE item performs its required functions within its tolerance limit(s)**. Given the remarks made in the preceding section, measurement reliability can be expressed as a function of time and referenced to a particular time of use. Principal factors affecting measurement reliability are inherent instrument stability, usage and storage environments, and degree and severity of usage.

Measurement reliability requirements may be based on application or purpose. These requirements are usually specified in terms of **reliability targets** established to achieve levels of measurement reliability consistent with mission/use requirements and logistic and economic constraints. The establishment of these targets is discussed later in this chapter.

5.4 Predictive Methods

5.4.1 Reliability Modeling and Prediction

Immediately following calibration, an equipment user typically has high confidence that his or her equipment conforms to specifications. As the equipment experiences the stresses of use and/or storage, this confidence decreases to a point where the conformance of the equipment to its specifications is placed in doubt. As the doubt increases to an uncomfortable level, the user feels compelled to recalibrate the equipment. This decreasing confidence in the conformance of the equipment to its specifications reflects the growing uncertainty that the equipment conforms to the required specifications.

Uncertainty growth is synonymous with the decline in measurement reliability for a given attribute as the number and/or duration of stresses applied to the attribute accumulate. It is important to note that in this description, the user is not becoming convinced that the accuracy of his equipment is degrading in response to stress, only that his *knowledge* of this accuracy is becoming increasingly uncertain. In some

circumstances, the equipment's accuracy could conceivably improve with stress, whereas the *uncertainty* with regard to this accuracy always increases.

It should also be noted that the policy employed for adjustment of attributes (e.g., center spec all calibrated attributes, center spec only out-of-tolerance attributes, etc.), referred to as the **renewal policy**, bears directly on the limits of this uncertainty immediately following calibration and, therefore, at any time thereafter, as does the calibration process uncertainty. This topic is discussed in Appendix G.

Whatever the nature or frequency of the stresses experienced by an item of equipment (see, for example, [IL07]), these stresses accumulate over time. For this reason, attribute uncertainty growth can be generally regarded as a non-decreasing function of time. In other words, the probability for an out-of-tolerance attribute increases or, at best, remains constant with time. Thus, immediately following calibration, attribute values can be regarded as being closely confined within a small neighborhood bounded by the limits of uncertainty of the calibration system. As time passes, and the uncertainty as to the value of each attribute increases, the size of this neighborhood expands until at some point it begins to fill the tolerance limits for the attribute. This situation, illustrated in Figure 5-1, forms the basis for measurement reliability modeling as applied to calibration interval analysis.

5.4.2 Reliability Modeling Objectives

The objective of modeling measurement reliability is the determination of the functional dependence between attribute uncertainty growth and time elapsed since calibration. Knowledge of this functional dependence enables the determination of a calibration interval that corresponds to a desired measurement reliability target. Methods that employ such modeling are seen to be predictive in nature in that they attempt to predict the period that corresponds to a measurement reliability of interest. Because measurement reliability modeling is concerned with the growth of uncertainty, it is by nature a statistical endeavor. Hence measurement reliability predictions are quantified in terms of probabilities, i.e., a 0.85 EOP (end of period) estimate of measurement reliability for a given calibration interval consists of a prediction that 85 % of instruments calibrated at the end of that interval will be in-tolerance. Statistical methods of analysis are usually required to determine the various underlying mechanisms that govern the measurement reliability behavior of a given item or type of equipment (see Appendices D and E).

Several uncertainty growth behavior mechanisms have been observed in practice. A sample of these mechanisms is represented in Figure 5-3. The mathematical expressions for these mechanisms are given in Appendix D. It is important to note that the applicability of these models to specific cases requires a certain degree of testing and validation.

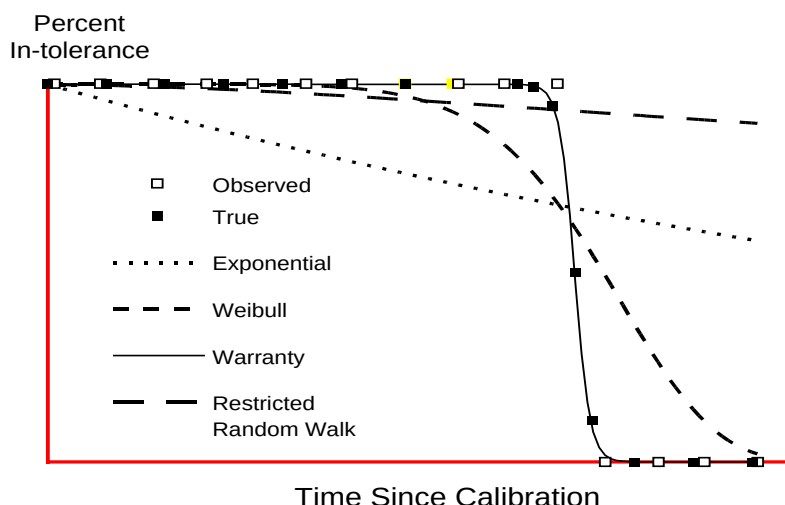


Figure 5-3. Measurement Uncertainty Growth Mechanisms. Several mathematical functions have been found applicable for modeling measurement uncertainty growth over time.

Statistical approaches that model uncertainty growth require fairly large quantities of representative data to yield accurate results. Facilities with limited inventories and/or limited access to calibration history data may find that such methods are beyond their reach. In these cases, calibration intervals are sometimes taken from external sources. The organization generating the selected external source should match as closely as possible with the interested facility with regard to such factors as usage, environmental stresses, equipment management policy and practice, calibration procedure, and technician skill level. In addition, if the measurement reliability target of the source organization differs from that of the requiring organization, the external interval will need to be adjusted to bring it in line with the requiring organization's target. These considerations are discussed in Chapters 2 and 4.

5.4.3 Observed Reliability

Test or calibration history consists of records of events in which MTE are calibrated and then recalled and recalibrated after various intervals. By grouping observed intervals into “sampling windows,” history data can take on the appearance of experimental life data [NM74].

Grouping historical data into sampling windows produces a time series. The time series consists of events (observed measurement reliabilities), governed by probabilistic laws (whether an out-of-tolerance occurs), arranged chronologically. An example of such a time series is shown in Table 5-1. If the observed reliabilities are portrayed graphically, an x-y plot is obtained that suggests the underlying behavior of reliability vs. time. Reliability modeling is essentially the practice of fitting curves to observed reliability plots.

Table 5-1. Observed Reliability Time Series

Sampling Window (Time)	Number Calibrated	Number In-Tolerance	Observed Reliability
0 - 14	5	5	1.000
14 - 28	7	7	1.000
28 - 42	6	6	1.000
42 - 56	10	6	0.600
56 - 70	11	9	0.818
70 - 84	12	9	0.750
84 - 98	6	3	0.500
98 - 112	8	4	0.500
112 - 126	8	4	0.500
126 - 140	14	5	0.357
140 - 154	12	3	0.250
154 - 168	7	0	0.000
168 - 182	5	0	0.000
182 - 196	6	0	0.000
196 - 210	5	0	0.000
210 - 224	6	0	0.000
224 - 238	8	3	0.375

5.4.4 Type III Censoring

In the statistical analysis of time-series data, the term “censoring” refers to a situation in which failure time information needed to determine a reliability function of interest is incomplete [NM74].

In the customary literature on the subject, two types of censoring are usually identified. They are type I censoring, in which the data gathering process is stopped after a certain period has elapsed, and type II censoring, in which the process is stopped after a preset number of failures has been observed.

In 1976, a third type of censoring was identified [HC76].⁵ This censoring, referred to as type III censoring, applies to cases where failure times are unknown. All that is known in analyzing type III censored data is the condition of the variable under study at the beginning and end of an interval. If a failure is observed at the end of the interval, it is assumed that the time of failure lies at some point within the interval.

Type III censoring describes the state of knowledge in analyzing calibration history data for purposes of modeling measurement reliability behavior. Methods of type III data analysis are given in Appendices C, D and E.

⁵ Type III censoring was later formally reported in 1987 by Jackson and Castrup [DJ87b] and by Morris [MM87].

In general, type III data analysis assumes that the probability density of the likelihood function is binomial, with the independent variable being the interval between successive calibrations.

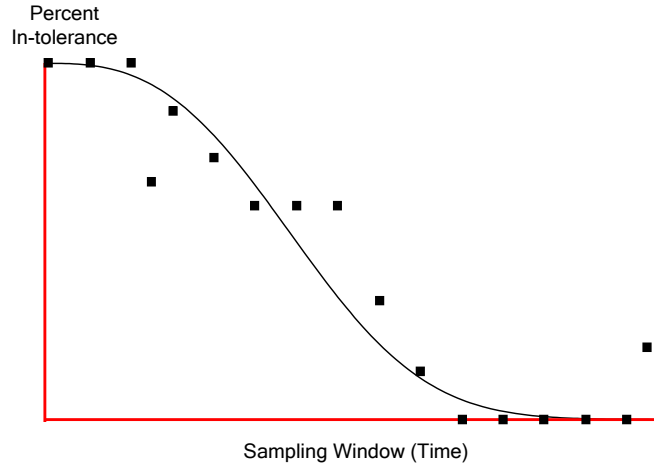


Figure 5-4. Observed Measurement Reliability. The filled squares represent observed percent in-tolerance vs. time elapsed since calibration. Time is quantized into sampling windows for the accumulation of samples of calibration results. The solid curve represents a reliability model adjusted to fit the observed data.

To see how the method works, consider Table 5-1. Label the midpoints of the sampling windows of the time series $t_1, t_2, \dots, t_k$, the number calibrated $n_1, n_2, \dots, n_k$ and the number observed in-tolerance $g_1, g_2, \dots, g_k$. Next, let $\hat{R}(t, \hat{\theta})$ represent the mathematical reliability function used to model the uncertainty growth process. In this function, the quantity $\hat{\theta}$ is a vector whose components are the parameters of the reliability model. If the “renew always” practice is in effect, the likelihood function is given by

$$L = \prod_{i=1}^k \frac{n_i!}{g_i!(n_i - g_i)!} \hat{R}(t_i, \hat{\theta})^{g_i} [1 - \hat{R}(t_i, \hat{\theta})]^{n_i - g_i}.$$

If the “renew-if-failed” practice is followed, then we let τ_i represent the time elapsed since the date of the last renewal of an item or attribute and the endpoint of the calibration interval in which the i th observed out-of-tolerance occurred and write⁶

$$L = \prod_{i=1}^X [\hat{r}(\tau_i) - \hat{R}(\tau_i)]$$

where X is the total number of observed out-of-tolerances and

$$\hat{r}(\tau_i) \equiv \hat{R}(\tau_i - I_i).$$

⁶ See Appendix D.

In this expression the variable I_i is the duration of the calibration interval in which the i th out-of-tolerance occurred.

If the “renew-as-needed” practice is followed, then the likelihood function becomes

$$L = \prod_{i=1}^N \hat{R}(\tau_i)^{x_i} \left[\hat{r}(\tau_i) - \hat{R}(\tau_i) \right]^{1-x_i}$$

where τ_i is the i th renewal time, N is the total number of observed renewals, and

$$x_i = \begin{cases} 1, & \text{if the } i\text{th renewal is for an in-tolerance item} \\ 0, & \text{otherwise} \end{cases}.$$

The function $\hat{r}(\tau_i)$ is defined as in the renew-if-failed case, except that the interval I_i is now the calibration interval immediately preceding the date at which the i th renewal occurred.

Following its construction, the likelihood function is maximized with respect to the components of $\hat{\theta}$. The component values that bring about this maximization are the ones sought for the function $\hat{R}(t, \hat{\theta})$. The maximization process is described in Appendices D, E and F.

5.4.5 User Detectability

Periodic calibration cannot, in general, prevent out-of-tolerances from occurring. What periodic calibration instead attempts to do is *prevent the continued use* of out-of-tolerance attributes. If an out-of-tolerance attribute is user detectable, then, presumably, the user will discontinue usage of the attribute or will apply it to uses that are not negatively impacted by the out-of-tolerance condition.

For this reason, in compiling out-of-tolerance time-series data it is common to ignore out-of-tolerances that are user detectable. This does not mean that the renewal of a user detectable out-of-tolerance is ignored, merely that the “clock is reset” without counting the out-of-tolerance in the data.

The issue of user detectability is sometimes a deciding factor in determining whether periodic calibration is performed or not. Many users feel that they can tell by the way in which equipment operates whether attributes are in-tolerance or not. The argument is that, if this is the case, then periodic calibration is not required. Users should merely submit MTE for recalibration when out-of-tolerances are suspected.

Informal studies have shown, however, that users who believe they are capable of detecting MTE out-of-tolerance times can instead typically detect when attribute values exceed specifications by several multiples of the tolerance limits. The time at which attribute values traverse tolerance limits is not ordinarily detectable solely from equipment behavior during use. For example, shipment of measurement standards for calibration may cause shifts unknown to the user; therefore, cross-checks against standards of comparable uncertainty upon receipt may prevent use while out of tolerance. Cross-checks before shipment may detect some out-of-tolerances that might otherwise be attributed to shipment.

5.4.6 Equipment Grouping

Projective methods of analysis typically assemble data in homogeneous groupings to facilitate collecting sufficient data for analysis. The following groupings have been found productive:

5.4.6.1 Model Number

MTE of the same manufacturer/model number designation are homogeneous with respect to design, fabrication, application and specifications. Regardless of whether interval-analysis is performed at the model number level or at the attribute level, grouping by model number is desirable.

5.4.6.2 Instrument Class

Instrument classes are collections of model numbers that are homogeneous with respect to application, complexity, stability and technology. An example of an instrument class is a noun nomenclature (e.g., voltmeter, AC, digital) subdivided by technology, complexity and accuracy.

5.4.6.3 Similar Items

MTE may be grouped in instrument class subgroups that contain model numbers with close similarity to one another. Such a similarity is found, for example, between a model number and an earlier version, where differences are essentially minor or even cosmetic. In such cases, the new item should be expected to have performance characteristics similar to those of its predecessor model, and data from the two models can be grouped for analysis.

In addition to a direct model number relationship, other bases for similarity are possible. Basically, any two or more MTE models with essentially the same features and specifications can be considered similar.

5.4.7 Data Validation

Data validation is required to eliminate data that are not representative of the MTE under analysis. There are three yardsticks by which data representativeness are measured:

- Data Validity
- Data Consistency
- Data Continuity

Each of these yardsticks will now be discussed.

5.4.7.1 Data Validity

Prior to analysis, data are truncated to remove inordinately short and inordinately long resubmission times. These periods are recognized as being both uncharacteristic with regard to duration and at odds with reliability expectations. To elaborate, short resubmission times are expected to be associated with high reliability, and long resubmission times are expected to be associated with low reliability. Thus, short resubmission time samples with inordinately low observed reliability or long resubmission times with inordinately high observed reliability are truncated.

A short resubmission time may be defined as one that is less than one quarter of the mode resubmission time, determined in the usual way. A long resubmission time may be defined as one that exceeds twice the mode resubmission time. The sampled MTE reliabilities for short resubmission times are considered inordinate if they fall below the $1-\alpha$ lower confidence limit for an *a priori* expected reliability. The

sampled long resubmission times are considered inordinate if they exceed the upper $1-\alpha$ confidence limit for the *a priori* expected MTE reliability.

The *a priori* MTE reliabilities are determined from a simple straight-line fit to the data:

$$R_{a\text{ priori}} = a + bt$$

The straight-line fit and the upper and lower confidence limits are determined by regression analysis.⁷

5.4.7.2 Data Consistency

It is often possible to improve an interval estimate by combining calibration results data from different model numbers, date ranges, or other groupings. However, it is valid to combine only data of homogeneous data sets. In these instances, the data sets should be evaluated for homogeneity or “consistency.” First, there should be an engineering basis to expect homogeneity: for example, two data sets for the same model number over different periods or in different organizations with no known differences in maintenance or usage, or data sets for different model numbers with the same basic design for the measurement mechanism. There is always the possibility that unforeseen factors can cause inconsistent measurement performance, so a statistical test should also be performed.

5.4.7.3 Two Data Sets

The most commonly used data consistency test uses an F-test to compare summary statistics of two data sets defined by Method S1. For data set i , $i \in \{1, 2\}$, Method S1 computes T_i , the estimated time in tolerance, and the number of OOTs, r_i as follows:

$$T_i = \sum_{j=1}^{n_i} t_{ij} (1 - r_{ij} / 2),$$

and

$$r_i = \sum_{j=1}^{n_i} r_{ij},$$

where for data set i , t_{ij} is the j th time between calibrations and r_{ij} equals 1 if the j th calibration is reported out of tolerance and equals 0 otherwise.

The observed out-of-tolerance rate, λ_i , for data set i is calculated:

$$\lambda_i = r_i / T_i$$

The data sets are trivially consistent if $\lambda_1 = \lambda_2$. Otherwise, a statistical test should be performed. If $\lambda_1 < \lambda_2$, the calculated “observed” F -statistic, F_c , is computed as

$$F_c = \frac{r_2}{r_1 + 1} \cdot \frac{T_1}{T_2}$$

⁷ Ref. ND66 provides an excellent resource for regression analysis methods. It is cited at several points in this RP.

RP-1: Establishment and Adjustment of Calibration Intervals

To reject the homogeneity of the two data sets with $1 - \alpha / 2$ confidence, this statistic is compared against the characteristic F -statistic obtained from the F -distribution:

$$F_{1-\alpha/2} [2(r_1 + 1), 2r_2]$$

On the other hand, if $\lambda_1 > \lambda_2$, the “observed” F -statistic F_c is computed as

$$F_c = \frac{r_1}{r_2 + 1} \cdot \frac{T_2}{T_1},$$

and the comparison statistic is

$$F_{1-\alpha/2} [2(r_2 + 1), 2r_1]$$

If $F_c > F_{1-\alpha/2}$, the homogeneity of the groupings is rejected and the data sets are considered inconsistent.

5.4.7.4 Two Data Sets Using Method A3

In the special case that the time between calibrations is the same constant value for both data sets, an exact binomial test is possible. Let data set 1 have r_1 out-of-tolerance conditions in N_1 calibrations, and let data set 2 have r_2 out-of-tolerance conditions in N_2 calibrations. The null hypothesis that the two data sets are homogeneous would be rejected at the α level of significance if the cumulative hypergeometric distribution

$$\sum_{x=0}^{r_1} HG(N_1 + N_2, N_1, r_1 + r_2, x) < \alpha / 2,$$

or

$$\sum_{x=0}^{r_2} HG(N_1 + N_2, N_2, r_1 + r_2, x) < \alpha / 2.$$

5.4.7.5 Many Data Sets

If there are $M > 2$ data sets, the data can be combined in a pair-wise fashion. It is generally preferable to combine the data first for the two data sets with the strongest engineering justification that the data are similar, then to compare the combined set with the next most expected similarity. For example, let the model 100B be an update of the 100A, and let the 100BOPT01 be the updated model with an IEEE interface. Unless the latter model is used substantially different from the others, the 100B and the 100BOPT01 would normally be tested first, because the measurement mechanism is considered identical. If the test accepts the hypothesis that the data are consistent, the two data sets are combined. If the upgrade from 100A to 100B is expected to substantially affect measurement performance, then no further testing would be appropriate. However, if the upgrade is not expected to affect measurement performance, then an F -test would be performed between the 100A data and the combined data for the 100B and the 100BOPT01. If the test passes, then all the data are combined.

Because each test has probability α of failing even if the data are homogeneous, the pair-wise approach becomes less reliable as the number of data sets becomes large. If there are many data sets, including at

least several reported out-of-tolerance conditions, then the likelihood ratio test can be used to circumvent the problem of too many pair-wise tests. The test statistic is as follows:

$$LR = \frac{-2}{B} \left(\sum_{i=1}^M r_i \ln \left(\frac{\sum_{i=1}^M r_i}{\sum_{i=1}^M T_i} \right) - \sum_{i=1}^M r_i \ln \left(\frac{r_i}{T_i} \right) \right),$$

where B is the Bartlett Correction Factor:

$$B = 1 + \frac{\left(\left(\left(\sum_{i \in \{m|r_m > 0\}} \frac{1}{r_i} \right) - 1 \right) \right) \sum_{i=1}^M r_i}{6 \sum_{i \in \{m|r_m > 0\}} 1}.$$

Homogeneity is rejected if

$$LR > \chi^2(n_f, \alpha),$$

where n_f is the number of data sets with OOT conditions minus one. If homogeneity is rejected, then pair-wise F -tests can help identify which data sets are different. All the data sets can be combined if homogeneity is accepted.

5.4.7.6 Example

The following example simulates exponential calibration results data for three models, the 100A, 100B, and 100C. Model 100A has a simulated out-of-tolerance rate of 0.0100 out-of-tolerance conditions per month, and the other two models have a simulated rate of 0.0050. Each data set has 100 calibrations of new data at a 12-month interval. Model 100C has an additional 100 calibrations of old data at a 24-month calibration interval.

Table 5-2 shows the results of this simulation. The reliability displayed is the theoretical EOP reliability under the exponential model, and the OOT count is the actual number, r_i , generated by the simulation. T_i is estimated total time in tolerance calculated by use of method S1. The next row shows the observed out-of-tolerance rates, which may be taken as estimates of the true rates.

Table 5-2. Simulated Group Calibration Results.

Parameter \ Model	100C	100C	100B	100A	Total
Data	new	Old	new	new	
Sim. OOTs per month	0.0050	0.0050	0.0050	0.0100	
Interval (month)	12	24	12	12	

Simulated EOP Reliability	0.941765	0.886920	0.941765	0.886920	
Interval Count	100	100	100	100	
OOT Count	3	14	5	17	39
T_i	1182	2232	1170	1098	5682
Obs. OOTs per month	0.0025	0.0063	0.0043	0.0155	

The Bartlett Correction Factor calculated from this table is

$$B = 0.999461,$$

which is very close to unity, because this well-balanced simulation requires no significant correction. The likelihood ratio statistic,

$$LR = 14.43,$$

for these four data sets tested against the chi-square distribution with three degrees of freedom gives a statistical significance of

$$\alpha = 0.00237,$$

which easily detects that the out-of-tolerance rates are not all the same, with 95 percent confidence (i.e., $\alpha < 0.05$).

Table 5-3 shows the result of the pair-wise homogeneity tests. Note that the approximate F -test and the exact cumulative hypergeometric test give very similar results in this case. At 95 percent confidence, the tests correctly combine the homogeneous data and reject the hypothesis that model 100A should have the same interval as the others. The hypergeometric test is not applicable to the last two pairs because the data sets in each pair have different intervals.

Table 5-3. Example Homogeneity Test Results

Data Set 1	Data Set 2	r_1	r_2	F_c	α_F	Cumulative Hypergeometric Parameters	α_{HG}	Combine?
100B/new	100A/new	5	17	3.0191	0.0112	$HG(200,100,22,5)$	0.0115	No
100C/new	100A/new	3	17	4.5751	0.0015	$HG(200,100,20,3)$	0.0015	No
100C/new	100B/new	3	5	1.2628	0.7154	$HG(200,100,17,3)$	0.7209	Yes
100C/new	100C/old	3	14	1.8535	0.2172	N/A	N/A	Yes
100B/new	100C/all	5	17	0.9710	0.9879	N/A	N/A	Yes

5.4.7.7 Data Continuity

To evaluate data continuity over the life cycle of a given MTE attribute, a calibration history must be maintained [DW91]. This history should contain information on service dates and calibration results for each attribute calibrated. This information should be recorded each time the calibration history data are incremented for analysis. Total attribute resubmission times and out-of-tolerances are computed as in Appendix C. Required data elements are discussed in Chapter 6.

RP-1: Establishment and Adjustment of Calibration Intervals

From the resubmission times and out-of-tolerance totals for each attribute, a history of MTBFs is assembled. This history is used to determine MTBF as a function of equipment inventory lifetime. Denoting this lifetime by T , we model MTBF according to

$$\hat{M}(T) = M_0 + \lambda T + \beta T^2.$$

Standard regression methods are used to obtain M_0 , λ and β , and to determine confidence limits for $\hat{M}(T)$ [ND66].

The procedure for determining discontinuities in the calibration history data begins with identifying and excluding attribute MTBF values that lie outside statistical confidence limits for $\hat{M}(T)$ [ND66].

Following this weeding out process, M_0 , λ and β are recomputed, and a more representative picture of $\hat{M}(T)$ is obtained. Next, the slope of $\hat{M}(T)$, given by

$$m = \frac{\partial \hat{M}}{\partial t} = \lambda + 2\beta t,$$

is searched for points (if any) at which $|m| > 0.5$. The latest calendar date for which this occurs is denoted T_c .

Two cases are possible: $m > 0.5$ and $m < -0.5$. For cases where $m < -0.5$, data recorded prior to T_c are excluded from analysis. If $m > 0.5$, reliability estimates R_c and R' are computed according to

$$R_c = \exp \left[-\frac{I}{\hat{M}(T_c)} \right]$$

and

$$R' = \exp \left[-\frac{I}{\hat{M}(T')} \right]$$

where I is the current assigned interval and T' is the most current date for which calibration history are available. Defining $\Delta R \equiv (R_c - R')/R_c$, a discontinuity in calibration history is identified if

$$|\Delta R| > D,$$

where D is a predetermined parameter. The value of D is determined in accordance with the amount of data available and the degree of data homogeneity desired. For most cases, $D = 0.2$ has been found useful.

If the condition $|\Delta R| > D$ applies, attribute calibration history data prior to T_c are deleted from records used for interval analysis.

5.5 Setting Measurement Reliability Targets

Establishing measurement reliability targets involves a consideration of several trade-offs between the desirability of controlling measurement uncertainty growth and the cost associated with maintaining such control. This section discusses this concept further.

Establishing an appropriate measurement reliability target is a multifaceted process. Unfortunately, no handy “rule-of-thumb” guidelines are applicable to the problem. However, a few general precepts have been established that assist in identifying important factors to consider and in getting a sense of how these factors inter-relate [NA89, HC89, JM92, NA94, MK07, HC08, MK08, SD09, DH09]. *NCSL International RP-18, “Estimation and Evaluation of Measurement Decision Risk”* also provides methodologies for establishing reliability targets.

The guiding points in establishing a measurement reliability target are the following:

- MTE measurement reliability is a measure of MTE attribute uncertainty.
- MTE attribute uncertainty is often a major contributor to the uncertainty of a product test process.
- The uncertainty in an item test process impacts the uncertainty in the product attributes being tested.
- Product attribute uncertainty impacts product utility. Low utility costs money.
- On the other hand, periodic calibration also costs money. The higher the target, the more frequent the calibration, the higher the operating cost.

Given that the immediate objective of setting a measurement reliability target is the control of test process error, the above list provokes four central questions:

1. How much does MTE attribute uncertainty contribute to test process uncertainty?
2. How sensitive is product uncertainty to test process uncertainty?
3. How sensitive is product utility to product uncertainty?
4. How does the cost of controlling product uncertainty compare with the cost of a loss of product utility?

Test process uncertainties emerge from several sources [HC95a, HC95b, HC95c]:

- 1) Intrinsic bias in the measuring and subject attributes
- 2) Random errors arising from fluctuations in the reference attribute, the subject attribute, the measuring environment, operator instability, etc.
- 3) Error due to attribute resolution
- 4) Sampling errors accompanying analogue-to-digital and digital-to-analogue conversion processes
- 5) Operator bias
- 6) Numerical round-off and other computation errors
- 7) Bias introduced by environmental factors (e.g., temperature, humidity, electromagnetic fields, etc.)
- 8) Errors due to stresses introduced by shipping and handling
- 9) Other sources.

The impact of MTE uncertainty on total test process uncertainty can be established by considering the product attribute value distributions that result from testing with MTE exhibiting maximum uncertainty (the lowest level of MTE measurement reliability achievable in practice) and minimum uncertainty (measurement reliability = 1.0). If the range between these extremes is negligible, then MTE uncertainty is not a crucial issue and measurement reliability targets can be set at low levels. In certain cases, it may be determined that periodic recalibration of MTE is not required. If product uncertainty proves to be a sensitive function of MTE uncertainty, however, then the MTE measurement reliability target takes on more significance. Under these conditions, a high measurement reliability target may be required.

For many on-orbit and deep-space applications, the length of the calibration intervals of on-board MTE requires designing systems to tolerate low measurement reliability targets. From the foregoing, it is apparent that this can be achieved if the MTE system is “over-designed” relative to what is required to support product tolerances or end-use requirements. Such over-design may involve the incorporation of highly stable components and/or built-in redundancy in measurement subsystems. In some cases where product performance tolerances are at the envelope of high-level measurement capability, it may be necessary to reduce the scope of the product's performance requirements. This alternative may sometimes be avoided by employing new SPC measures [HC84, DJ86b, HC91, NA94, RC95].

5.5.1 System Reliability Targets⁸

In many applications, a multi-component system is regarded as out-of-tolerance if one or more of its components is out-of-tolerance. If the system components are independent of one another, then the reliability of an n -component system can be written

$$R_S(t) = R_1(t) R_2(t) \dots R_n(t), \quad (5-1)$$

where

$$R_S(t) = \text{probability that all components are in-tolerance at time } t,$$

and

$$R_i(t) = \text{measurement reliability of the } i\text{th component at time } t, i = 1, 2, \dots, n.$$

Eq. (5-1) is the simplest expression of $R_S(t)$. We now consider an alternative expression that is more useful for the present topic. In this, we will imagine that we have a two-component system, where both components are independent. Extension to more complicated cases is straightforward. The relevant expression is

$$1 - R_S(t) = [1 - R_1(t)] R_2(t) + [1 - R_2(t)] R_1(t) + [1 - R_1(t)][1 - R_2(t)]. \quad (5-2)$$

Multiplying out the terms in this expression shows that Eq. (5-2) is equivalent to Eq. (5-1) for a two-component system.

We now consider the cost, C_S , of using the system in an out-of-tolerance condition. There are several alternatives for doing this, ranging from simple to complex. In the following, we will employ a fairly

⁸ Taken from Reference [IM95].

simple method. In this method, C_S is the product of the cost of using an item, given that it is out-of-tolerance, and the probability for an out-of-tolerance. Denoting the former by $C_{S|OOT}$, we have

$$C_S = C_{S|OOT} (1 - R_S) . \quad (5-3)$$

The contribution of each component to this cost is the product of (a) the cost of using an out-of-tolerance component, (b) the probability that the component will be used (given that the system is used) and (c) the probability that the component will be out-of-tolerance. The first term in this product is the **criticality function**, the second term is the **demand function**, and the third term is the complement of the **reliability function**.

Letting

C_i = criticality function for the i th component in terms of cost ,

and

d_i = demand function for the i th component ,

we have

$$C_{S|OOT} = C_1 d_1 (1 - d_2) + C_2 d_2 (1 - d_1) + (C_1 + C_2) d_1 d_2 . \quad (5-4)$$

Eqs. (5-2) through (5-4) suggest a weighted expression for the system reliability that will be useful in arriving at an interval for the system. This expression is

$$\begin{aligned} C_{S|OOT} [1 - R_S(t)] &= C_1 d_1 (1 - d_2) [1 - R_1(t)] + C_2 d_2 (1 - d_1) [1 - R_2(t)] \\ &\quad + (C_1 + C_2) d_1 d_2 [1 - R_1(t)] [1 - R_2(t)] . \end{aligned} \quad (5-5)$$

Dividing both sides of Eq. (5-5) by $C_{S|OOT}$ gives

$$\begin{aligned} 1 - R_S(t) &= c_1 d_1 (1 - d_2) [1 - R_1(t)] + c_2 d_2 (1 - d_1) [1 - R_2(t)] \\ &\quad + (c_1 + c_2) d_1 d_2 [1 - R_1(t)] [1 - R_2(t)] , \end{aligned} \quad (5-6)$$

where

$$c_i = \frac{C_i}{C_{S|OOT}} , \quad i = 1, 2 . \quad (5-7)$$

5.5.1.1 Applying System Reliability Targets

There are cases where a reliability target may be set for an entire system. In these cases, it is usually assumed that if any component of a system is found out-of-tolerance at test or calibration, the system as a whole is pronounced out-of-tolerance, and recorded as such. A system interval can therefore be found by interval-analysis performed at the system level on system test or calibration history.

It is also possible to compute a system interval without conducting a separate system interval analysis. In this approach, the system reliability is set equal to the system reliability target R_S^* and the interval T is

solved for from a knowledge of the reliability functions for the system components. For a two-component system, Eq. (5-6) yields the relevant equation as

$$1 - R_S^* = c_1 d_1 (1 - d_2) [1 - R_1(T)] + c_2 d_2 (1 - d_1) [1 - R_2(T)] \\ + (c_1 + c_2) d_1 d_2 [1 - R_1(T)] [1 - R_2(T)]. \quad (5-8)$$

5.5.1.2 Applying a Uniform Reliability Target

Some organizations impose a uniform or default reliability target, denoted R^* , that is intended to control measurement decision risk.

Because the reliability target is intended to control measurement decision risk, and measurement decision risk occurs at the component level, we will apply the target R^* at the component level. Denoting the desired calibration interval by T in Eq. (6-6) yields

$$1 - R_S(T) = c_1 d_1 (1 - d_2) (1 - R^*) + c_2 d_2 (1 - d_1) (1 - R^*) + (c_1 + c_2) d_1 d_2 (1 - R^*)^2,$$

and

$$R_S(T) = 1 - [c_1 d_1 + c_2 d_2 - (c_1 + c_2) d_1 d_2] (1 - R^*) - (c_1 + c_2) d_1 d_2 (1 - R^*)^2. \quad (5-9)$$

From Eq. (5-7), we see that $c_1 + c_2 = 1$. This simplifies Eq. (5-9) to

$$R_S(T) = 1 - (c_1 d_1 + c_2 d_2 - d_1 d_2) (1 - R^*) - d_1 d_2 (1 - R^*)^2. \quad (5-10)$$

The interval T is obtained from Eq. (5-10) by taking the inverse reliability function of $R_S(T)$ on both sides of the equation (see Appendix H)

$$T = R_S^{-1} [1 - (c_1 d_1 + c_2 d_2 - d_1 d_2) (1 - R^*) - d_1 d_2 (1 - R^*)^2]. \quad (5-11)$$

5.5.1.3 Ferling's Method

The above treatment applies to systems where demand probabilities and criticality levels are known. For some systems, this will not be the case. In these instances, all that is usually known about a system is that it is composed of tested or calibrated components that each have a reliability target and are calibrated at assigned intervals.

A method for setting system intervals that addresses these cases is called Ferling's method. Ferling showed [JF87] that criticality and demand requirements were both taken into account by simply setting the recall interval for a system equal to the shortest individual component interval and calibrating all components of the system at each calibration.

This approach offers a moderation of the traditional extreme view that all components of a multi-component system must be in-tolerance for the system itself to be considered in-tolerance. By focusing

attention on the “least reliable” component, it does this without compromising the control of measurement uncertainty growth.

5.5.1.4 Stratified Calibration

For some systems, the components perform in an individual way in which component functions are separate and distinct. Such systems may be regarded as collections of instruments that support independent functions. Thus the performance of one component has no bearing on the performance of any other component.

For such compartmentalized systems, the optimal recall strategy is one in which the system interval is set equal to the shortest interval of any of its components, as in Ferling's method, and components are calibrated as-needed.

This means that not all components are calibrated at every system recall interval; i.e., components are serviced according to their respective calibration schedules. Because the recall of components is dictated by the recall schedule for the system, however, implementing an individual component calibration schedule would involve some synchronization of component intervals with the system recall cycle. Such a scheme is referred to as a **stratified calibration plan**.

In stratified calibration, the calibration schedules for components are set at whole-number multiples of the system interval. This ordinarily involves a certain amount of “rounding off” or approximating. Intervals established in this way are examined to determine whether the rounding off compromises the measurement reliability to an unacceptable extent. If so, then some fine tuning may be called for.

5.6 Interval Candidate Selection

Because analyses of calibration history will be conducted periodically, it is unreasonable to suppose that sufficient new information will be accumulated between successive analyses to warrant reevaluation of calibration intervals for each attribute, manufacturer/model, or instrument class in the calibration history data base at each analysis session. This implies that only certain attributes, model numbers and instrument classes will be singled out for reevaluation at any given analysis run. This results in analysis of only those attributes, models or classes with nontrivial data increments accumulated since the previous interval assignment or adjustment. This naturally includes all first cases, which have accumulated sufficient data for initial analysis.

In the identification of interval candidates, the following definitions apply for the attribute, model or class of interest:

N_{cal}	$\equiv$	total number of calibrations accumulated at the date of the previous interval adjustment or assignment
T	$\equiv$	total resubmission time at the date of the previous interval adjustment or assignment
N_{OOT}	$\equiv$	total number of out-of-tolerances accumulated at the date of the previous interval adjustment or assignment
n_{OOT}	$\equiv$	number of out-of-tolerances accumulated since the last interval adjustment or assignment

RP-1: Establishment and Adjustment of Calibration Intervals

- n_{cal} $\equiv$ number of calibrations accumulated since the last interval adjustment or assignment
- I $\equiv$ current assigned calibration interval.

By use of these quantities, a candidate identification parameter is determined according to

$$\delta = \frac{\eta_{cal} I / T - \eta_{OOT} / N_{OOT}}{1 + \eta_{OOT} / N_{OOT}}.$$

An attribute, model or class is identified as a candidate for analysis if either of the following conditions is met:

- If $T = 0$ and $N_{cal} + n_{cal} \geq 15, 25$ or 40 at the attribute, model or class level, respectively.
- If $T \neq 0$ and $|\delta| \geq 0.05$ and $N_{cal} + n_{cal} \geq 15, 25$ or 40 at the attribute model or class level, respectively.

5.7 Identifying Outliers

5.7.1 Performance Dogs and Gems

Two methods for identifying performance outliers, one method for identifying support cost outliers, and one method for identifying suspect activities are discussed in this section.

The first performance outlier identification method requires that a “first pass” analysis be performed to ascertain the appropriate reliability model and to estimate its parameters. By use of the results of this analysis, serial-number item dogs and gems are identified and their records are removed from the data. The data are then re-analyzed and a refined set of parameter estimates is determined.

The second performance outlier identification method consists of an *a priori* identification of MTE attribute dogs and gems based on certain summary statistics. By use of these statistics, serial-number item dogs and gems are identified and their records are removed from the data prior to analysis.

The first method is preferred if accurate individual dog and gem calibration intervals are desired. The second method is preferred if dogs and gems are managed collectively. The second method is considerably easier to implement and is the recommended method where system operating cost and run time are of prime concern.

5.7.1.1 Dog and Gem Identification - Method 1

If measurement reliability modeling is performed, the computed variance in the model (see Appendix C) can be used to identify dogs and gems at the MTE serial-number and MTE manufacturer/model levels. Serial-number level dogs are identified as follows:

Let $(y_{\mu\nu}, t_{\mu\nu})$, $\nu = 1, 2, 3, \dots, n_{\mu}$ represent the pairs of observations on the μ th serial-numbered item of a given manufacturer/model. The variable $t_{\mu\nu}$ is the resubmission time for the ν th recorded calibration of the μ th item; $y_{\mu\nu} = 0$ for an out-of-tolerance, and $y_{\mu\nu} = 1$ for an in-tolerance. A mean interval and observed reliability are computed according to

$$\langle t_\mu \rangle = \frac{1}{n_\mu} \sum_{\nu=1}^{n_\mu} t_{\mu\nu} ,$$

and

$$\bar{R}_\mu = \frac{1}{n_\mu} \sum_{\nu=1}^{n_\mu} y_{\mu\nu} .$$

A lower confidence limit for the *expected* reliability is computed from

$$\hat{R}_{L\mu} = \hat{R}(\langle t_\mu \rangle, \hat{\theta}) - z_\alpha \sqrt{\text{var}[\hat{R}(\langle t_\mu \rangle, \hat{\theta})]} ,$$

where z_α is obtained from

$$1 - \alpha = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{z_\alpha} e^{-\zeta^2/2} d\zeta ,$$

and $\sqrt{\text{var}[\hat{R}(\langle t_\mu \rangle, \hat{\theta})]}$ is given in Appendix D.

An upper $1 - \beta$ confidence limit $\tilde{R}_U$ can be obtained for the *observed* reliability from the expression

$$\beta = \sum_{x=0}^{n_\mu \bar{R}_\mu} \binom{n_\mu}{x} \bar{R}_U^x (1 - \bar{R}_U)^{n_\mu - x} .$$

The item is identified as a dog with $1 - \alpha\beta$ confidence if $\bar{R}_U < \hat{R}_{L\mu}$. Gems are identified in like manner. An upper confidence limit is first determined for the expected reliability:

$$\hat{R}_{U\mu} = \hat{R}(\langle t_\mu \rangle, \hat{\theta}) + z_\alpha \sqrt{\text{var}[\hat{R}(\langle t_\mu \rangle, \hat{\theta})]} ,$$

whereas, for the observed reliability, we have

$$\beta = \sum_{x=n_\mu \bar{R}_\mu}^{n_\mu} \binom{n_\mu}{x} \bar{R}_L^x (1 - \bar{R}_L)^{n_\mu - x} .$$

The item is identified as a gem with $\alpha\beta$ confidence if $\tilde{R}_L > \hat{R}_{U\mu}$.

Following the same treatment with “instrument class” in place of “manufacturer/model” and “manufacturer/model” in place of “item,” identifies dogs and gems at the manufacturer/model level.

5.7.1.2 Dog and Gem Identification - Method 2

In method 2, a comparison is made between a summary statistic taken on one MTE serial number and a corresponding summary statistic for the MTE manufacturer/model. Given that method 2 is applied without prior knowledge of the specific reliability model governing the stochastic process, the statistic chosen should be one that can be considered a good general standard for comparison. One statistic that meets this requirement is the observed **mean time before failure**, or MTBF. The MTBF for the μ th attribute of the MTE manufacturer/model is given by

$$MTBF_{\mu} = \frac{t_{\mu}}{1 - R_{\mu}^{\%}},$$

where $\langle t_{\mu} \rangle$ and $R_{\mu}^{\%}$ are given by

$$t_{\mu} = \frac{1}{n_{\mu}} \sum_{i=1}^{n_{\mu}} t_{\mu i}$$

and

$$R_{\mu}^{\%} = \frac{1}{n_{\mu}} \sum_{j=1}^{n_{\mu}} g_{j\mu} = \frac{g_{\mu}}{n_{\mu}}.$$

In these expressions, $t_{\mu i}$ is the i th failure time for the μ th instrument; and g_{μ} and n_{μ} are, respectively, the number observed in-tolerance and the total number of calibrations for the μ th instrument.

Again, letting k represent the number of instruments within the MTE manufacturer/model grouping of interest, the aggregate MTBF for the manufacturer/model is given by

$$MTBF = \frac{T}{X^{\%}},$$

where

$$T = \sum_{\mu=1}^k n_{\mu} \langle t_{\mu} \rangle,$$

and

$$X^{\%} = \sum_{\mu=1}^k n_{\mu} (1 - R_{\mu}^{\%}).$$

5.7.1.3 Dog Identification

The test for identifying a serial-number dog involves computing an F-statistic with $2(x_2+1)$ and $2x_1$ degrees of freedom, where x_1 and x_2 are defined by

$$x_1 = \begin{cases} n_\mu (1 - \frac{P_0}{R_\mu}), & \text{if } MTBF_\mu < MTBF \\ \frac{P_0}{R_\mu}, & \text{otherwise,} \end{cases}$$

and

$$x_2 = \begin{cases} \frac{P_0}{R_\mu}, & \text{if } MTBF_\mu < MTBF \\ n_\mu (1 - \frac{P_0}{R_\mu}), & \text{otherwise.} \end{cases}$$

To complete the statistic, total resubmission times T_1 and T_2 are determined according to

$$T_1 = \begin{cases} n_\mu \langle t_\mu \rangle, & \text{if } MTBF_\mu < MTBF \\ T, & \text{otherwise,} \end{cases}$$

and

$$T_2 = \begin{cases} T, & \text{if } MTBF_\mu < MTBF \\ n_\mu \langle t_\mu \rangle, & \text{otherwise.} \end{cases}$$

Once x_1 , x_2 , T_1 and T_2 have been determined, an “observed” F-statistic is computed as

$$\hat{F}_0 = \frac{x_1}{x_2 + 1} \frac{T_2}{T_1}.$$

To identify the μ th serial number as a dog with $1 - \alpha$ confidence, this statistic is compared against a characteristic F-statistic obtained from the F distribution:

$$F_c = F_{1-\alpha} [2(x_2 + 1), 2x_1].$$

If $\hat{F}_0 > F_c$, the serial number is considered a dog.

5.7.1.4 Gem Identification

The serial number is considered a gem if

$$\frac{x_2}{x_1 + 1} \frac{T_1}{T_2} > F_{1-\alpha} [2(x_1 + 1), 2x_2].$$

Again, identification of dogs and gems at the manufacturer/model level is performed by substituting “manufacturer/model” for “attribute” and “instrument class” for “manufacturer/model.”

5.7.2 Support Cost Outliers

MTE items can be identified as outliers on the basis of excessive calibration support costs. The identification of support cost outliers may assist in decisions regarding corrective administrative or engineering action and/or may supplement the identification of performance outliers.

For purposes of support cost outlier identification, the expectation of the support cost per calibration action for a manufacturer/model is estimated. If the support cost for the j th calibration of the i th instrument is denoted CS_{ij} , then this estimate is given by

$$CS_i = \frac{1}{n_i} \sum_{j=1}^{n_i} CS_{ij},$$

where n_i is the number of calibrations performed on the i th instrument. The corresponding standard deviation is computed in the usual way:

$$s_i = \sqrt{\frac{1}{n_i - 1} \sum_{j=1}^{n_i} (CS_{ij} - CS_i)^2}.$$

To identify a given instrument as a support cost outlier, one determines whether its support cost exceeds the mean support cost for the manufacturer/model to such an extent that its cost can be considered to lie outside the manufacturer/model support cost distribution. This determination is accomplished by first computing the lower support cost confidence limit for the instrument and the upper support cost limit for the instrument's manufacturer/model. These limits are obtained as follows:

A lower $1 - \beta$ confidence limit (LCL) for the instrument is given by

$$CS_i^L = CS_i - t_{\beta, \nu_i} s_i / \sqrt{n_i},$$

where $\nu_i = n_i - 1$. To obtain an upper $1 - \alpha$ confidence limit (UCL) for the instrument's manufacturer/model, the following quantities are first computed:

$$CS = \frac{1}{n} \sum_{i=1}^k \sum_{j=1}^{n_i} CS_{ij},$$

and

$$s = \sqrt{\frac{1}{n - 1} \sum_{i=1}^k \sum_{j=1}^{n_i} (CS_{ij} - CS)^2},$$

where k is the number of serial-numbered instruments within the manufacturer/model, and $n = \sum n_i$.

The UCL is computed from

$$CS^U = CS + t_{\alpha, \nu} s / \sqrt{n},$$

where $v = n - 1$. If $CS_i^L > CS^U$, the item is identified as a support cost outlier with a confidence of $1 - \alpha\beta$.

5.7.3 Suspect Activities

A given MTE user's requirements may exert greater stresses on the MTE than those exerted by other users. This may have the effect of yielding calibration history data on the equipment that are not representative of the behavior of the equipment under ordinary conditions. Similarly, data recorded by certain calibrating facilities or by a certain calibrating technician may not be representative of mainstream data. Organizations or individuals whose calibration data are outside the mainstream are referred to as **suspect activities** [IM95].

For instance, suppose that an activity of interest is a calibrating technician's performance. In this case, we would identify a suspect activity by comparing all calibrations on all MTE performed by the technician with all calibrations of these same MTE performed by all other technicians. If, on the other hand, the activity of interest is an equipment user, we would compare all calibrations of MTE employed by the user of interest against all other calibrations of these MTE employed by other users. Note that suspect activity may also be caused by a combination of factors; detecting such conditions requires subjecting the possible permutations of factors, rather than a single factor, to the following analysis.

5.7.3.1 High Failure Rate Outliers

Let the set of calibrations corresponding to the activity of interest be designated m and let M label the set of all other activities' calibrations corresponding to these MTE. With these identifications, an activity can be identified as suspect through the use of a variation of the **median test** described in many statistics texts. In applying this test, we evaluate whether out-of-tolerance rates (OOTRs) observed from calibrations of MTE corresponding to a given activity tend to be significantly greater than OOTRs for these MTE taken in aggregate.

An item's OOTR is the inverse of its MTBF:⁹

$$OOTR = \frac{1}{MTBF}.$$

The median test procedure is as follows: First, determine the median OOTR for m and M combined (i.e., the set $m \cup M$). Next, define the following

n_m	=	the number of cases in m
n_M	=	the number of cases in M
n_a	=	the total number of cases in $m \cup M$ that lie above the median
n_{ma}	=	the number of cases in m that lie above the median
N	=	$n_m + n_M$.

⁹ MTBFs are computed as in dog and gem testing.

RP-1: Establishment and Adjustment of Calibration Intervals

Given that, in the sample of size N , the number of OOTRs lying above the median is n_a , the probability of observing an OOTR above the median in the sample is given by

$$p = \frac{n_a}{N}.$$

Regarding the observation of an OOTR above the median as the result of a Bernoulli trial, the probability of observing n OOTRs above the median in a sample of size n_m is given by the binomial distribution:

$$P(n > n_{ma}) = \sum_{n=n_{ma}}^{n_m} \binom{n_m}{n} p^n (1-p)^{n_m-n}.$$

Substituting for p in this expression gives

$$P(n > n_{ma}) = \sum_{n=n_{ma}}^{n_m} \frac{n_m!}{n!(n_m-n)!} \frac{n_a^n}{N^{n_m}} (N-n_a)^{n_m-n}.$$

The median test attempts to evaluate whether this result is inordinately high in a statistical sense. In other words, if the chance of finding n_{ma} or more OOTRs in a sample of size n_m is low, given that the probability for this is n_a/N , then we suspect that the sampled value n_{ma} is not representative of the population, i.e., it is an **outlier**. Specifically, the activity is identified as a suspect activity with $1-\alpha$ confidence if the probability of finding n_{ma} or more OOTRs above the median is less than α , i.e., if

$$P(n > n_{ma}) < \alpha.$$

5.7.3.2 Example 1:

Suppose that the following out-of-tolerance rates have been observed for calibrations on a given set of MTE:

Table 5-4. Example Outlier Identification Data

Technician	User	Calibrating Facility	OOTR
Eddie Zittslaff	Gondwana Park	Bob's Cal Service	0.075
Eddie Zittslaff	G. Gordon Gurgle	Bob's Cal Service	0.074
Mel Fernmeyer	Gondwana Park	SWAG Technologies, Inc.	0.082
Mel Fernmeyer	Jack (Rip) Huggeboom	SWAG Technologies, Inc.	0.077
Wanda Swoose	Jack (Rip) Huggeboom	Windy Finger Labs	0.078
Guy Gitchemoli	G. Gordon Gurgle	OOTs-R-Us	1.151
Guy Gitchemoli	Gondwana Park	OOTs-R-Us	1.031
Guy Gitchemoli	Wally Ballou	OOTs-R-Us	0.925
Hap Halvah	G. Gordon Gurgle	Bob's Cal Service	0.076

The median OOTR for the combined calibration history is obtained by first sorting by OOTR. This yields table 5-5.

Table 5-5. Sorted Outlier Identification Data

Technician	User	Calibrating Facility	OOTR
Eddie Zittslaff	G. Gordon Gurgle	Bob's Cal Service	0.074
Eddie Zittslaff	Gondwana Park	Bob's Cal Service	0.075
Hap Halvah	G. Gordon Gurgle	Bob's Cal Service	0.076
Mel Fernmeyer	Jack (Rip) Huggeboom	SWAG Technologies, Inc.	0.077
Wanda Swoose	Jack (Rip) Huggeboom	Windy Finger Labs	0.078
Mel Fernmeyer	Gondwana Park	SWAG Technologies, Inc.	0.082
Guy Gitchemoli	Wally Ballou	OOTs-R-Us	0.925
Guy Gitchemoli	Gondwana Park	OOTs-R-Us	1.031
Guy Gitchemoli	G. Gordon Gurgle	OOTs-R-Us	1.151

In this table, we have $N = 9$ and a median value of 0.078. Accordingly, $n_a = 4$.

5.7.3.2.1 Technician Outlier Identification

For this outlier identification, the relevant values are:

Table 5-6. Technician Outlier Identification Data

Technician	n_m	n_{ma}
Eddie Zittslaff	2	0
Hap Halvah	1	0
Wanda Swoose	1	0
Mel Fernmeyer	2	1
Guy Gitchemoli	3	3

In evaluating the probability $p(n)$ of observing n OOTRs above the median, we define a probability density $p(n)$ given by

$$p(n) = \frac{n_m!}{n!(n_m - n)!} \frac{n_a^n}{N^{n_m}} (N - n_a)^{n_m - n}.$$

Suppose that we want to identify outlier technicians with 90 % confidence. Then $\alpha = 0.10$, and the following results are obtained:

Eddie Zittslaff:

$$n_m = 2, n_{ma} = 0$$

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=0}^2 p(n) = 1^{10}$$

Hap Halvah:

$$n_m = 1, n_{ma} = 0$$

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=0}^1 p(n) = 1$$

Wanda Swoose:

$$n_m = 1, n_{ma} = 0$$

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=0}^1 p(n) = 1$$

Mel Fernmeyer:

$$n_m = 2, n_{ma} = 1$$

¹⁰ In cases where the summation is taken from zero to n_m , the sum is equal to unity.

$$\begin{aligned}
 p(n) &= \frac{2!}{n!(2-n)!} \frac{4^n}{9^2} (9-4)^{2-n} \\
 &= \frac{2!}{n!(2-n)!} \frac{4^n}{81} 5^{2-n} \\
 p(1) &= \frac{2!}{1!(2-1)!} \frac{4^1}{9^2} (9-4)^{2-1} = (2)(4/81)(5) = 40/81 \\
 p(2) &= \frac{2!}{2!(2-2)!} \frac{4^2}{9^2} (9-4)^{2-2} = 16/81
 \end{aligned}$$

and

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=1}^2 p(n) = 56/81 = 0.691$$

Guy Gitchemoli:

$$\begin{aligned}
 n_m &= 3, n_{ma} = 3 \\
 p(n) &= \frac{3!}{n!(3-n)!} \frac{4^n}{9^3} 5^{3-n} \\
 p(3) &= \frac{3!}{3!(3-3)!} \frac{4^3}{9^3} 5^{3-3} = (4/9)^3 = 64/729 = 0.0878
 \end{aligned}$$

and

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=3}^3 p(n) = 0.0878$$

If we employ a significance level of $\alpha = 0.10$, then we see that the calibration performance of Guy Gitchemoli is identified as an outlier.

5.7.3.2.2 User Outlier Identification

For this outlier identification, the relevant values are

Table 5-7. User Outlier Identification Data

User	n_m	n_{ma}
G. Gordon Gurgle	3	1
Gondwana Park	3	2
Jack (Rip) Huggeboom	2	0
Wally Ballou	1	1

As with technician outlier evaluation, we have

$$p(n) = \frac{n_m!}{n!(n_m - n)!} \frac{n_a^n}{N^{n_m}} (N - n_a)^{n_m - n}.$$

With these relations, the following results are obtained:

G. Gordon Gurgle:

$$n_m = 3, n_{ma} = 1$$

$$p(n) = \frac{3!}{n!(3-n)!} \frac{4^n}{9^3} 5^{3-n}$$

$$p(1) = \frac{3!}{1!(3-1)!} \frac{4^1}{9^3} 5^{3-1} = \frac{3!}{2!} \frac{4}{9^3} 5^2 = 300 / 729$$

$$p(2) = \frac{3!}{2!(3-2)!} \frac{4^2}{9^3} 5^{3-2} = 240 / 729$$

$$p(3) = \frac{3!}{3!(3-3)!} \frac{4^3}{9^3} 5^{3-3} = 64 / 729$$

and

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=1}^3 p(n) = \frac{300 + 240 + 64}{729} = \frac{604}{729} = 0.829$$

Gondwana Park:

$$n_m = 3, n_{ma} = 2$$

$$p(n) = \frac{3!}{n!(3-n)!} \frac{4^n}{9^3} 5^{3-n}$$

$$p(2) = 240 / 729$$

$$p(3) = 64 / 729$$

and

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=2}^3 p(n) = \frac{240 + 64}{729} = 304 / 729 = 0.417$$

Jack (Rip) Huggeboom:

$$n_m = 2, n_{ma} = 0$$

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=0}^2 p(n) = 1$$

Wally Ballou:

$$n_m = 1, n_{ma} = 1$$

$$p(1) = \frac{1}{1!(1-1)!} \frac{4^1}{9^1} 5^{1-1} = 0.444$$

and

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=1}^1 p(n) = 0.444$$

If we employ a significance level of $\alpha = 0.10$, then we see that no user's calibration performance is identified as an outlier.

5.7.3.2.3 Servicing Facility Outlier Identification

For this outlier identification, the relevant values are

Table 5-8. Facility Outlier Identification Data

Cal Facility	n_m	n_{ma}
Bob's Cal Service	3	0
SWAG Technologies, Inc.	2	1
Windy Finger Labs	1	0
OOTs-R-Us	3	3

Again, as with technician and user outliers, we use

$$p(n) = \frac{n_m!}{n!(n_m - n)!} \frac{n_a^n}{N^{n_m}} (N - n_a)^{n_m - n}.$$

With these relations, the following results are obtained:

Bob's Cal Service:

$$n_m = 3, n_{ma} = 0$$

and

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=0}^3 p(n) = 1$$

SWAG Technologies, Inc.:

$$n_m = 2, n_{ma} = 1$$

$$p(1) = \frac{2!}{1!(2-1)!} \frac{4^1}{9^2} 5^{2-1} = 0.494$$

$$p(2) = \frac{2!}{2!(2-2)!} \frac{4^2}{9^2} 5^{2-2} = 0.198$$

and

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=1}^2 p(n) = 0.494 + 0.198 = 0.691$$

Windy Finger Labs:

$$n_m = 1, n_{ma} = 0$$

$$\sum_{n=n_{ma}}^{n_m} p(n) - \sum_{n=0}^1 p(n) = 1$$

OOTs-R-Us:

$$n_m = 3, n_{ma} = 3$$

$$p(3) = 64 / 729 = 0.088$$

and

$$\sum_{n=n_{ma}}^{n_m} p(n) = \sum_{n=3}^3 p(n) = 0.088$$

If we employ a significance level of $\alpha = 0.10$, then we see that OOTs-R-Us is identified as a cal facility outlier.

5.7.3.3 Low Failure Rate Outliers

A low-failure-rate outlier is one whose OOTR is inordinately low compared to the mainstream. We can easily justify the effort to identify high-failure-rate outliers. High failure rate outliers tend to skew the data in a way that may have a significant impact on interval analysis.

Low-failure-rate outliers tend to have a lesser impact, because we are usually trying to reach reliability targets higher than 0.5 – often considerably higher. For this reason, the occurrence of false *in*-tolerance observations do not usually increase significantly the already high numbers of in-tolerances we expect to observe. So, why identify low-failure-rate outliers?

RP-1: Establishment and Adjustment of Calibration Intervals

The reason is that, in many cases, a low failure rate is due to unusual usage or handling by an MTE user or to a misunderstanding of Condition Received codes by a testing or calibrating technician. These cases need to be identified for equipment management purposes or for personnel training purposes.

Again, let the set of calibrations corresponding to the activity of interest be designated m , and let the set of all other activities' calibrations corresponding to these MTE be designated M . We again use the variables

n_m	=	the number of cases in m
n_M	=	the number of cases in M
n_a	=	the total number of cases that lie above the median
n_{ma}	=	the number of cases in m that lie above the median;

$$\text{then, } N = n_m + n_M.$$

Given that, in the sample of size N , the number of OOTRs lying above the median is n_a , the probability of observing an OOTR *below* the median in the set $m \cup M$ is given by

$$p = \frac{N - n_a}{N}.$$

Regarding the observation of an OOTR below the median as the result of a Bernoulli trial, the probability of observing n OOTRs below the median in a sample of size n_m is given by the binomial distribution:

$$P(n > n_m - n_{ma}) = \sum_{n=n_m-n_{ma}}^{n_m} \frac{n_m!}{n!(n_m-n)!} \frac{n_a^{n_m-n}}{N^{n_m}} (N - n_a)^n.$$

The low-failure-rate median test attempts to evaluate whether this result is inordinately high in a statistical sense. In other words, if the chance of finding $n_m - n_{ma}$ or more OOTRs in a sample of size n_m is low, given that the probability for this is $(N - n_a) / N$, then we suspect that the sampled value n_{ma} is not representative of the population, i.e., it is an outlier. Specifically, the activity is identified as a suspect activity with $1 - \alpha$ confidence if the probability of finding $n_m - n_{ma}$ or more OOTRs below the median is less than α , i.e., if

$$P(n > n_m - n_{ma}) < \alpha.$$

5.7.3.4 Example 2:

We will use the same data to illustrate the identification of low-failure-rate outliers as we used in the example of high-failure-rate-outliers. Again, we have $N = 9$, a median value of 0.078 and $n_a = 4$. For technician outlier identification, the relevant values are, as before,

Table 5-9. Technician Low OOT Rate Data

Technician	n_m	n_{ma}
Eddie Zittslaff	2	0
Hap Halvah	1	0
Wanda Swoose	1	0
Mel Fernmeyer	2	1
Guy Gitchemoli	3	3

In evaluating the probability $p(n)$ of observing n OOTRs below the median, we define a probability density $p(n)$ given by

$$p(n) = \frac{n_m!}{n!(n_m - n)!} \frac{n_a^{n_m - n}}{N^{n_m}} (N - n_a)^n.$$

Suppose that we want to identify outlier technicians with 90 % confidence. Then $\alpha = 0.10$, and the following results are obtained:

Eddie Zittslaff:

$$n_m = 2, n_{ma} = 0$$

$$p(2) = \frac{5^2}{9^2} = 25 / 81$$

$$\sum_{n=n_m-n_{ma}}^{n_m} p(n) = \sum_{n=2}^2 p(n) = p(2) = \frac{25}{81} = 0.309$$

Hap Halvah:

$$n_m = 1, n_{ma} = 0$$

$$p(1) = \frac{1!}{1!(1-1)!} \frac{4^{1-1}}{9^1} 5^1 = \frac{5}{9}$$

$$\sum_{n=n_m-n_{ma}}^{n_m} p(n) = \sum_{n=1}^1 p(n) = p(1) = 0.556$$

Wanda Swoose:

$$n_m = 1, n_{ma} = 0$$

$$p(1) = \frac{1!}{1!(1-1)!} \frac{4^{1-1}}{9^1} 5^1 = \frac{5}{9}$$

$$\sum_{n=n_m-n_{ma}}^{n_m} p(n) = \sum_{n=1}^1 p(n) = p(1) = 0.556$$

Mel Fernmeyer:

$$n_m = 2, n_{ma} = 1$$

$$p(n) = \frac{2!}{n!(2-n)!} \frac{4^{2-n}}{9^2} 5^n$$

$$p(1) = \frac{2!}{1!(2-1)!} \frac{4^{2-1}}{9^2} 5^1 = \frac{40}{81}$$

$$p(2) = \frac{25}{81}$$

and

$$\sum_{n=n_m-n_{ma}}^{n_m} p(n) = \sum_{n=1}^2 p(n) = (40 + 25) / 81 = 0.803$$

Guy Gitchemoli:

$$n_m = 3, n_{ma} = 3$$

and

$$\sum_{n=n_m-n_{ma}}^{n_m} p(n) = \sum_{n=0}^3 p(n) = 1$$

If we employ a significance level of $\alpha = 0.10$, then we see that none of the technicians is identified as a low-failure-rate outlier.

The identification of **User** and **Calibrating Facility** low-failure-rate outliers proceeds in the same way as in the identification of high failure rate outliers, with the same substitutions as were used in the above example.

5.8 Engineering Analysis

Engineering analysis may also be used to predict calibration intervals that are commensurate with predetermined in-tolerance percentages. While these methods are predictive, they base their predictions on stability and other engineering parameters rather than on calibration history.

As stated earlier, the stability of an attribute relative to its tolerances is a principal driving influence in determining test/calibration intervals. If the response of an attribute to stress and the magnitude and frequency of stress are known, it may be possible to form a deterministic estimate of the length of time required for the attribute to go out-of-tolerance. Such an estimate would be the result of engineering analysis.

In engineering analysis, attention is focused at the attribute level. The extension of results at this level to a recommended calibration interval at the equipment level is not always obvious. One approach is to

determine an interval of time corresponding to a predetermined fraction of attributes for an item being in-tolerance. Another is to use Ferling's method and key the interval on the least stable attribute [JF87]. Still another involves weighting attributes according to criticality and usage demand. At present, there is no general agreement on the best practice. If in doubt, Ferling's method is recommended on the grounds that it presents an economical solution without sacrificing measurement reliability.

Engineering analysis can be a valid and effective methodological approach if conducted in an objective, structured manner, focusing on attribute stability relative to performance specifications. This is particularly evident in the process of establishing initial intervals. In this RP, the term “engineering analysis” refers only to analyses that are methodological, objective and key on attribute stability (i.e., measurement reliability) as opposed to maintenance or other considerations.

Engineering analysis is to be distinguished from **engineering judgment**. The latter refers to a process in which knowledge of the operational “quality” and reliability of an item is extrapolated to an impression of its measurement reliability from which a calibration interval is recommended. Because of the subjective nature of this process and because cognizance of the distinction between operational and measurement reliabilities may not always be clear in the mind of the practitioner, estimating intervals by engineering judgment is not a recommended methodology.

5.9 Reactive Methods

In this RP, “reactive methods” is a term used to label calibration interval adjustment methods that react to data from recent calibrations without attempting to model or “predict” measurement reliability behavior over time. Several such methods are currently in use, and others have been proposed in the literature. In this document, we describe three algorithms that illustrate the essentials of these methods. These descriptions are presented in Appendix B.

5.10 Initial Intervals

Initial interval methodologies are recommended below in descending order of preference. The ranking is based on considerations of objectivity, flexibility, accuracy and long-term cost effectiveness. In selecting a methodology, readers are encouraged to pick the highest recommendation commensurate with budget, available staff expertise and data processing capability, and data availability. The pros and cons of these methods are discussed in Chapters 2 and 4.

5.11 Similar Item Assignment

This is the preferred method if calibration intervals are available for a similar item grouping that the equipment in question can be assigned to. Similar items calibration intervals can be applied directly to the equipment in question using the same reliability target or, if interval adjustment methodologies S1 through S3 are employed, adjusted for a different reliability target. Methods S1 through S3 are described in Appendices C, D and E.

5.12 Instrument Class Assignment

Next to Similar Item Assignment, this is the preferred method if the equipment in question can be categorized in an existing class, and calibration intervals are available at the instrument class level. Instrument class calibration intervals can be applied directly to the equipment in question by use of the

same reliability target or, if interval adjustment methodologies S1 through S3 are employed, adjusted for a different reliability target.

5.13 Engineering Analysis

If calibration intervals by instrument class are not available, engineering analysis is the preferred method for obtaining initial intervals. To employ this method, expertise is required at the journeyman or senior engineering level in the measurement discipline(s) of interest. Little development capital is required to implement this method. The method does, however, require an operating budget, which may exceed that required for maintaining an instrument class analysis capability.

If engineering analysis is employed, inferences drawn from data on similar items maintained within the user's facility are likely to be superior to inferences drawn from design analysis. On the other hand, inferences made on the basis of design analysis are likely to be superior to inferences made from manufacturer recommendations.

5.14 External Intervals

If instrument class intervals are not available and engineering analysis is not feasible, external authority is recommended as a source of initial interval information. This method has several serious drawbacks, however, and the user is cautioned to read the relevant sections of Chapters 2 and 4 of this RP prior to its application. Conversion of an external interval to one consistent with the requiring organization's reliability targets is described in Appendix F.

5.15 General Interval

Assigning a uniform interval to all items new in inventory is recommended as a last resort. If this method is used, the interval selected should be short enough to accommodate equipment with poor measurement reliability characteristics and to quickly generate sufficient data to enable interval-analysis and adjustment using other methods.

Chapter 6 Required Data Elements

6.1 Introduction and Background

Most of the calibration interval assignment and adjustment methods discussed in this RP base calibration intervals on various technical and other data. In particular, calibration interval adjustment is based primarily on the results of calibration as documented in calibration history. Interval assignment and adjustment cannot be effective unless these data are complete, valid and standardized. In most organizations, the vehicle for ensuring that these criteria are met is the calibration procedure. Accordingly, the quality and effectiveness of calibration intervals depend on the quality and effectiveness of calibration procedures, and it is highly recommended that such procedures be developed and maintained in accordance with the best available practices. In this regard, the reader is encouraged to implement the principles and guidelines documented in *NCSL International RP-3, "Calibration Procedures"* [NC90].

A cornerstone of calibration interval assignment, adjustment and verification is a basic set of data elements composed of equipment identification, maintenance, and calibration history data. The following discussion reviews specific record-keeping requirements relating to these data. Data elements are described and classified by usage to help determine the data required for a given interval adjustment method or to realize other benefits. Note that though many of the data elements are discussed in terms of a name or textual description, the database should standardize the nomenclature via unique identifiers or other codes to eliminate multiple descriptions that represent the same information. A relational and properly "normalized" database with software that assigns values via approved and standardized pick lists or other controlled methods will serve well in this regard.

Maintaining data reliability is perhaps the most tedious aspect of an automated interval-analysis system. In practice, an organization will encounter abnormal events such as revised calibration certificates, cancelled calibrations, multiple calibration events occurring on the same item on the same day, and other anomalies. If contained in the history database, all such anomalies should be appropriately flagged or otherwise filtered before the system performs interval-analysis computations. In addition, because calibration intervals depend on measurement reliability, not functional reliability, not all data recorded during a calibration is relevant to the specified equipment accuracy and the calibration interval. Provisions should be made to include only measurement performance data when determining the in- or out-of-tolerance condition; attributes pertaining to functionality, damage, physical condition, appearance, etc., should be filtered out before the analysis. Therefore, the data collection mechanisms, data forms or database structures should be designed by engineering personnel familiar with the MTE requirements specifications. Also, functional failures in which no measurement data are obtained do not constitute an out-of-tolerance condition, but rather an indeterminate condition that the system should ignore. Finally, the system should analyze intervals, not calibrations per se, and therefore should ensure that all data analyzed represent a valid interval consisting of two consecutive calibrations, the first having been *issued* to a user, and the second having obtained *as-found accuracy-related* measurement results.

The organization that assigns the calibration interval (whether the user, the calibrating laboratory, or a third party (note that some calibration quality standards [Z540.3, ISO05] prescribe who may assign intervals and under what conditions) should have access to *all* the relevant data. Some form of data pooling will be helpful if, for example, the user assigns intervals but contracts with multiple calibration service providers who maintain the calibration data. More complicated and challenging scenarios are possible in which data for a particular instrument model is scattered over a network of users and vendors joined by multiple, non-exclusive service agreements. Lacking a solution that pools all data

(anonymously) for shared access, one should at least gather as much of the data for a particular user as is practical before attempting interval analysis.

6.2 Identification Elements

For purposes of identification, the following data elements are recommended.

Table 6-1. Identification Elements

Data Element	Application	Description and/or Purpose	Relevant Adjustment Methods
Class, Group, and Type Names	Data Pooling for Interval Analysis, Dog & Gem Analysis	General description such as “Multimeter, Digital” or “Thermometer, PRT.” A hierarchy of such descriptions that represent instrument classes, groups, families and types facilitates data pooling.	BI, EA, A3, S1, S2, S3, VDA
Manufacturer	Data Pooling, Identification, Dog & Gem Analysis	The item’s manufacturer.	BI, EA, A3, S1, S2, S3, VDA
Model or Part Number	Data Pooling, Identification, Dog & Gem Analysis	Designator assigned to the equipment by the manufacturer, or a military nomenclature. The manufacturer and the model or part number are the basic equipment identifiers required to allow data grouping for determination and analysis of calibration intervals.	BI, EA, A3, S1, S2, S3, VDA
Serial or Control Number	Identification, Dog & Gem Analysis	Unique, non-transferable identifier assigned to a specific piece of equipment to track individual instruments. Essential for identification of statistically better or worse performers. Should be assigned by the contractor if not assigned by the manufacturer. Often the manufacturer’s serial number is tracked but the contractor maintains a separate control number that serves as the unique identifier.	All except GI
Current Location	Off-Target Reliability Analysis ¹¹	Last known location of equipment. Primarily an administrative aid for recall notification, on-site calibration, problem notification, etc. With regard to interval analysis, it could also be used for outlier detection and failure analysis.	BI, EA, A3, S1, S2, S3, VDA
Attribute Name	Attribute Interval Analysis, Identification	Primary designator of a calibrated attribute. May have one or more qualifier fields to uniquely identify the range, function or ancillary attributes.	EA, A3, S1, S2, S3, VDA

¹¹ Off-target reliability analysis determines the cause of inappropriately high or low measurement reliability relative to a reliability target. In the case of low reliability this may be known also as failure mode analysis (FMA).

RP-1: Establishment and Adjustment of Calibration Intervals

*GI=General Interval, BI=Borrowed Intervals, EA=Engineering Analysis, VDA=Variables Data Analysis.

6.3 Technical Elements

For purposes of calibration interval and reliability analyses, the recommended technical data elements are given below.

Table 6-2. Technical Elements

Data Element	Application	Description and/or Purpose	Relevant Adjustment Methods
Date of Last Calibration	Interval Analysis	Date when the most recent calibration was completed.	A3, S1, S2, S3, VDA
Assigned Interval	Interval Adjustments	The current calibration interval. Having both the due date and the assigned interval allows a distinction between an interval adjustment and a “one-time” extended or short-cycled due date. May be assigned by the laboratory, the user, or an independent third party.	All
Date Due for Calibration	Data Continuity Evaluation, Resubmission Time Windows	To compare against date submitted for service to determine if the reason for submission was routine, inordinately late, or reflected possible user detection of an out-of-tolerance. May be assigned by the laboratory, the user, or an independent third party.	A3, S1, S2, S3, VDA
Date Submitted for Calibration	Interval Analysis	Date when the item was submitted by the user for calibration. Signals the end of the in-use period.	A3, S1, S2, S3, VDA
Calibration Start Date	Interval-analysis for multi-day calibrations	Date the calibration was started. Required to calculate the time elapsed since the last calibration. Same as either the date submitted or the date completed in a simplified system.	A3, S1, S2, S3, VDA
Date of Completion	Interval-analysis for multi-day calibrations	Date the calibration completed. Required to set recall date and to calculate time between current service and subsequent “Date Submitted for Service.” Same as the date of last calibration in a simplified system.	A3, S1, S2, S3, VDA
Custodian	Off-Target Reliability Analysis	Using organization responsible for the equipment. This identification could be broken down further by department, shop, laboratory, loan pool, etc.	BI, EA, A3, S1, S2, S3, VDA
Servicing Laboratory and Technician	Off-Target Reliability Analysis	For verification crosscheck of the service performed.	BI, EA, A3, S1, S2, S3, VDA

RP-1: Establishment and Adjustment of Calibration Intervals

Data Element	Application	Description and/or Purpose	Relevant Adjustment Methods
Procedure Used	Data Continuity Evaluation, Off-Target Reliability Analysis	Identification (with revision number) of the calibration procedure or technical manual used by the technician to perform the calibration. Needed to ensure consistency of data recorded from one calibration to the next. Not required if only one procedure is used for all calibrations of the item of interest.	BI, EA, A3, S1, S2, S3, VDA
Condition Received	Interval Analysis, System Evaluation	Condition of operable equipment when received for calibration expressed either as in-tolerance (all attributes performed within the tolerances required at all test points), out-of-tolerance (one or more of the attributes failed to meet the requirements at one or more test points, or indeterminate. (Inoperable equipment shall be noted but that data should not affect the analysis.)	All
Physical Condition	Interval Analysis, Off-Target Reliability Analysis	Condition Received may also include separate information regarding physical condition or storage environment that may have affected the equipment's in-tolerance status.	A1, A2, A3, S1, S2, S3, VDA
Renewal Action	MTBF calculations for Interval Analysis	Identify actual adjustment events and the periods between them; e.g., "not adjusted" or "adjusted,"	S1, S3
Adjustments or Repairs Made, Parts Replaced	Off-Target Reliability Analysis, Data Continuity	Document any modification or repair actions taken to return the instrument to in-tolerance or functional condition; e.g., "significant repair" or "minor service." Identify parts replaced or repaired.	S1, S3
Man-Hours to Calibrate / Repair	Cost and Dog / Gem Analysis	Time expended to calibrate or repair equipment. Used to permit cost trade-offs where appropriate as well as to pinpoint excessive costs and report cost savings.	BI, EA, A3, S1, S2, S3, VDA
As-Found and As-Left Measurement Results	Drift & Stability Analysis, Feedback Analysis ¹²	The actual measurement data recorded at the previous calibration (as-left) and the succeeding calibration (as-found). Required for drift rate analysis.	VDA
As-Found and As-Left Measurement Uncertainty	Drift & Stability Analysis, Feedback Analysis	The uncertainty of the as-found and as-left measurement results. Variables data methods may use this information in weighted regression techniques to improve interval estimates.	VDA

¹² A method for estimating the point during the interval at which an attribute became OOT based on the observed drift rate and uncertainty growth characteristics

RP-1: Establishment and Adjustment of Calibration Intervals

Data Element	Application	Description and/or Purpose	Relevant Adjustment Methods
Tolerance Limits	Drift & Stability Analysis, Feedback Analysis	The in- / out-of tolerance boundaries or specification limits. Used with predicted drift or confidence limits to compute an interval in variables data analysis. Although attributes data analysis methods do not require tolerance limits and as-found measurements, automated determination of the IT / OOT state via this data is often more reliable than manual OOT flagging.	VDA

While there is no specific requirement as to how long maintenance and calibration data should be kept in readily accessible records, it is good practice to retain all information on an item as long as the item type or its higher- level equipment groupings are used by the requiring organization. See “Data Retention” in Chapter 3

Chapter 7 No Periodic Calibration Required

“No periodic calibration required” (NPCR) status may be assigned to selected items in inventory. Some of the justifications that have been found useful for this assignment are the following [METRL]:

- 1) The instrument does not make measurements or provide known outputs.
- 2) The instrument is used as a transfer device whose measurement or output value is not explicitly used.
- 3) The instrument is a component of a calibrated system or function.
- 4) The instrument is fail-safe in that failure to operate within specified performance limits will be evident to the user.
- 5) The instrument makes measurements or provides known outputs, which are monitored by a calibrated device, meter, or gage during use.
- 6) The instrument makes measurements, which are required only to provide an indication of operational condition rather than a numerical value.
- 7) The instrument is disposed of after a short life cycle within which its measurement reliability holds to an acceptable level.
- 8) Fundamental (e.g., quantum-mechanical) standards.

NPCR items are exempt from calibration interval assignment and adjustment. They may, however, require initial calibration or adjustment at their introduction into use. Accordingly, the designation NPCR is not to be confused with the designation NCR (no calibration required).

The above justifications are general in nature and as implemented by one organization. Other organizations should consider the quality standard(s) and any other requirements by which they operate.

References

- 5300.4 - Jet Propulsion Laboratory. *Metrology and Calibration Provisions Guidelines*, NASA Handbook NHB 5300.4(1A). June 1990.
- 45662A - *Calibration Systems Requirement*, MIL-STD-45662A, 1988.
- AE54 - Eagle, A. "A Method for Handling Errors in Testing and Measuring." *Industrial Quality Control*, pp 10-15, March 1954.
- BW91 - Weiss, B. "Does Calibration Adjustment Optimize Measurement Integrity?" *Proceedings, NSCL Workshop and Symposium*, Albuquerque, NM, August 1991.
- DD93 - Deaver, D. "How to Maintain Your Confidence." *Proceedings, NSCL Workshop and Symposium*, Albuquerque, NM, July 1993.
- DD94 - Deaver, D. "Guardbanding with Confidence." *Proceedings, NSCL Workshop and Symposium*, Chicago, IL, July - August, 1994.
- DD95 - Deaver, D. "Using Guardbands to Justify TURs Less Than 4:1". *Proceedings, Measurement Science Conference*, Anaheim, CA, January 1995.
- DH09 - Huang, D, and S. Dwyer. "Test Instrument Reliability Perspectives and Practices: Interpreted within System Reliability Framework." *Proceedings, NSCL Workshop and Symposium*, San Antonio, 2009.
- DJ85 - Jackson, D. *Analytical Methods Used in the Computer Software for the Manometer Audit System*, SAIC Technical Report TR-830016-4M112/006-01, Computer Software Specification, Dept. of the Navy Contract N00123-83-D-0016, Delivery Order 4M112. 8 October, 1985.
- DJ86a - Jackson, D., J. Ferling, and H. Castrup. "Concept Analysis for Serial Number Based Calibration Intervals." *Proceedings, Measurement Science Conference*, Irvine, January 23-24, 1986.
- DJ86b - Jackson, D. *Instrument Intercomparison: A General Methodology*, Analytical Metrology Note AM 86-1. NWS Seal Beach, U.S. Navy Metrology Engineering Center, January 1, 1986.
- DJ87a - Jackson, D. *Instrument Intercomparison and Calibration*. *Proceedings, Measurement Science Conference*, Irvine, January 29 – 30, 1987.
- DJ87b - Jackson, D, and H. Castrup. *Reliability Analysis Methods for Calibration Intervals: Analysis of Type III Censored Data*. *Proceedings, NCSL International Workshop & Symposium*, Denver, CO July 1987.
- DJ03a - Jackson, D. *Calibration Intervals and Measurement Uncertainty Based on Variables Data*. *Proceedings, Measurement Science Conference*, Anaheim, CA, January 2003.
- DJ03b - Jackson, D. *Binary Data Calibration Interval-analysis Using Generalized Linear Models*. *Proceedings, NCSL International Workshop & Symposium*, Tampa, FL, August 2003.

RP-1: Establishment and Adjustment of Calibration Intervals

- DW91 - Wyatt, D. and H. Castrup. *Managing Calibration Intervals*. Proceedings, NCSL International Workshop & Symposium, Albuquerque, NM, August 1991.
- EH60 - Hannan, E. *Time Series Analysis*. Methuen, London, England, 1960.
- EP62 - Parzen, E. *Stochastic Processes*. Holden-Day, Inc., San Francisco, CA, 1962.
- FG54 - Grubbs, F. and H. Coon. "On Setting Test Limits Relative to Specification Limits." *Industrial Quality Control*, pp 15-20, March 1954.
- GIDEP - Government-Industry Data Exchange Program (GIDEP), <http://www.gidep.org>.
- GR82 - Reed, G. Report presented on recall control system. NCSL International Workshop and Symposium, Gaithersburg, MD, 1982.
- HC76 - Castrup, H. "Intermediate System for EMC Instrument Recall Interval Analysis," Interoffice Correspondence 76.2212.4-010. TRW Systems Group, August 6, 1976.
- HC78 - Castrup, H. *Equipment Recall Optimization System (EROS) System Manual*. TRW Defense & Space Systems Group, 1978.
- HC80 - Castrup, H. "Evaluation of Customer and Manufacturer Risk vs. Acceptance Test Instrument In-Tolerance Level." Proceedings, NCSL Workshop & Symposium, Gaithersburg, MD, September 1980.
- HC84 - Castrup, H. *Intercomparison of Standards: General Case*, SAI Comsystems Technical Report, U.S. Navy Contract N00123-83-D-0015, Delivery Order 4M03. March 16, 1984.
- HC88 - Castrup, H. "A Calibration Interval-analysis System Case Study." Proceedings, NCSL Workshop & Symposium, Washington, D.C., August 1988.
- HC89 - Castrup, H. "Calibration Requirements Analysis System." Proceedings, NCSL Workshop & Symposium, Denver, CO, 1989.
- HC91 - Castrup, H. "Analytical Metrology SPC for ATE Implementation." Proceedings, NCSL Workshop & Symposium, Albuquerque, NM, August 1991.
- HC92 - Castrup, H. "Practical Methods for Analysis of Uncertainty Propagation." Proceedings, Annual Instrumentation Symposium, Las Vegas, NM, April 1992.
- HC94 - Castrup, H. and K. Johnson. "Techniques for Optimizing Calibration Intervals." Proceedings, ASNE Test & Calibration Symposium, Arlington, VA, November - December 1994.
- HC95a - Castrup, H. "Uncertainty Analysis for Risk Management." Proceedings, Measurement Science Conference, Anaheim, CA, January 1995.
- HC95b - Castrup, H. "Analyzing Uncertainty for Risk Management." Proceedings, ASQC- 49th Annual Quality Congress, Cincinnati, OH, May 1995.

RP-1: Establishment and Adjustment of Calibration Intervals

- HC95c - Castrup, H. "Uncertainty Analysis and Parameter Tolerancing." Proceedings, NCSL Workshop & Symposium, Dallas, TX, July 1995.
- HC05 - Castrup, H. "Calibration Intervals from Variables Data." Proceedings, NCSL International Workshop & Symposium, Washington, DC, August 2005.
- HC07 - Castrup, H. "Risk Analysis Methods for Complying with Z540.3." Proceedings, NCSL International Workshop & Symposium, St. Paul, MN, August 2007.
- HC08 - Castrup, C. "Applying Measurement Science to Ensure End Item Performance." Proceedings, Measurement Science Conference, Anaheim, CA, March 2008.
- HH61 - Hartley, H. "The Modified Gauss-Newton Method for the Fitting of Non-Linear Regression Functions by Least Squares." *Technometrics* 3, No. 2, (1961): 269.
- HP95 - Agilent Technologies. "The Adjustment Dilemma." <http://metrologyforum.tm.agilent.com/adjustment.shtml>.
- HW54 - Wold, H. *A Study in the Analysis of Stationary Time Series*, 2nd Ed. Upsala, Sweden, 1954.
- HW63 - Wold, H. "Forecasting by the Chain Principle," *Time Series Analysis*, ed. by M. Rosenblatt, pp 475-477, John Wiley & Sons, Inc., New York, 1963.
- IE08 - Integrated Sciences Group. "Free Analytical Metrology Software Tools: Method A3 Interval Testor". <http://www.isgmax.com/freeware.asp>. (formerly called *Interval-analysis System Evaluator*)
- IL07 - *Guidelines for the Determination of Calibration Intervals of Measuring Instruments*, ILAC-G24:2007 / OIML D 10:2007 (E), 2007.
- IM95 - Integrated Sciences Group, *How IntervalMAX Works*, ISG, 1995.
- ISO90 - *General Requirements for the Competence of Calibration and Testing Laboratories*, ISO/IEC Guide 25, 1990.
- ISO95 - *Guide to the Expression of Uncertainty in Measurement*, ISO/TAG 4/WG 3, BIPM, IEC, IFCC, ISO, IUPAC, IUPAP, OIML; 1995.
- ISO03 - *Measurement Management Systems - Requirements for Measurement Processes and Measuring Equipment*, ISO 10012-2003, 2003.
- ISO05 - *General Requirements for the Competence of Calibration and Testing Laboratories*, ANSI/ISO/IEC 17025:2005, 2005.
- IT05 - Integrated Sciences Group. "ISG Method A3 Interval Tester, Description of the Methodology", *isg.max.com*. 2005. <http://www.isgmax.com/Freeware/A3%20Method.pdf>.
- JF84 - Ferling, J. "The Role of Accuracy Ratios in Test and Measurement Processes." Proceedings, Measurement Science Conference, Long Beach, CA, January 1984.

RP-1: Establishment and Adjustment of Calibration Intervals

- JF87 - Ferling, J. "Calibration Intervals for Multi-Function Test Instruments, A Proposed Policy." Proceedings, Measurement Science Conference, Irvine, CA, January 1987.
- JF95 - Ferling, J. "Uncertainty Analysis of Test and Measurement Processes." Proceedings, Measurement Science Conference, Anaheim, CA, January 1995.
- JG70 - Glassman, J. "Intervals by Exception." Proceedings, NCSL International Workshop & Symposium, July 1970.
- JH55 - Hayes, J. "Factors Affecting Measurement Reliability." Technical Memorandum No. 63-106, October 1955. U.S. Naval Ordnance Laboratory, Corona, CA.
- JH81 - Hilliard, J. *Development and Analysis of Calibration Intervals for Precision Measuring and Test Equipment*, Technical Report prepared under NBS Order No. NB81NAAG8825, Request No. 512-021. 1981.
- JL87 - Larsen, J. *A Handy Approach to Examine and Analyze Calibration Decision Risks and Accuracy Ratios*, Analytical Metrology Note (AMN) 87-2. Navy Metrology Engineering Dept., Corona, CA, 1987.
- JM92 - Miche, J. "Bayesian Calibration Specifications and Intervals." Proceedings, NCSL International Workshop & Symposium, Washington, D.C., August 1992.
- KB65 - Brownlee, K. *Statistical Theory and Methodology in Science and Engineering*, 2nd Ed. John Wiley & Sons, New York, 1965.
- KC94 - Chhongvan, K., and J. Larsen. "Analysis of Calibration Renewal Policies." Proceedings, Test & Calibration Symposium, 1994.
- KC95 - Chhongvan, K. "Analysis of Calibration Adjustment Policies for Electronic Test Equipment," M.S. Thesis. Cal State Dominguez Hills, 1995.
- KK84 - Kuskey, K. "New Capabilities for Analyzing METCAL Technical Decisions." Proceedings, Measurement Science Conference, Long Beach, CA, January 1984.
- MB55 - Bartlett, M., *An Introduction to Stochastic Processes*, Cambridge University Press, London, 1955.
- MK07 - Kuster, M. "Balancing Risk to Minimize Testing Costs." Proceedings, Measurement Science Conference, Long Beach, CA, January 2007.
- MK08 - Kuster, M. "Optimizing the Measurement Chain." Proceedings, Measurement Science Conference, Anaheim, CA, March 2008.
- MK09 - Kuster, M., G. Cenker, and H. Castrup. "Calibration Interval Adjustment: The Effectiveness of Algorithmic Methods." Proceedings, NCSL International Workshop & Symposium, San Antonio, TX, July 2009.

RP-1: Establishment and Adjustment of Calibration Intervals

- ML94 - DoDMIDAS, *Department of Defense Metrology Information & Document Automation System*, Measurement Science Directorate, Naval Warfare Assessment Division, Corona, CA.
- MTRL - NAVAIR 17-35-MTL-1 SPAWARS P4734-310-0001, NAVSEA OD 45845, USMC TI-4733-15/13, U.S. Navy Metrology Requirements List (METRL), Measurement Science Directorate, Naval Warfare Assessment Division, Corona, CA.
- MM87 - Morris, M. "A Sequential Experimental Design for Estimating a Scale Parameter from Quantal Life Testing Data." *Technometrics*, 29, (1987): 173-181.
- NA89 - Dept. of the Navy. *ETS Methodology*, Navy Metrology Research & Development Program Technical Report, Metrology Engineering Center, NWS, Seal Beach, March 1989.
- NA94 - *Metrology-Calibration and Measurement Processes Guidelines*, NASA Reference Publication 1342, Jet Propulsion Laboratory, Pasadena, CA, 1994.
- NC90 - NCSL International, *RP-3: Calibration Procedures*, November 1990, last revised October, 2007.
- NC94 - NCSL International, *Glossary of Metrology-Related Terms*, Boulder, CO, August 1994, last revised September 1999.
- NC94a - NCSL International, *RP-12: Determining and Reporting Measurement Uncertainty*, February 1994, last revised April 2013
- ND66 - Draper, N. and H. Smith. *Applied Regression Analysis*, John Wiley & Sons, Inc., New York, NY, 1966.
- NH75 - Hastings, N., and J. Peacock. *Statistical Distributions*. Butterworth & Co (Publishers) Ltd, London, 1975.
- NIST94 - NIST. *Guidelines for Evaluating and Expressing the Uncertainty of NIST Measurement Results*, Technical Note 1297. September 1994.
- NM74 - Mann, N., R. Schafer and N. Singpurwalla. *Methods for Statistical Analysis of Reliability and Life Data*. John Wiley & Sons, New York, 1974.
- PH62 - Hoel, P. *Introduction to Mathematical Statistics*. 3rd Ed. John Wiley & Sons, Inc., New York, 1962.
- RC95 - Cousins, R. "Why Isn't Every Physicist a Bayesian." *American Journal of Physics*, 63, No. 5, May 1995.
- RJ68 - Jennrich, R. and P. Sampson. "Application of Stepwise Regression to Non-Linear Estimation." *Technometrics*, 10, No. 1,(1968): 63.
- RK95 - Kacker, R. "Calibration of Industrial Measuring Instruments." Proceedings, Measurement Science Conference, Anaheim, CA January 1995.

RP-1: Establishment and Adjustment of Calibration Intervals

- SD09 - Dwyer, S. "Test Instrument Reliability Perspectives and Practices: Cost Structure for an Optimal Calibration Recall Plan." Proceedings, NCSL International Workshop & Symposium, San Antonio, 2009.
- SW84 - Weber, S. and A. Hillstrom. *Economic Model of Calibration Improvements for Automatic Test Equipment*, NBS Special Publication 673. April 1984.
- TM01 - Rowe, M. "Here Come the Lawyers." *Test & Measurement World*, Issue 4, (2006).
- TR5 - Measurement Science Directorate. *Technical Requirements for Calibration Interval Establishment for Test and Monitoring Systems (TAMS)*, NAVAIR 17-35TR-5. Dept. of the Navy Metrology and Calibration Program, Naval Warfare Assessment Division, Corona, CA 31 December 1986, latest revision 31 May 1992.
- UG63 - Grenander, U. and M. Rosenblatt. *Statistical Analysis of Stationary Time Series*. John Wiley & Sons, New York, 1963.
- VIM3 - *International Vocabulary of Metrology — Basic and General Concepts and Associated Terms*, ISO/IEC Guide 99-12:2007 (E/F), VIM.
- WM76 - Meeker, W. and W. Nelson. "Weibull Percentile Estimates and Confidence Limits from Singly Censored Data by Maximum Likelihood." IEEE Transactions on Reliability, R-25, No. 1. April 1976.
- WS75 - Scratchley, W. "Kearfott Calibration Scheduling System and Historical File." Kearfott Division, The Singer Co.
- Z540-1 - *Calibration Laboratories and Measuring and Test Equipment General Requirements*, ANSI/NCSL Z540-1-1994, October 1995.
- Z540.3 - *Requirements for the Calibration of Measuring and Test Equipment*, ANSI/NCSL Z540.3-2006. See also *Handbook for the Application of ANSI/NCSL Z540.3-2006*, NCSLI, 2009.

For Additional Reading:

Bishop, Y., S. Feinberg, and P. Holland. *Discrete Multivariate Analysis: Theory and Practice*. MIT Press, Cambridge, 1975.

Appendix A Terminology and Definitions

The definitions given in this section were either developed by the NCSL International Calibration Interval Committee for use in this RP or were taken from other sources. Where possible, they have been structured to be consistent with standard metrology definitions, such as are given in [NC94] and the VIM [VIM3]. These terms and definitions are specific to the context of this RP and therefore may vary from other sources in specificity, generality, or usage.

Accuracy: The closeness of the agreement between the measured or stated value of an attribute and the attribute's true value.

Adjustment Limit: See Guardband Limit.

ADP: Automated Data Processing. Refers to the hardware and software involved in processing data by computer or computing system.

Artifact: A physical entity characterized by measurable features.

Attribute: A quantifiable feature of a device or other artifact

Note 1: — May be characterized by a nominal value bounded by performance specifications

Note 2: — Other documents may use terms such as parameter, measurement quantity, etc.

Attribute Interval: The calibration interval for an individual equipment attribute.

Attributes Data: Data indicating the state (e.g., “in-tolerance” or “out-of-tolerance”) of an attribute.

Average over period (AOP) Reliability: The measurement reliability of an item averaged over its calibration interval.

Beginning of period (BOP) Reliability: The measurement reliability of an item at the beginning of its calibration interval.

Calibration: The set of operations that establish, under certain specified conditions, the relationship between the documented value of a measurement reference and the corresponding value of an attribute. In this Recommended Practice, the relationship is used to ascertain whether the attribute is in-tolerance.

Calibration Interval: The period between successive, scheduled calibrations for a given item of equipment or designated attribute set.

Confidence Limits: Limits that bound a range of values that contains a particular value with a specified probability.

Control Number: A unique identifier assigned by an owning or controlling organization to an individual item of equipment. Once assigned, it cannot be assigned to any other item of equipment of the owning or controlling organization, regardless of the status of the item to which the identifier is originally assigned.

End of period (EOP) Reliability: The measurement reliability of an item at the end of its calibration interval.

Failure Time: The time elapsed since calibration for the occurrence of an out-of-tolerance event.

Guardband: A region of attribute values subtracted from a tolerance limit to reduce false-accept decisions.

Guardband Limit: A limit for observed values of an attribute that indicates whether corrective action (adjustment, repair, etc.) should be performed. Same as adjustment limits.

Instrument Class: A grouping of model manufacturer items characterized by similar accuracy, performance criteria, and application.

In-Tolerance (Observed): (1) A condition in which the observed difference between a measured value and a reference value lies within its documented tolerance limit(s). (2) A state in which all attributes of an item of equipment are in conformance with documented tolerances.

In-Tolerance (True): A condition in which the bias of an attribute lies within its documented tolerance limit(s).

Maximum Likelihood Estimation (MLE): A method of estimating the parameters of a reliability model from calibration history or other life data.

Measurand: The quantity whose value is estimated by measurement.

Measurement Reliability: The probability that a designated set of attributes of an item of equipment is in conformance with performance specifications. (A fundamental assumption of calibration interval-analysis is that measurement reliability is a function of time between calibrations.)

Measurement Reliability Model: A mathematical function and a set of parameters used to model measurement reliability over time.

Measurement Reliability Model Parameter: One of a set of coefficients used to fit a measurement reliability model to observed reliability data at corresponding intervals after calibration.

Measurement Reliability Target: (1) A specified level of measurement reliability commensurate with quality, cost and logistic objectives. (2) The minimum acceptable probability that an MTE item or designated set of MTE items or attributes will be in-tolerance during use.

Measurement Standard: A device employed as a measurement reference.

Measuring and Test Equipment (MTE): Those devices used to measure, gage, test, inspect, or otherwise examine items to determine compliance with specifications. Sometimes designated M&TE, TAMS (Test and Monitoring Systems), TMDE (Test, Measuring and Diagnostic Equipment) or TME (Test and Measuring Equipment).

MLE: See Maximum Likelihood Estimation.

Model Number: A designation for a grouping of equipment characterized by a unique design, set of performance specifications, fabrication, materials, warranty and application and expected to have the same measurement reliability characteristics.

MTE: See Measuring and Test Equipment.

Outlier: Observed values that are deemed unrepresentative of values sampled from a given population.

Out-of-Tolerance (Observed): (1) A condition in which the observed difference between a measured value and a reference value lies beyond the attribute's documented tolerance limit(s). (2) A state in which one or more of an item's attributes are observed to be not in conformance with documented tolerances.

Out-of-Tolerance (True): A condition in which the bias of an attribute lies outside its documented tolerance limit(s).

Out-of-Tolerance Rate (OOTR): (1) The rate at which an attribute transitions from in-tolerance to out-of-tolerance. (2) The negative of the time derivative of the measurement reliability divided by the measurement reliability.

Parameter: In this RP, Parameter is used exclusively to refer to Measurement Reliability Model Parameter. (See Attribute for Equipment or Measurement Parameter.)

Performance Specifications: Specifications that bound the range of values of an attribute considered indicative of acceptable performance.

Reference Attribute: An attribute of a measurement standard whose indicated or stated value is taken to represent a reference during measurement.

Regulated Interval: An interval directly or indirectly constrained by regulation, contractual agreement, or other external or internal policy. The constraint is often a maximum interval, but may also be a minimum interval or a single fixed value. The constraint may also be indirect, such as an imposed reliability target or unit of measurement.

Renew Always: An equipment management policy or practice in which MTE attributes are adjusted or otherwise optimized (where possible) at every calibration.

Renew-if-Failed: An equipment management policy or practice in which MTE attributes are adjusted or otherwise optimized (if possible) only if found out-of-tolerance at calibration.

Renew-as-Needed: An equipment management policy or practice in which MTE attributes are adjusted or otherwise optimized (if necessary) if found outside "safe" adjustment limits.

Reporting Limit: A limit for observed values of an attribute that indicates whether the attribute should be reported as in-tolerance or out-of-tolerance.

Requiring Organization: The company, agency or other organization that requires calibration intervals for MTE or other equipment. Usually the organization that estimates the required intervals.

Resolution: The smallest change in a quantity being measured that causes a perceptible change in the corresponding indication.

Resubmission Time: The time elapsed between successive calibrations.

Serial-Numbered Item: A single, identifiable unit of equipment, usually identified by a unique serial or property number. (See also: Control Number.)

Similar Items: MTE model number families whose function, complexity, accuracy, design and stability are similar. The homogeneity of similar items lies between that of the model number grouping and the instrument class grouping.

Stability: The magnitude of the response of an attribute to a given stress (e.g., activation, shock, time, etc.) divided by the magnitude of its tolerance limit(s). Roughly, the tendency of an attribute to remain within tolerance.

Stratified Calibration: A practice in which MTE attributes or sets thereof are assigned individual calibration intervals. Only those attributes due for calibration at a given service date are calibrated.

Subject Attribute: An attribute whose value is sought by measurement.

Tolerance Limit: A limit for values of an attribute that defines acceptable performance. Values that fall beyond the limit are said to be out-of-tolerance.

Uncertainty: The parameter associated with the result of a measurement that characterizes the dispersion of the values of the measurand.

Uncertainty Growth: The increase in the uncertainty of a measured or reported value of an attribute as a function of time elapsed since measurement.

Uncertainty Growth Process: The underlying mechanism that governs uncertainty growth. Instrumental in determining a reliability model describing uncertainty growth vs. time.

Variables Data: Data indicating the numerical value of a measured attribute.

Appendix B Reactive Methods

B.1 Introduction

In this RP, **reactive methods** are those in which calibration intervals are adjusted in response to data from recent calibrations without any attempt to model or “predict” measurement reliability behavior over time.

Most reactive methods are, in general, less effective than statistical methods in terms of establishing intervals to meet reliability objectives. Additionally, reactive methods usually require long times (up to sixty years) to reach a steady state where the average in-tolerance rate attains a desired level. Despite these shortcomings, reactive methods are intuitively appealing and easy to use. Consequently, they will be around until equally appealing yet more effective methods are found to replace them.

Several reactive methods are currently in use, and others have been proposed in the literature. In this RP, we describe two algorithms that illustrate the essentials of these methods. A third method, differing from the others in its use of statistical criteria, is also described.

B.2 Method A1 - Simple Response Method

Method A1 is one of the simplest algorithms in use, sometimes called “automatic” or “staircase” adjustment [IL07]. In this method [GR82], an interval is increased by an amount a if an item is in-tolerance when received for calibration, and decreased by an amount b if the item is out-of-tolerance. The values of a and b are set to achieve a given measurement reliability target. For example, if a is set equal to 0.1 and b is set equal to 0.55, simulation studies show [DJ86a] that a measurement reliability target of about 90 % is achieved. More generally, b can be chosen to achieve any long-term average reliability R_t by use of the following equation [MK09]

$$b = 1 - (1 + a)^{-R_t / (1 - R_t)}.$$

In the variation described above, the new interval, I_1 , is calculated from the previous interval, I_0 , as follows:

$$I_1 = I_0 \begin{cases} 1 + a & \text{if in tolerance} \\ 1 - b & \text{if OOT} \end{cases}$$

There is a tradeoff in selecting the parameter a . The greater the value selected for a , the faster Method A1 will approach the correct interval from an initial value. The smaller the value selected for a , the closer Method A1 will maintain the interval around the correct interval once it is achieved. Unfortunately with Method A1, one does not know when the correct interval has been reached. Furthermore, Method A1 achieves the long-term average reliability only over an impractically large number of calibrations; even the average reliability achieved for one given instrument will vary considerably from the target.

B.2.1 Method A1 Pros and Cons

Method A1 is characterized by the following pros and cons:

B.2.1.1 Pros

Method A1 is attractive primarily because it is cheap and easy to implement. No specialized knowledge is required and startup costs are minimal.

B.2.1.2 Cons

Method A1 suffers from the following drawbacks:

1. Interval changes are responses to single calibration events. It can be easily shown that any given calibration result is a random event. Adjusting an interval to a single calibration result is, accordingly, equivalent to attempting to control a process by adjusting to random fluctuations. Such practices are inherently futile.
2. Method A1 makes no attempt to model underlying uncertainty growth mechanisms. Consequently, if an interval change is required, the appropriate magnitude of the change cannot be determined.
3. If an interval is attained with Method A1 that is consistent with a desired level of measurement reliability, the results of the next calibration will invariably cause a change away from the correct interval. For example, suppose that an item is assigned an interval that is consistent with a particular organization's reliability target of 90 %, i.e., its interval is "correct." This means that, at the end of the assigned interval, the item has a 90 % chance of being in-tolerance. Method A1 causes an interval extension if the current calibration finds an item to be in-tolerance prior to calibration. But with a 90 % in-tolerance probability, there is a 90 % chance that this will occur. In other words, nine calibrations out of ten will cause an increase in the interval, even though the interval is correct. Thus, Method A1 causes a change away from a correct interval in response to events that are highly probable if the interval is correct.
4. Although a correct interval cannot be maintained, a given time-averaged steady-state measurement reliability can be targeted. However, Method A1 requires considerable time to achieve a steady-state average measurement reliability. The typical time required ranges from fifteen to sixty years [DJ86a].
5. Because Method A1's interval changes are ordinarily computed manually by calibrating technicians, rather than established via automated methods, operating costs can be high.

B.3 Method A2 - Incremental Response Method

Method A2 is a variation of Method A4 of the second edition of this RP. It is a modification of an algorithm proposed by Hilliard [JH81] in 1981. In this method, the magnitude of each interval adjustment is a function of prior adjustments. If the behavior of an item of interest is stable over the adjustment process, then adjustments become successively smaller until a final "correct" interval is reached (if one exists). By making interval changes incrementally smaller with each change, negative consequences associated with adjustments away from a correct interval are ameliorated.

In addition, Method A2 directly accommodates designated EOP reliability targets. There are two variations of Method A2. Variation 1 applies if there are administrative restrictions on interval increases (as is often the case with DoD contracts or in DoD programs), while Variation 2 applies if increases are viewed as neither more nor less attractive than decreases. The algorithms are

$$I_{m+1} = I_m \left[1 + \Delta_{m+1} (y_{m+1} - R) \right], \quad (\text{Variation 1})$$

and

$$I_{m+1} = I_m \left[1 + \Delta_{m+1} (-R)^{1-y_{m+1}} (R)^{y_{m+1}} \right]. \quad (\text{Variation 2})$$

where

m = iteration counter

I_m = interval at the m th calibration

R = reliability target

$y_m = \begin{cases} 1, & \text{if in-tolerance at the } m\text{th calibration} \\ 0, & \text{if out-of-tolerance at the } m\text{th calibration} \end{cases}$

$$\Delta_{m+1} = \frac{\Delta_m}{2^{|y_{m+1}-y_m|}}, \Delta_0 = 1, y_0 = 1$$

The parameter Δ_m is a positive function that shrinks in magnitude in response to an altered condition (rather than just to a succeeding iteration). The factor “2” in the denominator for this function gives the interval adjustment algorithm the flavor of the familiar bisection method widely used in numerical analysis. The initial interval in the iteration is labeled I_0 ; i.e., $m = 0$ at the start of the process.

B.3.1 Example:

Suppose that the calibration history for an item of interest is as follows:

Calibration	Result
1	out-of-tolerance
2	in-tolerance
3	in-tolerance
4	in-tolerance
5	in-tolerance
6	out-of-tolerance
7	in-tolerance
8	in-tolerance

If the initial interval is 45 days, then the initial conditions are

$$I_0 = 45 \text{ days}$$

$$y_0 = 1$$

$$\Delta_0 = 1$$

Suppose we use **Variation 2** with a reliability target $R = 0.9$. Then the interval adjustments for the item will be as follows:

RP-1: Establishment and Adjustment of Calibration Intervals

First interval (out-of-tolerance):

$$y_1 = 0, \Delta_1 = \frac{1}{2^{|0-1|}} = 0.5$$

and

$$\begin{aligned} I_1 &= 45 \left[1 + 0.5(-0.9)^{1-0} (0.9)^0 \right] \\ &= 45(0.55) \\ &= 24.75 \cong 25 \text{ days} \end{aligned}$$

Second interval (in-tolerance)

$$y_2 = 1, \Delta_2 = \frac{0.5}{2^{|1-0|}} = 0.25$$

and

$$\begin{aligned} I_2 &= 25 \left[1 + 0.25(-0.9)^{1-1} (0.9)^1 \right] \\ &= 30.625 \cong 31 \text{ days} \end{aligned}$$

Third interval (in-tolerance)

$$y_3 = 1, \Delta_3 = \frac{0.25}{2^{|1-1|}} = 0.25$$

and

$$\begin{aligned} I_3 &= 31 \left[1 + 0.25(-0.9)^{1-1} (0.9)^1 \right] \\ &= 37.975 \cong 38 \text{ days.} \end{aligned}$$

Fourth interval (in-tolerance)

$$y_4 = 1, \Delta_4 = \frac{0.25}{2^{|1-1|}} = 0.25$$

and

$$\begin{aligned} I_4 &= 38 \left[1 + 0.25(-0.9)^{1-1} (0.9)^1 \right] \\ &= 46.55 \cong 47 \text{ days} \end{aligned}$$

Fifth interval (in-tolerance)

RP-1: Establishment and Adjustment of Calibration Intervals

$$y_5 = 1, \Delta_5 = \frac{0.25}{2^{|1-1|}} = 0.25$$

and

$$\begin{aligned} I_5 &= 47 \left[1 + 0.25(-0.9)^{1-1} (0.9)^1 \right] \\ &= 57.575 \cong 58 \text{ days} \end{aligned}$$

Sixth interval (out-of-tolerance)

$$y_6 = 0, \Delta_6 = \frac{0.25}{2^{|0-1|}} = 0.125$$

and

$$\begin{aligned} I_6 &= 58 \left[1 + 0.125(-0.9)^{1-0} (0.9)^0 \right] \\ &= 51.475 \cong 51 \text{ days} \end{aligned}$$

Seventh interval (in-tolerance)

$$y_7 = 1, \Delta_7 = \frac{0.125}{2^{|1-0|}} = 0.0625$$

and

$$\begin{aligned} I_7 &= 51 [1 + 0.0625(-0.9)^{1-1} (0.9)^1] \\ &= 53.869 \cong 54 \text{ days.} \end{aligned}$$

Eighth interval (in-tolerance)

$$y_8 = 1, \Delta_8 = \frac{0.0625}{2^{|1-1|}} = 0.0625$$

and

$$\begin{aligned} I_8 &= 54 [1 + 0.0625(-0.9)^{1-1} (0.9)^1] \\ &= 57.037 \cong 57 \text{ days.} \end{aligned}$$

B.3.2 Method A2 Pros and Cons

B.3.2.1 Pros

1. Compared to statistical predictive methods (Appendices C, D and E), the implementation of Method A2 is inexpensive and requires no specialized knowledge.
2. Method A2 attempts to adjust intervals to meet specified reliability targets.
3. Method A2 can attain “equilibrium.” If the uncertainty growth character of a given serial-numbered MTE or MTE attribute remains constant over its life span, intervals can eventually be found that are resistant to spurious changes.

B.3.2.2 Cons

1. Interval changes are responses to isolated calibration results. As discussed under Method A1, single data points are inherently insufficient for making interval change decisions.
2. Method A2 makes no attempt to model underlying uncertainty growth mechanisms. Consequently, if an interval change is triggered, the appropriate magnitude of the change cannot be determined.
3. Although Method A2 may eventually settle on an interval, considerable interval fluctuation is experienced in the process. In other words, until interval increments become small, Method A2 is little better than Method A1 in holding to an interval.
4. Although Method A2 attempts to achieve a specified reliability target, simulation studies [MK09] show that the resulting intervals, including the final interval, vary considerably from the correct interval.
5. Method A2 requires considerable time to settle on an interval. The typical time required ranges from ten to sixty years [DJ86a].
6. In the time required to reach a correct interval, the uncertainty growth character of an MTE item or attribute is likely to change. Such changes should reset the incremental interval search process. There is no provision in Method A2 that identifies when this reset should occur. The same problem exists when Method A2 settles on an incorrect interval: it will not respond to any further data regardless of observed reliability.
7. If Method A2's interval changes are computed by calibrating technicians, operating costs can be high.

B.4 Method A3 - Interval Test Method

Method A3 employs accumulated calibration history for a given item to test statistically whether the item's assigned interval is appropriate. The result of a test is whether to adjust or not based on whether or not calibration results are consistent with expectations. For instance, in-tolerance events are expected if an interval is believed to be associated with a high reliability target. In this case, an in-tolerance event is not likely to trigger an interval change.

Because Method A3 bases adjustments on statistically significant results, it does not suffer from many of the drawbacks of Methods A1 and A2.

B.4.1 Interval Change Criteria

In Method A3, if the percentage of calibrations observed in-tolerance at a given interval proves to be significantly different from the desired reliability, an interval change is made. Interval changes may require extrapolation or interpolation. Because Method A3 repeatedly applies an interval test in what is essentially closed-loop feedback control, the choice of interval adjustment methods is flexible. When a change is made, any algorithm that lengthens (shortens) the interval when the observed reliability is higher (lower) than the reliability target is potentially viable because the new interval will be subsequently tested and rejected if incorrect.

B.4.2 Interval Extrapolation

Two commonly used extrapolation methods are mentioned here.

B.4.2.1 Exponential Extrapolation

Though an extrapolated interval may be computed by use of any reliability model believed to apply, one of the simplest and most widely used is the exponential reliability model. In computing the new interval, the observed measurement reliability is first computed for the existing interval I_0 . This reliability, denoted R_0 , is set equal to the number of observed in-tolerance calibrations at the assigned interval divided by the total number of calibrations at that interval:

$$R_0 = \frac{\text{number in-tolerance at } I_0}{\text{number calibrated at } I_0}$$

A revised interval I_1 is computed from this quantity by use of an equation derived from the exponential model's reliability function:

$$I_1 = \frac{\ln R}{\ln R_0} I_0$$

where R is the reliability target. Note that, if R_0 is lower than R , then I_1 is smaller than I_0 (i.e., the interval is shortened). If R_0 is higher than R , then I_1 is larger than I_0 (i.e., the interval is lengthened). However, care should be taken with this method because a small range of observed reliability may produce large interval adjustments and the cases when R_0 is equal to one or zero are undefined. The following two heuristic methods avoid these problems by bounding the revised interval. The first requires $aI_0 \leq I_1 \leq bI_0$, where the user sets the parameters a and b , say 0.5 and 2.0. The second method has bounds dependent upon the reliability target:

$$(1) \quad I_1 = I_0 \begin{cases} 1 + \frac{1}{R} & \text{if } R_0 > R^{\frac{R}{R+1}} \\ 1 - \frac{R}{2} & \text{if } R_0 < R^{\frac{2}{2-R}} \\ \frac{\ln(R)}{\ln(R_0)} & \text{otherwise} \end{cases} \quad (2) \quad I_1 = I_0 \begin{cases} b & \text{if } R_0 > R^{\frac{1}{b}} \\ a & \text{if } R_0 < R^{\frac{1}{a}} \\ \frac{\ln(R)}{\ln(R_0)} & \text{otherwise} \end{cases}$$

B.4.2.2 Confidence-Compensated Extrapolation

Exponential extrapolation can produce extreme interval adjustments (especially without bounding) even when only a small adjustment is warranted. If the statistical test rejects the existing interval, exponential

extrapolation adjusts the interval in full, without regard to how strongly the interval was rejected. Confidence-compensated extrapolation attempts to rectify this problem by varying the interval adjustment according to the confidence with which the statistical test rejected the existing interval.

In this method [IT05] the revised interval is calculated by use of

$$I_1 = I_0 \begin{cases} \text{if } R_0 > R \\ \begin{cases} b \text{ if } Q = 1 \\ \text{otherwise} \\ \begin{cases} b \text{ if } w > b \\ w \text{ otherwise} \end{cases} \end{cases} \\ \text{otherwise} \\ \begin{cases} a \text{ if } v < a \\ v \text{ otherwise} \end{cases} \end{cases}$$

where

$$v = 10^{(R_0 - R)Q}$$

and

$$w = 10^{\frac{R_0 - R}{1 - Q}}, Q \neq 1$$

The rejection confidence Q , is the probability with which the interval was rejected (explained later), and a and b are the same user-chosen bounding parameters as above, typically 0.5 and 2.0. Note that a higher Q produces a larger interval adjustment, whether the adjustment is an increase or a decrease.

B.4.3 Interval Interpolation

Following an interval change, calibration history is accumulated at the new interval. If this history indicates that the interval was overcorrected, the interval is regressed to a point midway between the prior interval and the new interval. Thus, if the interval had been lengthened, and the observed reliability at the new interval is significantly lower than the desired target, the interval is shortened to a value midway between its present value and its prior value. If the interval had been shortened, and the observed reliability at the new interval is significantly higher than the desired target, the interval is lengthened in the same way. (What is meant by significantly lower and significantly higher will be discussed later.)

The regressed interval, denoted I_2 , is computed from the prior interval I_0 and the present interval I_1 from the relation

$$I_2 = \frac{I_0 + I_1}{2}.$$

RP-1: Establishment and Adjustment of Calibration Intervals

If the regressed interval later fails its test, then depending on whether further regression or reversed regression is indicated, a new interval, I_3 , is computed from

$$I_3 = \frac{I_0 + I_2}{2}$$

or

$$I_3 = \frac{I_1 + I_2}{2}.$$

The process continues in this way until an interval is found that is commensurate with the reliability target.

B.4.4 Interval Change Procedure

The results of all adjustment methods are normally rounded to the nearest interval unit (e.g., day). The extrapolation and interpolation processes are implemented as follows:

B.4.4.1 Initial Interval Changes

If, following the assignment of an initial interval the recorded calibration history indicates that an interval should change, the interval is extrapolated. Extrapolation is applied for both interval decreases and increases.

B.4.4.2 Subsequent Interval Changes

When making subsequent interval changes, the same method (interpolation or extrapolation) is used as was previously used if the new interval adjustment is in the same direction as the previous adjustment. If the interval change is reversing direction, then the interpolation method is used for the new adjustment. For example, when lengthening an interval previously shortened by extrapolation, the new change would be done by interpolation. It is always possible that the newly accumulated history may indicate an adjustment in the wrong direction. If interpolation converges by chance to the wrong interval, extrapolation can again be employed.

B.4.5 Significant Differences

Because the occurrence of an in- or out-of-tolerance condition is a random event, it is not advisable to adjust calibration intervals in response to a single in- or out-of-tolerance condition.

Under certain circumstances, it may not even be advisable to adjust intervals in response to the occurrence of two or even three or more successive in- or out-of-tolerance conditions. Given the specific reliability target and the number of observed calibrations, it may be that such combinations of events are expected to occur fairly frequently at the correct interval. Whether to adjust a calibration interval or not depends on whether in- or out-of-tolerance events occur in a way that is highly *unlikely*, i.e., in a way that is not consistent with the assumption that the interval is correct.

Method A3 uses a statistical test to evaluate whether calibration results are consistent with a correct interval. If the test shows that the observed measurement reliability is significantly different from the target reliability, then an interval change is required. That is, if the observed measurement reliability is found to be significantly higher or lower than the reliability target, the interval is lengthened or shortened.

What is meant by “significantly higher” or “significantly lower” is that the observed rate of occurrence of out-of-tolerance events causes a rejection of the notion that the calibration interval is correct. This rejection is made with a predetermined level of statistical significance, hence, the use of the term “significant.”

For example, suppose that all interested parties have agreed to reject a calibration interval if the observed out-of-tolerance behavior had less than a 30 % chance of occurring if the interval were correct. Another way of saying this is that the calibration interval would not be adjusted (up or down) unless the out-of-tolerance rate observed at an interval fell outside statistical 70 % confidence limits.

Table B-1. Example Method A3 Interval Adjustment Criteria

Reliability Target = 80 %				Level of Significance = 0.30	
Number of Calibrations	Number In-Tolerance	Lower 70% Confidence Limit	Upper 70% Confidence Limit	Adjust Interval	Adjustment
1	0	0.0000	0.7000	yes	decrease
	1	0.3000	1.0000	no	
2	0	0.0000	0.4523	yes	decrease
	1	0.0780	0.9220	no	
	2	0.5477	1.0000	no	
3	0	0.0000	0.3306	yes	decrease
	1	0.0527	0.7556	yes	decrease
	2	0.2444	0.9473	no	
	3	0.6694	1.0000	no	
4	0	0.0000	0.2599	yes	decrease
	1	0.0398	0.6265	yes	decrease
	2	0.1794	0.8206	no	
	3	0.3735	0.9602	no	
5	4	0.7401	1.0000	no	
	0	0.0000	0.2140	yes	decrease
	1	0.0320	0.5321	yes	decrease
	2	0.1419	0.7101	yes	decrease
	3	0.2899	0.8581	no	
	4	0.4679	0.9680	no	
	5	0.7860	1.0000	no	

To illustrate the process, suppose that the reliability target is 80 %. If so, then some criteria for accepting or rejecting an interval are shown in Table B-1. (The confidence level of 70 % was picked for this discussion because, for a reliability target of 80 %, this level of significance precludes interval increases after only one calibration.)

In using a decision table such as Table B-1, an adjustment is called for if the reliability target of 0.80 (i.e., 80 %) lies outside the confidence limits. For a 70 % confidence level and an 80 % reliability target and for sample sizes less than or equal to five, no interval increases occur. In fact, for an 80 % reliability target, interval increase decisions do not occur until a sample size of sixteen is reached if one calibration is out-of-tolerance. The pattern is shown in Table B-2.

Table B-2. Example Interval Increase Criteria

Reliability Target = 80 %
Level of Significance = 0.30

Number of Calibrations	Increase Interval if Number In-tolerance $\geq$
12	12
13	13
14	14
15	15
16	15
17	16
18	17
19	18
20	19
30	27
40	36

A different decision profile would apply if the confidence level or the reliability target were different. For example, if the reliability target were equal to 70 %, interval increases would be recommended with 70 % confidence if none out of four or more, or one out of ten or more calibrations were out-of-tolerance.

From the foregoing, it can be appreciated that, with Method A3, a key objective is obtaining good initial interval estimates. With high reliability targets (e.g., 80 % or higher), it takes a considerable number of calibrations to justify a longer interval, on the grounds that it yields an observed reliability significantly higher than the reliability target.

B.4.6 Speeding up the Process

Combining calibration results from a grouping of individual items can reduce the period required to obtain sufficient numbers of calibrations for making interval adjustment decisions. In combining data in this way, it is important to bear in mind that what is being statistically tested is a particular calibration interval for given physical characteristics, usage, operating environment, tolerance limits, calibration uncertainty, etc. This means that applying Method A3 to a group of items is most effective if all items are on the same calibration interval and homogeneous with respect to physical characteristics, usage, operating environment, tolerance limits, calibration uncertainty, etc.

Figure B-1, based on simulation, shows the mean time to reach an interval commensurate with reliability within ± 2 % of the target reliability by use of unbounded exponential extrapolation for significance level and reliability choices ranging from 50 % to 95 % in 5 % steps¹³. As can be seen, lowering the significance level also shortens the time required to reach the correct interval. However, there is a tradeoff between the time required and stability once the correct interval is reached.

¹³ Computed for initial intervals twice the correct interval, assuming exponential reliability behavior. The simulation ignored cases in which interpolation settled at an interval significantly different from the correct interval. Results are fitted to a quadratic surface.

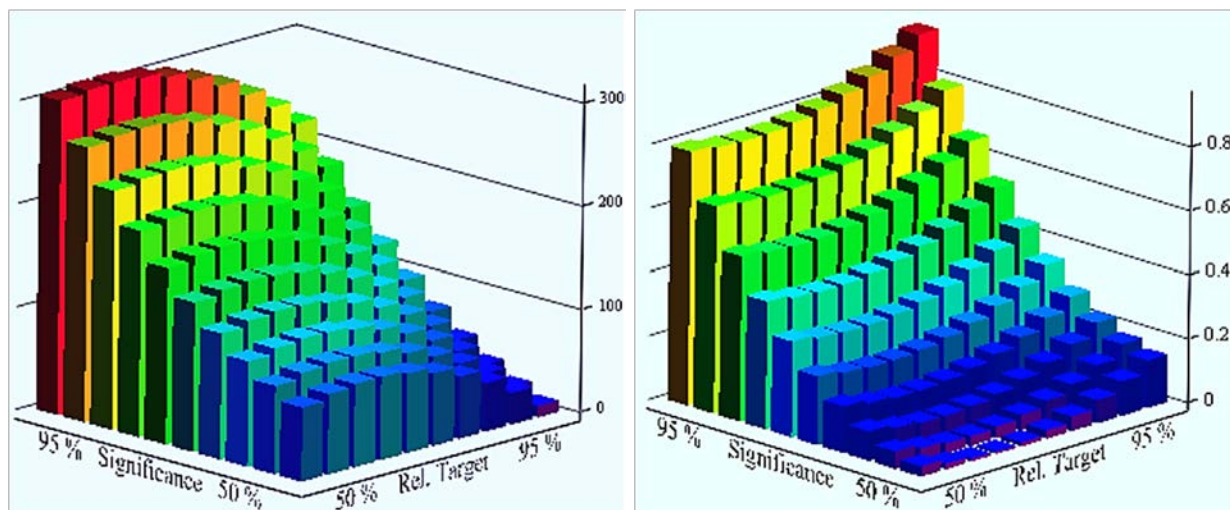


Figure B-1. Time (in Correct Intervals) to Arrive at Correct Interval
and
Figure B-2. Stability at the Correct Interval

B.4.7 Stability

The chosen significance level and reliability target also affect the stability at the correct interval. Figure B-2 depicts the probability that Method A3 will maintain the correct interval (once reached) for the next 50 calibrations of like items for significance level and reliability choices ranging from 50 % to 95 % in 5 % steps based on simulation. As would be expected, selecting a higher significance level provides more stability. Lowering the significance level too far may degrade Method A3's stability to that of the less favorable reactive methods; randomly hitting the correct interval once in a series of intervals is ineffective. Note that higher reliability targets also increase stability at the correct interval, though to a lesser degree.

B.4.8 Determining Significance Limits and Rejection Confidence

Significance limits are limits that are said to contain the underlying or "true" measurement reliability associated with an interval under consideration. The "significance" is the level of confidence or probability that this is so. Accordingly, significance limits are computed as confidence limits around a given observed reliability. If the observed reliability differs from the desired reliability enough that the significance limits do not contain the desired reliability, then it is surmised that the underlying reliability associated with the interval in question, differs significantly from the desired reliability.

Significance limits are obtained as follows. Let

I	=	the assigned interval
R_U	=	upper significance limit for R_0
R_L	=	lower significance limit for R_0
n	=	number of calibrations at I
g	=	number observed in-tolerance at I

RP-1: Establishment and Adjustment of Calibration Intervals

R^*, R_t = the desired reliability
 α = the significance level of the interval test.

Because there are only two possible outcomes in a given calibration, in-tolerance or out-of-tolerance, the observed measurement reliability R_0 is binomially distributed. Consequently, significance limits for this variable are obtained by use of the binomial distribution. The appropriate expressions are [PH62].

$$\sum_{k=0}^g \frac{n!}{k!(n-k)!} R_U^k (1-R_U)^{n-k} = \alpha$$

and

$$\sum_{k=g}^n \frac{n!}{k!(n-k)!} R_L^k (1-R_L)^{n-k} = \alpha .$$

Solving for the limits R_U and R_L obtained from these expressions, we state that the range $[R_L, R_U]$ contains the underlying reliability R with $(1 - 2\alpha) \times 100$ % confidence. If R^* is not within $[R_L, R_U]$, then it is asserted that R^* is significantly different from R , and the interval I is rejected.

The rejection confidence, Q , after allowing for the special cases in which R_0 is equal to one or zero, or very close to R_t , can be calculated by use of the expressions

$$Q = 1 - 2 \sum_{k=0}^g \left[\frac{n!}{k!(n-k)!} R_t^k (1-R_t)^{n-k} \right] \text{ if } R_0 < R_t$$

or

$$Q = 1 - 2 \sum_{k=g}^n \left[\frac{n!}{k!(n-k)!} R_t^k (1-R_t)^{n-k} \right] \text{ if } R_0 > R_t$$

Because computing factorials by brute force will cause numeric overflow when data sets of any size are analyzed, it is helpful to use logarithms for the intermediate values, as in

$$\ln \left[\binom{n}{k} R_t^k (1-R_t)^{n-k} \right] = \ln f(n) - \ln f(k) - \ln f(n-k) + k \ln(R_t) + (n-k) \ln(1-R_t),$$

with the factorial approximation,¹⁴ which also increases computation speed without significantly impacting accuracy,

¹⁴ This is an alternative to Stirling's approximation attributed to Srinivasa Ramanujan. See Raghavan, S. and S. S. Rangachari (eds.) "S. Ramanujan: The lost notebook and other unpublished papers," Springer, 1988, p. 339. Stirling's approximation appears in most engineering math texts, e.g., Kreyszig, E., "Advanced Engineering Mathematics," John Wiley & Sons, 1979, p. 861.

$$\ln f(n) = \begin{cases} \ln(n!) & \text{if } n < 10 \\ n \ln(n) - n + \frac{\ln\left[n\left(1 + \left\lceil \frac{4n(1+2n)}{6} \right\rceil\right)\right]}{6} + \frac{\ln(\pi)}{2} & \text{otherwise} \end{cases}$$

When looping to compute summations, iterating over the distribution from the peak toward the tails and terminating when the probability density falls below some chosen error level will create additional speed gains. A computation environment that provides binomial distribution functions simplifies calculating the rejection confidence. Similarly, access to the inverse beta function or inverse F-distribution function simplifies the upper and lower confidence limit computations.

B.4.9 Considerations for Use

Because Method A3 concerns itself with testing a current assigned interval, all items contributing data to the statistical significance test described above should be at or near the same interval. This could restrict the method in some organizations to application at the individual serial-number level. However, as a preparatory step to collecting history at a single interval in cases where intervals of interest are widely divergent, the average interval may be tested in place of a single assigned interval and the individual item intervals set to either the average interval (if no change is indicated) or an interval adjusted from the average (if a change is indicated). It is prudent in this case to limit an increased interval to 1.2 times the longest resubmission time in the group [IT05].

If applied at the serial-number level, it may take so long to accumulate enough data to justify an interval change that historically older data are not homogeneous with recent data. This would be the case if the stability of an item were to change as the item aged. If so, the older data should be excluded on the grounds that it is no longer relevant to the statistical test. The upshot of this is that, for items whose stability changes over a period of less than ten or twenty calibration intervals, there may *never* be enough representative data to justify an interval change.

Another consideration in the use of Method A3, though not unique to it, is that data taken prior to any change that bears on an item's in-tolerance probability cannot be used to evaluate the current interval. Such a change might be a calibration procedure revision or a modification of tolerance limits. Whatever the variable, the behavior of an item prior to the change may not be relevant to the item's current situation.

For example, suppose that an item's tolerance limits are cut in half. Clearly, with half the original tolerances, it could require substantially less time for the item to drift out-of-tolerance than it did prior to the change. Thus, if the item's prior history consists of a string of in-tolerance observations, these observations cannot be taken to have any relevance to current tendencies for in- or out-of-tolerance. It may be that, with the new limits, a string of *out-of-tolerances* are on the horizon, even if the current interval is maintained. Under these circumstances, if the current interval is lengthened on the strength of past behavior, the likelihood for out-of-tolerances may increase dramatically.

When a process change warrants ignoring historical data, the existing interval should be treated as an initial interval with regard to the interval change procedure.

B.4.10 Criteria for Use

Given the foregoing, Method A3 achieves best results when the following criteria are met:

- Data used to test a given calibration interval consist of calibration results taken at the end of a period of use equal to (or nearly equal to) the interval in question.
- Data used to test a given calibration interval of an item are homogenous with respect to the current stability of the item.
- Data used to test a given calibration interval are homogenous with respect to calibration procedure, tolerance limits, and other variables that impact measurement reliability over time.

B.4.11 Method A3 Pros and Cons

B.4.11.1 Pros

1. Method A3 adjusts intervals to meet specified reliability targets.
2. Method A3 is resistant to spurious interval changes. Intervals are adjusted only if a change is justified on the basis of statistical significance.
3. Compared to statistical predictive methods (Appendices C - E), the design and implementation of Method A3 are inexpensive.
4. Operating costs are low.
5. Method A3 is a convenient and useful backup method for statistical predictive methods when the predictive method requires more data than are available.

B.4.11.2 General Comment:

Method A3 provides most of the advantages of statistical predictive methods at a fraction of the development cost of such methods.

B.4.11.3 Cons

Method A3 suffers from the following drawbacks:

1. Compared to other reactive methods, the design and implementation of Method A3 is relatively expensive.
2. Except for interval extrapolation, Method A3 makes no attempt to model underlying uncertainty growth mechanisms. Consequently, if an interval change is required, the appropriate magnitude of the change may not be accurately determined.
3. If initial intervals are grossly incorrect, Method A3 may require substantial time to arrive at correct intervals.

B.4.11.4 General Comment:

Method A3 requires strict control of calibration intervals and is sensitive to the validity of initial interval estimates.

B.4.11.5 Final Note

Readers should be advised that selecting some other reactive method over Method A3 should not be made on the grounds that the other method is “more responsive.” This is often a deficiency of reactive methods rather than a strength.

Appendix C Method S1 - Classical Method

C.1 Background

In Method S1, an attempt is made to estimate the time at which out-of-tolerances occur. In particular, if an out-of-tolerance is observed at the end of a calibration interval, the time of occurrence of the out-of-tolerance is estimated at the midpoint of the interval. Calibration history is accumulated that consists of observed intervals or “resubmission times,” coupled with recorded in- or out-of-tolerance observations [TR5, JG70].

In assembling data for analysis, note is made of “start times” and “stop times.” A start time marks the point immediately following a renewal (adjustment). A stop time occurs when one of the following happens

- A renewal takes place.
- A final recorded calibration is encountered.
- A break in the continuity of calibration history occurs.

Method S1 employs the simple exponential function to model measurement reliability vs. interval. Method S1 employs both a reliability function and a failure-time probability distribution function (pdf) in constructing the likelihood function. These functions are designated $R(t)$ and $f(t)$, respectively, where t represents a “stop” time.

C.2 Renew-Always Version

If the renew-always policy is in effect, then start times are at the beginning of each observed calibration interval, and stop times are at the end of each interval. The likelihood function is written

$$L = \prod_{i=1}^n \left[f(I_i / 2) \right]^{X_i} \left[R(I_i) \right]^{1-X_i}$$

where n is the total number of observed calibrations (resubmissions), I_i is the i th observed resubmission time and

$$X_i = \begin{cases} 1, & \text{if the } i\text{th calibration record is out-of-tolerance} \\ 0, & \text{otherwise.} \end{cases}$$

For the exponential reliability model, the reliability function and failure time pdf are

$$R(I_i) = e^{-\lambda I_i}$$

and

$$f(I_i / 2) = \lambda e^{-\lambda I_i / 2}$$

The log of the likelihood function is

$$\begin{aligned}\ln L &= \sum_{i=1}^n X_i \ln [f(I_i / 2)] + \sum_{i=1}^n (1 - X_i) \ln [R(I_i)] \\ &= \sum_{i=1}^n X_i \ln \lambda + \frac{1}{2} \lambda \sum_{i=1}^n X_i I_i - \lambda \sum_{i=1}^n I_i\end{aligned}$$

Taking the partial derivative of $\ln L$ with respect to λ yields

$$\frac{\partial}{\partial \lambda} \ln L = \frac{1}{\lambda} \sum_{i=1}^n X_i + \frac{1}{2} \sum_{i=1}^n X_i I_i - \sum_{i=1}^n I_i.$$

Setting this quantity to zero to maximize L with respect to λ gives

$$\lambda = \frac{X}{I - \frac{1}{2} \sum_{i=1}^n X_i I_i}, \quad (\text{C-1})$$

where X is the total number of observed out-of-tolerances given by

$$X = \sum_{i=1}^n X_i, \quad (\text{C-2})$$

and I is the sum of the observed resubmission times

$$I = \sum_{i=1}^n I_i. \quad (\text{C-3})$$

C.3 Renew-As-Needed Version

In the renew-as-needed version, we represent a stop time by the variable t_i . A stop time occurs when an attribute adjustment occurs. An adjustment takes place when an attribute value falls outside predetermined **adjustment limits**. The likelihood function is written

$$L = \prod_{i=1}^N [f(t_i - I_i / 2)]^{X_i} [R(t_i)]^{1-X_i},$$

where N is the observed number of stop times, and I_i is the interval at which the adjustment took place, i.e., the end of the interval preceding the stop time. Performing the same maximization as with the renew-always method yields

$$\lambda = \frac{X}{T - \frac{1}{2} \sum_{i=1}^N X_i I_i} \quad (\text{C-4})$$

where X is given in Eq. (C-2), and T is the sum of the observed stop times given by

$$T = \sum_{i=1}^N t_i. \quad (C-5)$$

Note that Eqs. (C-4) and (C-5) become Eqs. (C-1) and (C-3) if stop times occur at the end of each interval, i.e., if the renew-always practice is in effect, and $t_i = I_i$.

C.3.1 Time Series Formulation

It is sometimes expedient to write Eq. (C-4) in a slightly different form by use of the observed time series approach discussed earlier. In this approach, resubmission times in which attribute adjustments occurred are grouped into sampling windows. The variable X_{ij} represents whether an out-of-tolerance occurred during the j th resubmission time in the i th sampling window, labeled T_i .

$$X_{ij} = \begin{cases} 1, & \text{if an out-of-tolerance occurred within the } j\text{th resubmission time of the } i\text{th sampling window} \\ 0, & \text{otherwise.} \end{cases}$$

If the sampling windows are labeled T_i , the summation in the denominator of Eq. (C-4) can be written

$$\begin{aligned} \sum_{i=1}^N X_i I_i &= \sum_{i=1}^k T_i \sum_{j=1}^{n_i} X_{ij} \\ &= \sum_{i=1}^k x_i T_i \end{aligned} \quad (C-6)$$

where x_i is the number observed out-of-tolerance in the i th sampling window, and k is the number of sampling windows. Substituting Eq. (C-6) in Eq. (C-4) gives

$$\lambda = \frac{X}{T - \frac{1}{2} \sum_{i=1}^k x_i I_i} \quad (C-7)$$

C.4 Renew-If-failed Version

The renew-if-failed version is a specialized form of the renew-as-needed version in which the attribute adjustment limits are synonymous with the tolerance limits. In the renew-if-failed version, a stop time occurs when one of the following happens

- An out-of-tolerance is observed.
- A final recorded calibration is encountered.
- A break in the continuity of calibration history occurs.

The mathematical expressions are the same as for the renew-as-needed version.

C.5 Method S1 Pros and Cons

C.5.1 Pros

1. Method S1 adjusts intervals to meet specified reliability targets.

2. Method S1 is inexpensive to operate.
3. Method S1 complements the statistical identification of dogs and gems (see Chapter 5).

C.5.2 Cons

1. Reliability modeling in Method S1 is restricted to the use of the exponential model. As has been discussed previously, reliance on a single reliability model can lead to significant errors in interval estimation.
2. Method S1 is moderately expensive to design and implement.
3. To be effective, method S1 requires an inventory of moderate to large size.

Appendix D Method S2 - Binomial Method

D.1 Mathematical Description

This appendix provides the mathematical and detailed methodology needed to implement and optimize Method S2 identified in Chapters 2, 4 and 5. In the development of the methodology, it will be worthwhile to review the concepts of measurement reliability and optimal calibration intervals.

D.2 Measurement Reliability

For a given MTE attribute population,¹⁵ the out-of-tolerance probability can be measured in terms of the fraction of observations on the attribute that correspond to out-of-tolerance conditions. It is shown later that the fraction of observations on a given MTE attribute that are classified as out-of-tolerance at calibration is a *maximum likelihood estimate* (MLE) of the out-of-tolerance probability for the attribute. Thus, because out-of-tolerance probability is a measure of test process uncertainty, the percentage of calibrations that yield out-of-tolerance observations is a measure of this uncertainty. This leads to using “percent observed out-of-tolerance” as a variable by which test process uncertainty can be monitored.

The complement of percent observed out-of-tolerance is the percent observed *in-tolerance*. The latter is referred to as **measurement reliability**. Measurement reliability is defined as: *the probability that an attribute of an item of equipment conforms to performance specifications*.

An effective approach to determining and implementing a limit on test process uncertainty involves defining a *minimum measurement reliability target* for MTE attributes. In practice, many organizations have found it expedient to manage measurement reliability at the instrument level rather than the attribute level. In these cases, an item of MTE is considered out-of-tolerance if one or more of its attributes is found out-of-tolerance. Variations on this theme are possible.

D.3 The Out-of-Tolerance Process

Periodic MTE calibration is motivated by the fact that the confidence that MTE are operating in an in-tolerance state diminishes with time since last calibrated. This presupposes that there is some process by which MTE attributes transition from in-tolerance to out-of-tolerance.

Because of the complexity of many instrument types, deterministic descriptions of this process are often difficult or impossible to achieve. This is not to say that the behavior of an individual instrument cannot in principle be described in terms of physical laws with predictions of specific times of occurrence for out-of-tolerance conditions, but rather that such descriptions are typically beyond the scope of equipment management programs. Such descriptions become overwhelmingly impractical when attempted for populations of instruments subject to diverse conditions of handling, environment and application.

¹⁵ A population may be identified at several levels. Those which are pertinent to calibration interval-analysis are (1) all observations taken on serial-numbered items of a given model number or other homogeneous grouping, (2) all observations taken on model numbers within an instrument class, (3) all observations on an MTE parameter of a model number or other homogeneous grouping, and (4) all observations on an MTE parameter of a serial-numbered item.

Variations in these conditions are usually unpredictable. This argues for descriptions of the in-tolerance to out-of-tolerance process for populations of like instruments to be *probabilistic* rather than deterministic in nature. This point is further supported by the notion, commonly accepted, that each individual instrument is characterized by random inherent differences, which arise from the vagaries of fabrication and subsequent repair and maintenance. Moreover, for MTE managed via an equipment pool system, the conditions of handling, environment and application may switch from instrument to instrument in a random way due to the stochastic character of equipment demand and availability in such systems. For these reasons, the failure of an individual MTE attribute to meet a set of performance criteria (i.e., the occurrence of an out-of-tolerance state) is considered a random phenomenon, that is, one that can be described in terms of probabilistic laws.

D.4 The Out-of-Tolerance Time Series

As indicated earlier, a relatively high degree of confidence can be placed on the supposition that attributes are in conformance with performance specifications immediately following calibration. As the equipment experiences random stresses due to use and/or storage, this confidence decreases. Unless subsequent recalibration is performed, the confidence in the in-tolerance status (measurement reliability) of attributes decreases monotonically with time. A random phenomenon that arises through a process that is developing in time in a manner described by probabilistic laws is referred to as a *stochastic process*.

One method of analysis by which stochastic processes of this kind are described is *time series analysis*. A time series is a set of observations arranged chronologically. Suppose that the observations composing the time series are made over an interval T and that the observations have been taken at random times t . Let the observed value of the variable of interest at time t be labeled $\tilde{R}(t)$. The set of observations $\{\tilde{R}(t), t \in T\}$ is then a time series, which is a realization of the stochastic process $\{R(t), t \in T\}$. Time-series analysis is used to infer from the observed time series the probability law of the stochastic process [HW54, MB55, UG63, EH60]. Time-series analysis is applied to the calibration interval-analysis problem by letting $\tilde{R}(t)$ represent the observed measurement reliability corresponding to a calibration interval of duration t .

$\tilde{R}(t)$ is obtained by taking a sample of in- or out-of-tolerance observations recorded after a time interval t has elapsed since the previous calibrations. Representing the number of in-tolerance observations in the sample by $g(t)$ and the size of the sample by $n(t)$, the observed measurement reliability associated with a calibration interval of duration t is given by $\tilde{R}(t) = g(t)/n(t)$. The observed measurement reliability, based on a sample of observations, represents the theoretical or expected measurement reliability $R(t)$ in the sense that

$$R(t) = \lim_{n(t) \rightarrow \infty} \frac{g(t)}{n(t)}, \text{ or}$$

$$R(t) = E[\tilde{R}(t)]$$

where the function $E(x)$ represents the statistical expectation value for the argument x .

D.5 Analyzing the Time Series

Discovering and describing the stochastic process underlying the transition from in tolerance to out of tolerance can be thought of as an experiment in which samples are taken of times between calibrations

RP-1: Establishment and Adjustment of Calibration Intervals

paired with calibration results. To provide visibility of the time series, the samples are arranged chronologically. Data can be either measured values (variables data) or observed in- or out-of-tolerances (attributes data). The former lead to models of the stochastic process that describe MTE attribute value vs. time. The latter lead directly to probability models, which represent attribute measurement reliability.

Traditionally, nearly all calibration recall systems used only attributes data, so the treatment in this RP is applicable primarily to attributes data systems. Variables data systems have since become much more prevalent however. Considering that the main handicap to interval-analysis systems is the time required to collect adequate data to accurately estimate an interval, and the fact that attributes data systems essentially discard most of the information contained in the measurement results by reducing a measurement set to a single binary value (pass / fail), variables data systems and analysis promise to deliver intervals more quickly. Work has been done on this topic specifically for interval-analysis [DJ03a, HC05] and there are existing applications for devices considered to have predictable drift, such as Zener voltage references and frequency standards. The next edition of this RP should contain detailed variables data analysis methodology.

With attributes data systems, the observed time series looks something like Table D-1. Note that the sampled data are grouped in two-week sampling intervals, and that these sampling intervals are not spaced regularly. This reflects the “take it where you can find it” aspect of gathering data in sufficient quantity to infer with reasonable confidence the out-of-tolerance stochastic process. Ordinarily, data are too sparse at the individual MTE serial-number level to permit this inference. Consequently, serial number histories are typically accumulated in homogeneous groupings, usually at the manufacturer/model level. More will be said on this later.

Note that, for many MTE management programs, the conditions “in-tolerance” and “out-of-tolerance” are applied at the instrument level rather than at the attribute level. Although this leads to less accurate calibration interval determinations than can be obtained by tracking at the attribute level, the practice is still workable. The observed time series is constructed the same way, regardless of the level of refinement of data collection. A plot of the observed time series of Table D-1 is shown in Figure D-1.

Table D-1. Typical Out-of-Tolerance Time Series

Weeks Between Calibrations	Number Calibrations Recorded	Number In-Tolerances Observed	Observed Measurement Reliability
t	$n(t)$	$g(t)$	$R(t)$
2-4	4	4	1.0000
5-7	6	5	0.83333
8-10	14	9	0.6429
11-13	13	8	0.6154
19-21	22	12	0.5455
26-28	49	20	0.4082
37-40	18	9	0.5000
48-51	6	2	0.3333

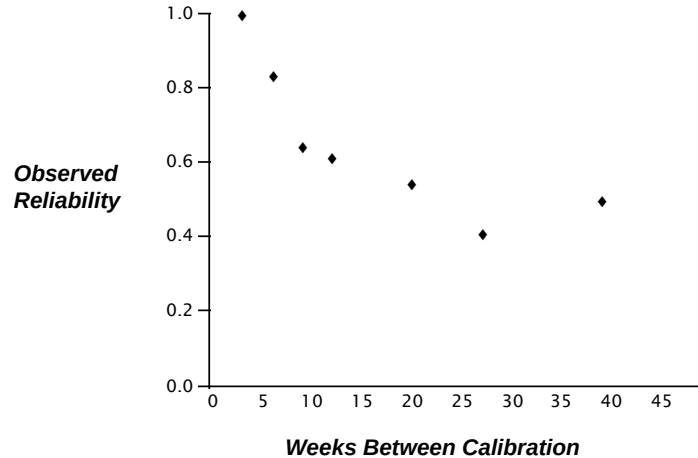


Figure D-1. Hypothetical Observed Time Series. The observed measurement reliabilities for the time series tabulated in Table D-1.

To analyze the time series, a model is assumed for the stochastic process (*Stochastic Processes* (EP62)). The model is a mathematical function characterized by parameters. The functional form is specified while the parameters are estimated on the basis of the observed time series $\{\tilde{R}(t), t \in T\}$. The problem of determining the probability law for the stochastic process thus becomes the problem of selecting the correct functional form for the time series and estimating its parameters.

The method used to estimate the parameters involves choosing a functional form that yields meaningful predictions of measurement reliability as a function of time. By its nature, the function cannot precisely predict the times at which transitions to out-of-tolerance occur. Instead, it predicts measurement reliability expectation values, given the times since calibration. Thus the analysis attempts to determine a predictor $\hat{R}(t, \hat{\theta}) = R(t) + \varepsilon$, where the random variable ε satisfies $E(\varepsilon) = 0$. It can be shown that the method of maximum likelihood estimation provides consistent reliability model parameter estimates for such predictors [HW63].

D.6 Measurement Reliability Modeling

Whether the aim is to ensure measurement integrity assurance for periodically calibrated MTE or to design MTE to tolerate extended periods between calibrations, the uncertainty growth stochastic process is described in terms of mathematical models, characterized by two features: (1) a functional form, and (2) a set of numerical parameters. Figure D-2 models the time series of Table D-1 with an exponential reliability model characterized by the parameters $R_0 = 1$ and $\lambda = 0.03$. Determination as to which mathematical form is appropriate for a given stochastic process and what values are to be assigned the parameters are discussed in the following sections.

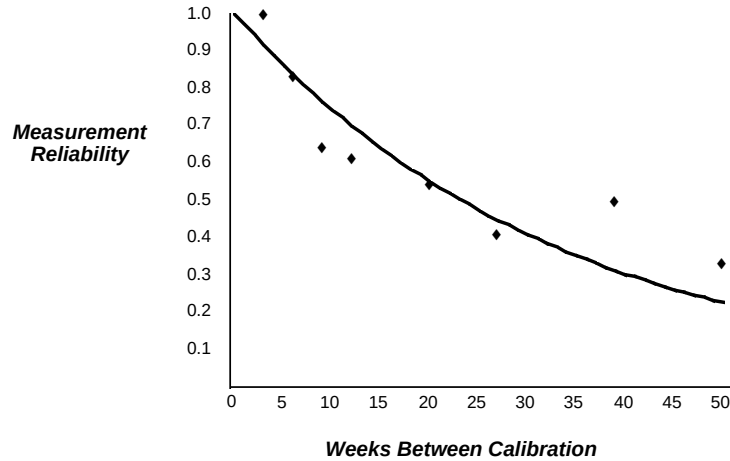


Figure D-2. Out-of-Tolerance Stochastic Process Model. The stochastic process underlying the time series is modeled by an exponential function of the form $R(t) = R_0 e^{-\lambda t}$.

D.6.1 The Likelihood Function

Maximum-likelihood parameter estimation for measurement reliability modeling is somewhat different from parameter estimation employed in “classical” reliability modeling. In the latter, each item in a sample from a population of items is monitored at specified intervals, spaced closely enough together to enable the detection and recording of accurate times to failure. These failure times are inserted into a likelihood function [NM74] incorporating the probability density function of the model of the failure time distribution given by

$$\hat{f}(t, \hat{\theta}) = -\frac{1}{\hat{R}(t, \hat{\theta})} \frac{d\hat{R}(t, \hat{\theta})}{dt}. \quad (\text{D-1})$$

where $\hat{\theta}$ is a vector whose components are the parameters used to characterize the reliability model. To construct the likelihood function, let the observed times to failure be labeled $t_i, i = 1, 2, 3, \dots, m$, and let the times for which sample members were observed to be operational and in-tolerance be labeled $t_j, j = m+1, m+2, m+3, \dots, n$. Then the likelihood function is given by

$$L = \prod_{i=1}^m \hat{f}(t_i, \hat{\theta}) \prod_{j=m+1}^n \hat{R}(t_j, \hat{\theta}) \quad (\text{D-2})$$

By use of Eq. (D-2), the parameters of the model are obtained by differentiating the logarithm of L with respect to each component of $\hat{\theta}$, setting the derivatives equal to zero and solving for the component values [NM74].

In measurement reliability modeling, constructing a likelihood function by use of recorded failure times is not feasible in that “failures” are defined as out-of-tolerance conditions whose precise, actual times of occurrence are undetected and unrecorded. This means that any attempt to model the distribution function for out-of-tolerance times would be far from straightforward. Yet this is precisely the function that

classical reliability modeling methods attempt to fit to observed data. At first sight, then, the fact that the failure times are unknown might be viewed as an insurmountable obstacle.

Fortunately, however, we *can* attempt to fit a model that represents what is known, namely the percent or fraction out-of-tolerance observed at the ends of calibration intervals. The observed in- or out-of-tolerance conditions constitute what are called “Bernoulli trials.” As is well known, the outcomes of such trials are distributed according to the binomial distribution. Then, if we go “back to basics” with regard to maximum-likelihood analysis, we can construct likelihood functions using the Binomial distribution with in- or out-of-tolerance probabilities modeled by reliability functions. By performing maximum likelihood fits of these functions to observed data, we can uncover the time-dependence of the distribution of the Bernoulli trials [HC78, DJ87b, MM87]. In other words, we can discover the functional relationship between in- or out-of-tolerance probability and calibration interval. The procedure is as follows.

D.6.2 Maximum Likelihood Modeling Procedure

First, subdivide the domain of observations on an instrument model or type under study into sampling intervals in such a way that each sampling interval contains some minimum number of observations. Let n be the total number of observations and let k , n_i and b_i denote the number of sampling intervals, the sample size of the i th sample, and the number of failures observed in the i th sample, $i = 1, 2, 3, \dots, k$. Let t_i represent the interval (time) corresponding to the i th sampling interval, and let $P(t_i)$ be the probability that an out-of-tolerance will have occurred by time t_i . The reliability at time t_i is defined as $R(t_i) = 1 - P(t_i)$. Let y_{ij} be the j th observation for the i th sample of size n_i , such that $y_{ij} = 1$ for an observed in-tolerance and $y_{ij} = 0$ for an observed out-of-tolerance. Using the density function for Bernoulli trials, the likelihood function for the i th sample is written

$$L_i = \prod_{j=1}^{n_i} R(t_i)^{y_{ij}} [1 - R(t_i)]^{1-y_{ij}} \quad (D-3)$$

Maximizing this function with respect to $R(t_i)$ yields the maximum-likelihood binomial estimate for the sample in-tolerance probability:

$$\hat{R}_i^o \equiv \frac{1}{n_i} \sum_{j=1}^{n_i} y_{ij} \quad (D-4a)$$

The number observed in-tolerance for the i th sample, denoted g_i , is given by

$$g_i = \sum_{j=1}^{n_i} y_{ij} \quad (D-4b)$$

which yields, after combining with Eq. (D-4a),

$$\hat{R}_i^o = \frac{g_i}{n_i} \quad (D-4c)$$

The estimates $\hat{R}_i$, $i = 1, 2, 3, \dots, k$ are binomially distributed random variables with means $R(t_i)$ and variances $R(t_i)[1 - R(t_i)]/n_i$.

Having identified the distribution of the observed variables, the probability law of the stochastic process $\{\hat{R}(t), t \in T\}$ can be determined by maximizing the likelihood function

$$L = \prod_{i=1}^k \frac{n_i!}{g_i!(n_i - g_i)!} \hat{R}(t_i, \hat{\theta})^{g_i} [1 - \hat{R}(t_i, \hat{\theta})]^{n_i - g_i}. \quad (D-5)$$

with respect to the components of the parameter vector $\hat{\theta}$.

D.6.3 Steepest Descent Solutions

For measurement reliability modeling, the functional forms are usually nonlinear with respect to the parameters that characterize them. Consequently, closed-form solutions for the components of $\hat{\theta}$ are not obtainable in general, and iterative techniques are used. To introduce these techniques, a simplified method is discussed. Practitioners of numerical modeling will recognize the method as a variation of the method of steepest descent.

D.6.3.1 The Normal Equations

If the theoretical reliability model $\hat{R}(t, \hat{\theta})$ is characterized by an m -component parameter vector, then maximizing $\log(L)$ in Eq. (D-5) leads to m simultaneous equations

$$\sum_{i=1}^k \frac{n_i [\hat{R}_i - \hat{R}(t_i, \hat{\theta})]}{\hat{R}(t_i, \hat{\theta}) [1 - \hat{R}(t_i, \hat{\theta})]} \left(\frac{\partial \hat{R}(t_i, \hat{\theta})}{\partial \theta_v} \right) = 0, \quad v = 1, 2, 3, \dots, m, \quad (D-6)$$

which are nonlinear in the parameters. These m simultaneous equations are solved for $\hat{\theta}$ by use of an iterative process.

D.6.3.2 The Iterative Process

As indicated above, iterative methods are used to solve for the vector $\hat{\theta}$. The method of steepest descent begins by “linearizing” the nonlinear model $\hat{R}(t, \hat{\theta})$. This linearization is accomplished by expanding $\hat{R}(t, \hat{\theta})$ in a first-order Taylor series approximation at each iteration:

$$\hat{R}(t_i, \hat{\theta}^{r+1}) = \hat{R}(t_i, \hat{\theta}^r) + \sum_{v=1}^m \left(\frac{\partial \hat{R}(t_i, \hat{\theta})}{\partial \theta_v} \right)_{\hat{\theta}=\hat{\theta}^r} (\theta_v^{r+1} - \theta_v^r) \quad (D-7)$$

where $r+1$ and r refer to the $(r+1)$ th and r th iterations. Substitution of Eq. (D-7) in (D-6) gives

$$\sum_{i=1}^k W_i^r \left[R_i - R(t_i, \theta^r) \right] D_{iv}^r = \sum_{i=1}^k W_i^r \left(\sum_{\mu=1}^m D_{i\mu}^r [\theta_\mu^{r+1} - \theta_\mu^r] \right) D_{iv}^r, \quad v = 1, 2, 3, \dots, m \quad (D-8)$$

where the quantities W_i^r and D_{iv}^r are defined by

$$W_i^r \equiv \frac{n_i}{\hat{R}(t_i, \hat{\theta}^r) [1 - \hat{R}(t_i, \hat{\theta}^r)]} \quad (\text{D-9})$$

and

$$D_{iv}^r \equiv \left(\frac{\partial \hat{R}(t_i, \hat{\theta})}{\partial \theta_v} \right)_{\hat{\theta} = \hat{\theta}^r} \quad (\text{D-10})$$

D.6.3.3 Matrix Notation

Eq. (D-8) can be written in matricial form by defining the vectors $\hat{\mathbf{R}}^r$, $\hat{\mathbf{R}}^r$ and $\hat{\mathbf{b}}^r$, with components $\hat{R}_i^r$, $\hat{R}_i^r = \hat{R}(t_i, \hat{\theta}^r)$, and $b_v^r = \theta_v^{r+1} - \theta_v^r$, respectively, and the matrices $\mathbf{W}$ and $\mathbf{D}$ with elements D_{iv}^r , and $W_{ij}^r = W_i^r \delta_{ij}$:¹⁶

$$(\mathbf{D}^r)^T \mathbf{W}^r (\hat{\mathbf{R}}^r - \hat{\mathbf{R}}^r) = (\mathbf{D}^r)^T \mathbf{W}^r \mathbf{D}^r \mathbf{b}^r \quad (\text{D-11})$$

where the T superscript indicates transposition. Solving Eq. (D-11) for $\mathbf{b}^r$ gives

$$\begin{aligned} \mathbf{b}^r &= \left[(\mathbf{D}^r)^T \mathbf{W}^r (\mathbf{D}^r)^T \right]^{-1} (\mathbf{D}^r)^T \mathbf{W}^r (\hat{\mathbf{R}}^r - \hat{\mathbf{R}}^r) \\ &= \hat{\theta}^{r+1} - \hat{\theta}^r, \end{aligned}$$

and

$$\hat{\theta}^{r+1} = \hat{\theta}^r + \left[(\mathbf{D}^r)^T \mathbf{W}^r (\mathbf{D}^r)^T \right]^{-1} (\mathbf{D}^r)^T \mathbf{W}^r (\hat{\mathbf{R}}^r - \hat{\mathbf{R}}^r) \quad (\text{D-12})$$

The iterations begin ($r = 0$) with initial estimates for the parameter vector components and continue until some desired convergence is reached, i.e., until $\hat{\theta}^{r+1} \cong \hat{\theta}^r$.

If the process converges, the first-order expansion in Eq. (D-7) becomes increasingly appropriate. Problems arise when the process diverges, as will often occur if the initial parameter estimates are substantially dissimilar to the maximum-likelihood values. To alleviate such problems, a modification of the steepest-descent method described above has been developed by Hartley [HH61]. This modification is the subject of the next section.

¹⁶The symbol δ_{ij} is the Kronecker delta symbol defined by $\delta_{ij} = \begin{cases} 1, & \text{if } i = j \\ 0, & \text{otherwise} \end{cases}$.

D.6.3.4 Modified Gauss-Newton Iteration Method

The Hartley method of obtaining consistent maximum-likelihood parameter estimates is a modified Gauss-Newton technique. The approach utilizes Eq. (D-12) but departs from the method described in the previous section by introducing a convergence parameter $\lambda \in [0,1]$:

$$\hat{\theta}^{r+1} = \hat{\theta}^r + \lambda \mathbf{b}^r. \quad (\text{D-13})$$

The modified technique employs the integral of Eq. (6) with respect to θ_v^{r+1} given by

$$\begin{aligned} Q(t, \hat{\theta}^{r+1}) &= \sum_{i=1}^k W_i^r \left[R_i^0 - \hat{R}(t_i, \hat{\theta}) \right]^2 \\ &= \left(\mathbf{R}^0 - \hat{\mathbf{R}}^r \right)^T \mathbf{W} \left(\mathbf{R}^0 - \hat{\mathbf{R}}^r \right) \end{aligned} \quad (\text{D-14})$$

The method assumes a parabolic $Q(t, \hat{\theta}^{r+1})$ in the parameter subspace, which composes the domain corresponding to the local minimum of $Q(t, \hat{\theta}^{r+1})$. Different values of λ are used to search the parameter space in a grid in an attempt to locate a region that contains this local minimum. Hartley uses the values $\lambda = 0, 1/2$ and 1 to get

$$\lambda_{min} = \frac{1}{2} + \frac{1}{4} \frac{Q(0) - Q(1)}{Q(1) - 2Q\left(\frac{1}{2}\right) + Q(0)}$$

where

$$Q(\lambda) = Q(t, \hat{\theta}^r + \lambda \mathbf{b}^r) \quad (\text{D-16})$$

Hartley's method works by using the value λ_{min} for λ in Eq. (D-13). Unfortunately, for multiparameter reliability models, Hartley's method as described in the foregoing does not invariably lead to convergence.

To ensure convergence, a stepwise Gauss-Jordan pivot is employed. With this technique, λ_{min} is sought in a restricted neighborhood of the parameter subspace. The restriction comes from user-defined bounds on the components of the parameter vector. The upshot of the restriction is that pivots that correspond to boundary violations are undone. In this way, if the iteration begins to diverge, the process is partially “reversed” until things are back on track. For a detailed treatment of the technique, the reader is referred to the benchmark article by Jennrich and Sampson, [RJ68].

D.7 Reliability Model Selection

A variety of mathematical reliability models have been identified as useful for modeling uncertainty growth processes. In instances where a process can be inferred from an engineering analysis of MTE design, component stabilities and user applications, determination of the appropriate reliability model is straightforward. In most instances, such analyses are unavailable. In these cases, the appropriate reliability model may be determined by comparing a set of viable “candidate” models against the

observed out-of-tolerance time series and selecting the model that best fits the data. Unfortunately, the reliability model selection procedures found in the literature consist primarily of tests of *applicability* rather than correctness. Moreover, such tests are usually applied to the parameter vector rather than the model itself. These tests are useful only if the model is correct in the first place.

The recommended method is one that attempts to test for correctness of the model. The method is based on the practice of determining whether $R(t, \hat{\theta})$ follows the observed data well enough to be useful as a predictive tool.

It should be noted that the subject of reliability models is an area of current research.

D.7.1 Reliability Model Confidence Testing

The recommended test of $R(t, \hat{\theta})$ is a confidence test constructed by use of statistical machinery developed for treating $N(\mu, \sigma^2)$ random variables. The validity of this approach derives from the approximately similar statistical properties of binomial and normal distributions [NH75].

The test compares the error that arises from the disagreement between $R(t, \hat{\theta})$ and $\hat{R}(t_i)$, $i = 1, 2, 3, \dots, k$, referred to as the “lack of fit” error, and the error due to the inherent scatter of the observed data around the sampled points, referred to as the “pure error” [KB65].

D.7.1.1 Pure Error Sum of Squares

Pure error will be considered first. Returning to the Bernoulli variables defined earlier, the dispersion for the i th sampling interval is given by $\sum (y_{ij} - \hat{R}_i^0)^2$, $i = 1, 2, 3, \dots, k$. The total dispersion of the observed data, referred to as the *pure error sum of squares* (ESS) is accordingly given

$$ESS = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \hat{R}_i^0)^2 \quad (D-17)$$

Because $y_{ij}^2 = y_{ij}$, and $\sum_j y_{ij} = n_i \hat{R}_i^0$, Eq. (D-17) can be written

$$ESS = \sum_{i=1}^k n_i \hat{R}_i^0 (1 - \hat{R}_i^0) \quad (D-18)$$

ESS has $n-k$ degrees of freedom, where $n = \sum n_i$. Thus the *pure error*, denoted by s_E^2 , is estimated by

$$s_E^2 = \frac{1}{n-k} \sum_{i=1}^k n_i \hat{R}_i^0 (1 - \hat{R}_i^0) \quad (D-19)$$

The estimate s_E^2 is a random variable which, when multiplied by its degrees of freedom, behaves approximately like a χ^2 random variable.

D.7.1.2 Residual Sum of Squares

The dispersion of the model is given by the *residual sum of squares*

$$RSS = \sum_{i=1}^k \sum_{j=1}^{n_i} (y_{ij} - \hat{R}_i)^2 \quad (D-20)$$

which can be written as

$$RSS = \sum_{i=1}^k n_i \left[(\hat{R}_i^0 - \hat{R}_i)^2 + \hat{R}_i^0 (1 - \hat{R}_i^0) \right] \quad (D-21)$$

RSS , which has $n-m$ degrees of freedom, contains the dispersion due to lack of fit, together with the pure error.

D.7.1.3 Lack of Fit Sum of Squares

The dispersion due to lack of fit, referred to as the *lack of fit sum of squares (LSS)* is obtained by subtracting ESS from RSS . From Eqs. (D-18) and (D-21), we have

$$LSS = RSS - ESS = \sum_{i=1}^k n_i (\hat{R}_i^0 - \hat{R}_i)^2 \quad (D-22)$$

LSS has $(n - m) - (n - k) = k - m$ degrees of freedom, and the error due to lack of fit is given by

$$s_L^2 = \frac{1}{k - m} \sum_{i=1}^k n_i (\hat{R}_i^0 - \hat{R}_i)^2 \quad (D-23)$$

The variable s_L^2 , when multiplied by its degrees of freedom, follows an approximate χ^2 distribution. This fact, together with the χ^2 nature of $(n - k) s_E^2$ and the fact that s_E^2 and s_L^2 are independently distributed, means that the random variable $F = s_L^2 / s_E^2$ follows an approximate F-distribution with $v_1 = k - m$ and $v_2 = n - k$ degrees of freedom.

If the lack of fit is large relative to the inherent scatter in the data (i.e., if s_L^2 is large relative to s_E^2), then the model is considered inappropriate. Because an increased s_L^2 relative to s_E^2 results in an increased value for F , the variable F provides a measure of the appropriateness of the reliability model. Thus the model can be rejected on the basis of an F-test to determine whether the computed F exceeds some critical value, corresponding to a predetermined rejection confidence level, e.g., 0.95.

D.7.2 Model Selection Criteria

D.7.2.1 (a) Statistical Criterion

Once the rejection confidence levels for the trial failure models are computed, it remains to select the one that best describes the stochastic process $\{R(t), t \in T\}$. At first, it might be reasonable to suppose that the best model in this regard would be the one with the lowest rejection confidence. However, while rejection confidence should certainly be an important factor in the selection process, there are other considerations. One such consideration is the interval recommended by a given model, that is, the interval whose predicted reliability equals the target reliability.

D.7.2.2 (b) Economic Criterion

For example, suppose two models have nearly equal rejection confidences but one yields an interval several times longer than the interval recommended by the other. The question in this instance is: How does one choose between two, apparently equally “good,” reliability models with markedly dissimilar behavior? Unless the MTE whose reliability is being modeled supports a critical product application, economic considerations dictate that the model corresponding to the longest interval should be selected.

While an economic criterion in conjunction with a rejection confidence criterion may be viewed as an improvement over using a rejection criterion alone, there still lingers a suspicion that perhaps some additional criteria be considered. This arises from the fact that, in the above example, for instance, two seemingly appropriate models yield very different reliability predictions. If this is the case, which one is *really* the correct model? For that matter, is *either* one the correct model?

D.7.2.3 (c) “Democratic” Criterion

One way out of the dilemma is to resolve the issue democratically by having each candidate model “vote” for its choice of a recommended interval. In this approach, the intervals recommended by the candidate models are grouped according to similarity. Intervals belonging to the largest group tend to be regarded more favorably than others. This tendency stems from a presumed belief that, given an infinite number of “wrong” solutions, agreement among intervals is not likely to be accidental. This belief has been corroborated in simulation studies (unpublished).

D.7.2.4 Model Figure of Merit

So, there are three criteria for reliability model selection. Using these criteria, a figure of merit G is computed for each trial reliability model:

$$G = \frac{N_G}{C} t_R^{1/4}, \quad (\text{D-24})$$

where C is the rejection confidence for the model, N_G is the size of the group that the model belongs to and t_R is obtained from

$$\hat{R}(t_R, \hat{\theta}) = 1 - R^* \quad (\text{D-25})$$

where R^* is the reliability target.

The figure of merit in Eq. (D-24) is not derived from any established decision theory paradigms. Instead, it has emerged from experimentation with actual cases and is recommended for implementation on the basis that it yields decisions that are in good agreement with decisions made by expert analysts.

D.7.3 Variance in the Reliability Model

In many applications (e.g., dog or gem identification), the variance of $\hat{R}(t, \hat{\theta})$ for any given t is a useful statistic. This variance may be computed in a manner similar to that employed in linear regression analysis by imagining that the parameter vector of the next-to-last iteration is a fixed quantity, independent of the k -tuple of the time series $\{R(t), t \in T\}$, but still very close to the final parameter vector. While this construct may seem arbitrary, it leads to results that are at least qualitatively valid.

Extension of linear regression methods [ND66] to the nonlinear maximum likelihood estimation problem at hand gives the variance-covariance matrix for the model parameter vector $\mathbf{b}$ as

$$\mathbf{V}(\mathbf{b}^r) = \left[(\mathbf{D}^r)^T \mathbf{W}^r \mathbf{D}^r \right]^{-1}. \quad (\text{D-26})$$

Then, defining a vector $\mathbf{d}$ with components

$$d_v(t, \hat{\theta}) = \left(\frac{\partial \hat{R}(t, \theta)}{\partial \theta_v} \right)_{\hat{\theta} = \hat{\theta}^r}, \quad v = 1, 2, 3, \dots, m \quad (\text{D-27})$$

permits the variance in $\hat{R}(t, \hat{\theta})$ for any t to be written

$$\text{Var}[\hat{R}(t, \hat{\theta}^{r+1})] = \mathbf{d}^T(t, \hat{\theta}^r) \left[(\mathbf{D}^r)^T \mathbf{W}^r \mathbf{D}^r \right]^{-1} \mathbf{d}(t, \hat{\theta}^r) \quad (\text{D-28})$$

For a converging process, the parameter vector corresponding to the next-to-last iteration is nearly equal to that of the final iteration, and the two can be used interchangeably with little difficulty. Thus, letting $\hat{\theta}^f$ denote the final parameter vector, Eq. (D-28) can be rewritten as

$$\text{Var}[\hat{R}(t, \hat{\theta}^f)] = \mathbf{d}^T(t, \hat{\theta}^f) \left[(\mathbf{D}^f)^T \mathbf{W}^r \mathbf{D}^r \right]^{-1} \mathbf{d}(t, \hat{\theta}^f) \quad (\text{D-29})$$

D.8 Measurement Reliability Models

Ten reliability models are proposed for modeling out-of-tolerance stochastic processes. Except for the drift model and the longevity model, all have been found useful in practice. The drift model is included because of its intuitive appeal and because it offers some unique benefits. The longevity model has been proposed because of its ability to reach steady-state reliability, which may be applicable in some cases. These will be briefly described following the model listing. Each of the proposed models corresponds to a particular out-of-tolerance mechanism. The mechanisms are as follows:

- 1) Constant out-of-tolerance rate (exponential model).
- 2) Constant-operating-period out-of-tolerance rate with a superimposed burn-in or wear-out period (Weibull model).
- 3) System out-of-tolerances resulting from the failure of one or more components, each characterized by a constant failure rate (mixed exponential model).
- 4) Out-of-tolerances due to random fluctuations in the MTE attribute (random walk model).
- 5) Out-of-tolerances due to random attribute fluctuations confined to a restricted domain around the nominal or design value of the attribute (restricted random-walk model).
- 6) Out-of-tolerances resulting from an accumulation of stresses occurring at a constant average rate (modified¹⁷ gamma model).

¹⁷ The true gamma model is an infinite sum, whereas this modified gamma model truncates to third order.

- 7) Monotonically increasing or decreasing out-of-tolerance rate (mortality drift model).
- 8) Out-of-tolerances occurring after a specific interval (warranty model).
- 9) Systematic attribute drift superimposed over random fluctuations (drift model).
- 10) Out-of-tolerances occurring on a logarithmic time scale (lognormal model).

These processes are modeled by the mathematical functions listed below, illustrated by plots. Derivatives with respect to the parameters are included for purposes of maximum likelihood estimation [see Eqs. (D-10) and (D-27)]. The time scales in the model graphs are arbitrary.

D.8.1 Exponential Model

The exponential model is derived from the “survival” equation in which the number of survivors declines at a constant rate. The model and its derivative with respect to the rate parameter are

$$R(t, \hat{\theta}) = e^{-\hat{\theta}_1 t}$$

$$\frac{\partial R}{\partial \hat{\theta}_1} = -te^{-\hat{\theta}_1 t}$$

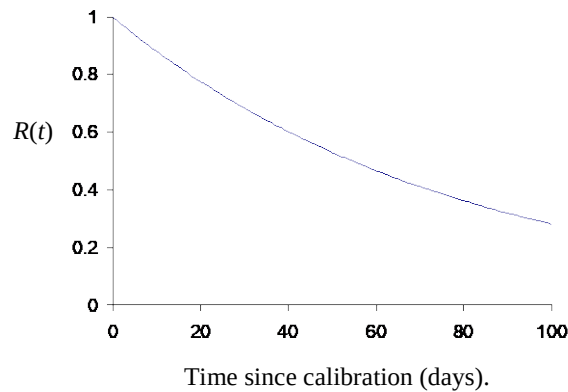


Figure D-3. Exponential Measurement Reliability Model

$$(\theta_1 = 0.01271)$$

D.8.2 Weibull Model

The Weibull model has a form similar to the exponential model except that, instead of a constant failure rate, provision is made for either a “burn-in” or a “wear out” mechanism. Hence, the model accommodates a constant operating period failure rate θ_1 with a superimposed burn-in or wear-out characterized by a shape parameter θ_2 .

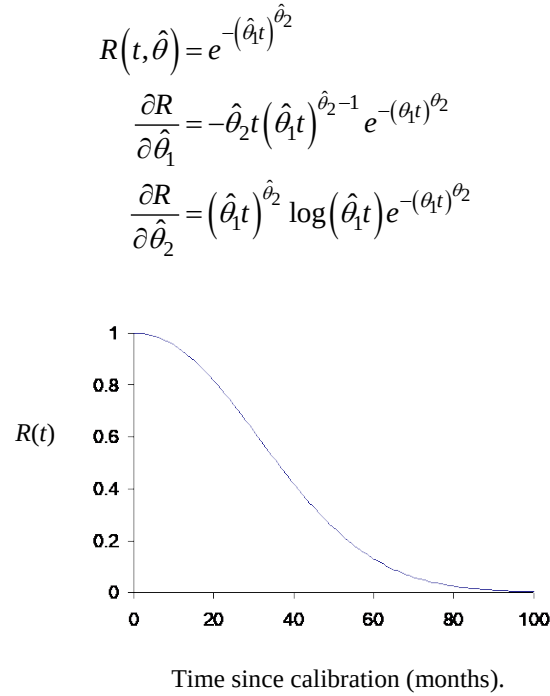


Figure D-4. Weibull Measurement Reliability Model

$$(\theta_1 = 0.02338 \text{ and } \theta_2 = 2.09880)$$

D.8.3 Mixed Exponential Model

The mixed exponential model applies to multiparameter items where each attribute is described by an exponential model and where the attribute failure rates are gamma-distributed. Hence, the i th attribute follows a reliability model of the form

$$R_i(t) = e^{-\lambda_i t}.$$

Assuming a large number of attributes, the distribution of failure rate parameters can be considered to be approximately continuous. Then, for gamma-distributed failure rates, the pdf is

$$f(\lambda) = \frac{(a\lambda)^{(\nu-2)/2} e^{-a\lambda/2}}{2^{\nu/2} \Gamma(\nu/2)},$$

and the item reliability model is then given by

$$\begin{aligned}
 R(t) &= \int_0^{\infty} e^{-\lambda t} f(\lambda) d(a\lambda) \\
 &= \frac{1}{2^{v/2} \Gamma(v/2)} \int_0^{\infty} e^{-\lambda t} (a\lambda)^{(v-2)/2} e^{-a\lambda/2} d(a\lambda) \\
 &= \frac{1}{2^{v/2} \Gamma(v/2)} \int_0^{\infty} e^{-(1/2+t/a)x} x^{(v-2)/2} dx \\
 &= \frac{1}{2^{v/2} \Gamma(v/2)} \frac{\Gamma[(v-2)/2+1]}{(1/2+t/a)^{(v-2)/2+1}} \\
 &= \frac{1}{(1+2t/a)^{v/2}},
 \end{aligned}$$

where a and v are the parameters of the model. Setting $\theta_1 = 2/a$, and $\theta_2 = v/2$, we have

$$\begin{aligned}
 R(t, \hat{\theta}) &= (1 + \hat{\theta}_1 t)^{-\hat{\theta}_2} \\
 \frac{\partial R}{\partial \hat{\theta}_2} &= -\hat{\theta}_2 t (1 + \hat{\theta}_1 t)^{-\hat{\theta}_2 - 1} \\
 \frac{\partial R}{\partial \hat{\theta}_1} &= -\log(1 + \hat{\theta}_1 t) (1 + \hat{\theta}_1 t)^{-\hat{\theta}_2}
 \end{aligned}$$

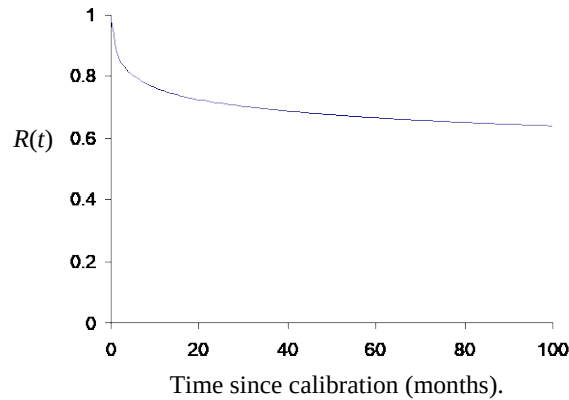


Figure D-5. Mixed Exponential Measurement Reliability Model

$$(\theta_1 = 3.06860 \text{ and } \theta_2 = 0.07779)$$

D.8.4 Random-Walk Model

The random-walk model is derived from the assumptions that (1) attribute biases x change randomly with time t , (2) the probabilities for positive changes and negative changes are equal, and (3) the magnitude of each change is a random variable. These conditions lead to the diffusion equation

$$\frac{\partial f(x,t)}{\partial t} = D \nabla^2 f(x,t)$$

with the solution

$$f(x,t) = (4\pi Dt)^{-1/2} \exp\left(-\frac{x^2}{4Dt}\right).$$

For nonzero variance at $t = 0$, the variance at time t is given by

$$\sigma^2 = 2D(t + \lambda)$$

where λ is a parameter expressed in units of time. The solution then becomes

$$f(x,t) = \frac{1}{\sqrt{2\pi(\sigma_0^2 + \kappa t)}} \exp\left(-\frac{x^2}{2(\sigma_0^2 + \kappa t)}\right),$$

where $\sigma_0^2 = 2D\lambda$, and $\kappa = 2D$. Let $\pm L$ be the tolerance limits for x . Then the probability for an in-tolerance condition at time t is given by

$$\begin{aligned} R(t) &= \int_{-L}^L f(x,t) dx \\ &= \frac{1}{\sqrt{2\pi(\sigma_0^2 + \kappa t)}} \int_{-L}^L \exp\left(-\frac{x^2}{2(\sigma_0^2 + \kappa t)}\right) dx \\ &= 2\Phi\left(\frac{L}{\sqrt{\sigma_0^2 + \kappa t}}\right) - 1, \end{aligned}$$

where σ_0 , L and κ are the parameters of the model. Out-of-tolerances then occur due to random fluctuations in the MTE attribute measurement bias whose standard deviation grows with the square root of the time elapsed since test or calibration. Setting $\theta_1 = (\sigma_0 / L)^2$ and $\theta_2 = \kappa / L$, we have¹⁸

$$R(t) = 2\Phi[Q(t, \hat{\theta})] - 1,$$

where Φ is the normal distribution function and

¹⁸ Because L is a constant and not a parameter of the model, statistical independence between θ_1 and θ_2 is not compromised.

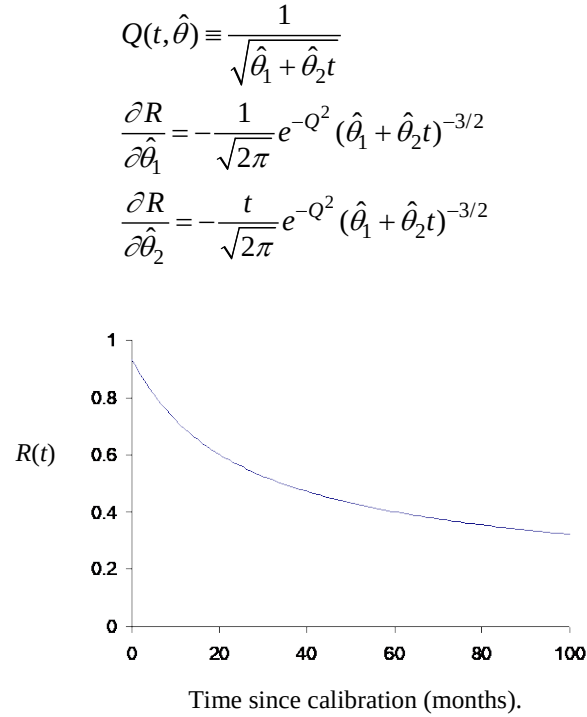


Figure D-6. Random-Walk Measurement Reliability Model

$$(\theta_1 = 0.30144 \text{ and } \theta_2 = 0.05531)$$

D.8.5 Restricted Random-Walk Model

The restricted random-walk model is essentially the random-walk model in which changes in attribute bias are restricted to a neighborhood around a value of zero. This restriction is enforced by adding the condition that the probability of an attribute bias change away from zero is lower than the probability of a change toward zero.

$$R(t, \hat{\theta}) = 2\Phi[Q(t)] - 1$$

where

$$Q(\hat{\theta}) \equiv \frac{1}{\sqrt{\hat{\theta}_1 + \hat{\theta}_2 (1 - e^{-\hat{\theta}_3 t})}}$$

$$\frac{\partial R}{\partial \hat{\theta}_1} = \frac{1}{\sqrt{2\pi}} e^{-Q^2} \left[\hat{\theta}_1 + \hat{\theta}_2 (1 - e^{-\hat{\theta}_3 t}) \right]^{-3/2}$$

$$\frac{\partial R}{\partial \hat{\theta}_2} = -\frac{1}{\sqrt{2\pi}} e^{-Q^2} (1 - e^{-\hat{\theta}_3 t}) \left[\hat{\theta}_1 + \hat{\theta}_2 (1 - e^{-\hat{\theta}_3 t}) \right]^{-3/2}$$

$$\frac{\partial R}{\partial \hat{\theta}_3} = -\frac{1}{\sqrt{2\pi}} e^{-Q^2} \hat{\theta}_2 t e^{-\hat{\theta}_3 t} \left[\hat{\theta}_1 + \hat{\theta}_2 (1 - e^{-\hat{\theta}_3 t}) \right]^{-3/2}$$

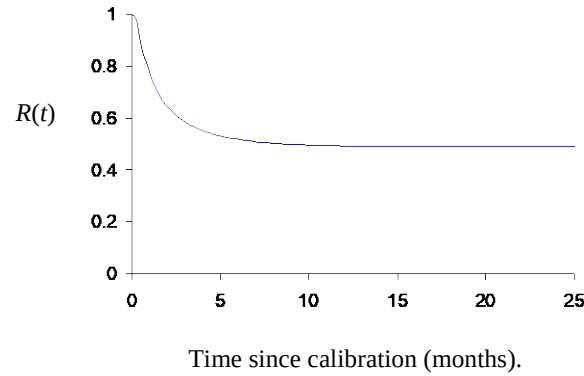


Figure D-7. Restricted Random-Walk Measurement Reliability Model

$$(\theta_1 = 0, \theta_2 = 2.29342, \text{ and } \theta_3 = 0.35761)$$

D.8.6 Modified Gamma Model

Assume that events take place at some mean rate λ . Let t_i be the waiting time for the i th event. Then the probability that the number of events will be less than or equal to some number of events n is given by

$$P[N(t) \leq n] = P[t_{n+1} > t].$$

If the waiting times are gamma-distributed, then the probability $P[N(t) = n]$ is given by

$$P[N(t) = n] = 1 - e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!}.$$

To place this in a reliability modeling context, we take n to be the average number of events that correspond to causing an out-of-tolerance condition. Hence, $P[N(t) = n]$ is the failure probability, with corresponding reliability function

$$R(t) = e^{-\lambda t} \sum_{k=0}^{n-1} \frac{(\lambda t)^k}{k!}.$$

From experience in fitting the model to observed out-of-tolerance time series, it turns out that setting $n = 4$ is applicable to a wide variety of instrumentation with different failure rates. Setting $\theta_1 = \lambda$, we have

$$R(t, \hat{\theta}) = e^{-\hat{\theta}_1 t} \sum_{n=0}^3 \frac{(\hat{\theta}_1 t)^n}{n!}$$

$$\frac{\partial R}{\partial \hat{\theta}_1} = -t e^{-\hat{\theta}_1 t} \frac{(\hat{\theta}_1 t)^3}{3!}$$

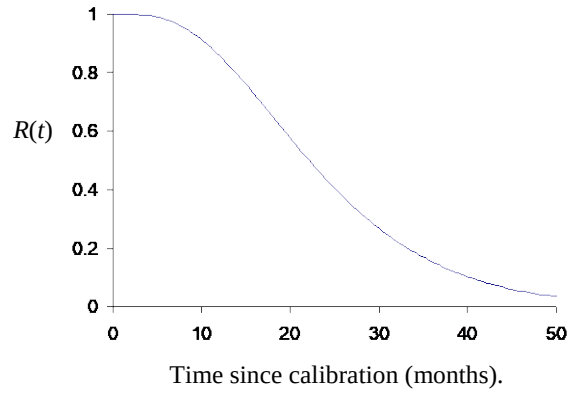


Figure D-8. Modified Gamma Measurement Reliability Model

$$(\theta_1 = 0.16599)$$

D.8.7 Mortality Drift Model

The mortality drift model is essentially the exponential model with the constant failure rate replaced with a failure rate that varies slowly in time. Hence, we replace λ with $\lambda + \beta t$ to get

$$R(t) = e^{-(\lambda t + \beta t^2)},$$

where $|\beta| \ll \lambda$. Setting $\theta_1 = \lambda$ and $\theta_2 = \beta$, we have

$$\begin{aligned} R(t, \hat{\theta}) &= e^{-(\hat{\theta}_1 t + \hat{\theta}_2 t^2)} \\ \frac{\partial R}{\partial \hat{\theta}_1} &= -t e^{-(\hat{\theta}_1 t + \hat{\theta}_2 t^2)} \\ \frac{\partial R}{\partial \hat{\theta}_2} &= -t^2 e^{-(\hat{\theta}_1 t + \hat{\theta}_2 t^2)} \end{aligned}$$

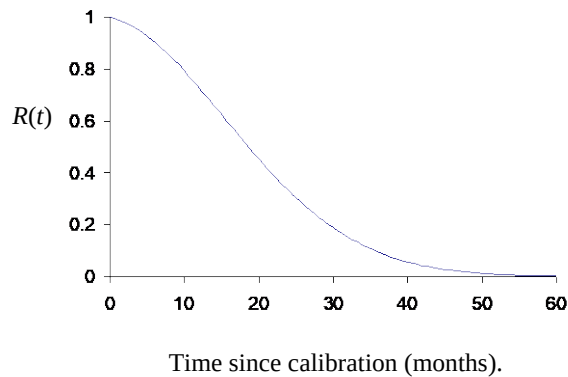


Figure D-9. Mortality Drift Measurement Reliability Model

$$(\theta_1 = 0.00682 \text{ and } \theta_2 = 0.00163)$$

D.8.8 Warranty Model

The warranty model is suitable for cases where the measurement reliability is nearly one until some “cut-off” time is reached, after which the measurement reliability drops to zero. The mathematical form of the model is taken from the distribution function of Fermi-Dirac statistics,

$$f(\varepsilon) = \frac{1}{1 + e^{(\varepsilon - \mu)/kT}},$$

where ε is the energy of electrons in an electron gas, k is Boltzmann’s constant and T is the absolute temperature of the gas. The parameter μ is the energy at which the probability that the energy ε is equal to one-half. Using the form of the Fermi-Dirac distribution function, we write the measurement reliability as

$$R(t) = \frac{1}{1 + e^{\lambda(t - \beta)}}.$$

Setting $\theta_1 = \lambda$ and $\theta_2 = \beta$, we have

$$\begin{aligned} R(t, \hat{\theta}) &= \frac{1}{1 + e^{\hat{\theta}_1(t - \hat{\theta}_2)}} \\ \frac{\partial R}{\partial \hat{\theta}_1} &= -(t - \hat{\theta}_2) e^{\hat{\theta}_1(t - \hat{\theta}_2)} \left[1 + e^{\hat{\theta}_1(t - \hat{\theta}_2)} \right]^{-2} \\ \frac{\partial R}{\partial \hat{\theta}_2} &= \hat{\theta}_1 e^{\hat{\theta}_1(t - \hat{\theta}_2)} \left[1 + e^{\hat{\theta}_1(t - \hat{\theta}_2)} \right]^{-2} \end{aligned}$$

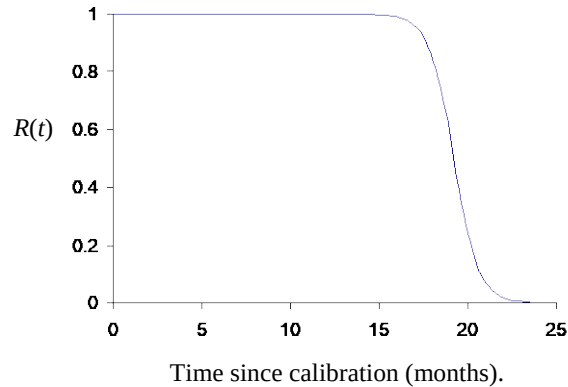


Figure D-10. Warranty Measurement Reliability Model

$$(\theta_1 = 1.43869 \text{ and } \theta_2 = 19.20568)$$

D.8.9 Drift Model

$$R(t, \hat{\theta}) = \Phi(\hat{\theta}_1 + \hat{\theta}_3 t) + \Phi(\hat{\theta}_2 - \hat{\theta}_3 t)$$

$$\frac{\partial R}{\partial \hat{\theta}_1} = \frac{1}{\sqrt{2\pi}} e^{-(\hat{\theta}_1 + \hat{\theta}_3 t)^2 / 2}$$

$$\frac{\partial R}{\partial \hat{\theta}_2} = \frac{1}{\sqrt{2\pi}} e^{-(\hat{\theta}_2 - \hat{\theta}_3 t)^2 / 2}$$

$$\frac{\partial R}{\partial \hat{\theta}_3} = \frac{t}{\sqrt{2\pi}} \left[e^{-(\hat{\theta}_1 + \hat{\theta}_3 t)^2 / 2} - e^{-(\hat{\theta}_2 - \hat{\theta}_3 t)^2 / 2} \right]$$

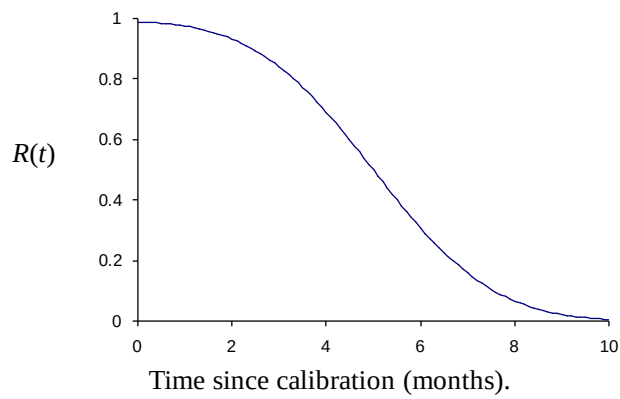


Figure D-11. Drift Measurement Reliability Model

$$(\theta_1 = 2.5, \theta_2 = 0.5, \text{ and } \theta_3 = 0.5)$$

D.8.10 Lognormal Model

The lognormal model is given by

$$R(t, \hat{\theta}) = 1 - \Phi\left[\frac{\ln(\hat{\theta}_1 t)}{\hat{\theta}_2}\right]$$

$$\frac{\partial R}{\partial \hat{\theta}_1} = -\frac{1}{\sqrt{2\pi} \hat{\theta}_1 \hat{\theta}_2} e^{-[\ln(\hat{\theta}_1 t)]^2 / 2 \hat{\theta}_2^2}$$

$$\frac{\partial R}{\partial \hat{\theta}_2} = \frac{\ln(\hat{\theta}_1 t)}{\sqrt{2\pi} \hat{\theta}_2^2} e^{-[\ln(\hat{\theta}_1 t)]^2 / 2 \hat{\theta}_2^2}$$

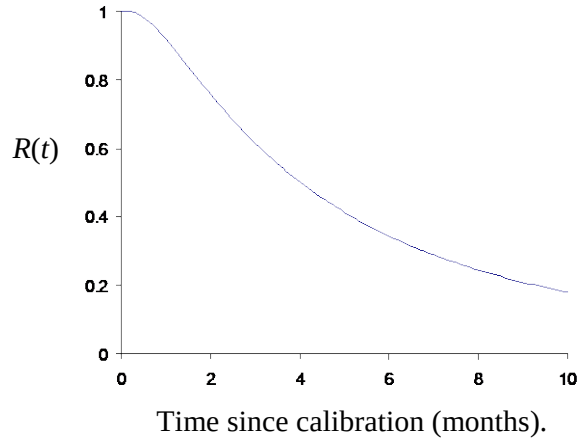


Figure D-12. Lognormal Measurement Reliability Model

$$(\theta_1 = 0.25 \text{ and } \theta_2 = 1.0)$$

In the expressions for both the drift model and the lognormal model, we employ the usual notation

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-\zeta^2/2} d\zeta$$

D.8.11 Renewal Policy and the Drift Model

In the drift model, if the conditions $|\hat{\theta}_3 t \ll \hat{\theta}_1|$ and $|\hat{\theta}_3 t \ll \hat{\theta}_2|$ hold, then the measurement reliability of the attribute of interest is not sensitive to time elapsed since calibration. This is equivalent to saying that if the parameter $\hat{\theta}_3$ is small enough, the attribute can essentially be left alone, i.e., not periodically adjusted.

Interestingly, the parameter $\hat{\theta}_3$ is the rate of attribute value drift divided by the attribute value standard deviation:

$$\hat{\theta}_3 = \frac{m}{\sigma},$$

where m = attribute drift rate, and σ = attribute standard deviation. From this expression, we see that the parameter $\hat{\theta}_3$ is the ratio of the systematic and random components of the mechanism by which attribute values vary with time. If the systematic component dominates, then $\hat{\theta}_3$ will be large. If, on the other hand, the random component dominates, then $\hat{\theta}_3$ will be small. Putting this observation together with the foregoing remarks concerning attribute adjustment leads to the following axiom:

NOTE 1: — If random fluctuation is the dominating mechanism for attribute value changes over time, then the benefit of periodic adjustment is minimal.

As a corollary, it might also be stated that

NOTE 2: — If drift or other systematic change is the dominating mechanism for attribute value changes over time, then the benefit of periodic adjustment is high.

Obviously, use of the drift model can assist in determining which adjustment practice to employ for a given attribute. By fitting the drift model to an observed out-of-tolerance time series and evaluating the parameter $\hat{\theta}_3$, it can be determined whether the dominant mechanism for attribute value change is systematic or random. If $\hat{\theta}_3$ is small, then random changes dominate, and a renew-only- if-failed practice should be considered. If $\hat{\theta}_3$ is large, then a renew-always practice should perhaps be implemented.

D.9 Calibration Interval Determination

D.9.1 Interval Computation

Once the failure model is selected, the computation of the calibration interval T , corresponding to the prescribed EOP reliability target R , is obtained from¹⁹

$$\hat{R}(T, \hat{\theta}) = R \quad (D-30)$$

The recommended method for obtaining T involves a two-step process. First, attempt to solve for T by use of the Newton-Raphson method. If this fails to converge, then obtain T by trial and error, in which t is incremented until a value is found for which $\hat{R}(t, \hat{\theta}) > R$.

D.9.2 Interval Confidence Limits

Upper and lower confidence limits for T are computed to indicate the bounds beyond which the assigned interval becomes questionable. While explicit methods exist for computing these limits for certain specified reliability models (for example, the exponential and Weibull models [WM76], no general method is available for computing these limits for arbitrary models applied to the analysis of censored data [NM74]. Because calibration history data are in this category, an alternative approach is required.

Rather than attempt to formulate a general method directly applicable to interval confidence limit determination, an indirect approach will be followed involving the determination of confidence limits for the reliability function $\hat{R}(t, \hat{\theta})$. This enables the determination of upper and lower bounds for T that are related to interval confidence limits (indeed, for single-parameter reliability functions, these bounds are synonymous with interval confidence limits).

Upper and lower bounds for T , denoted τ_u and τ_l , respectively, are computed for $1-\alpha$ confidence from the relations

$$\hat{R}(\tau_u, \hat{\theta}) + z_{\alpha} \sqrt{\text{var}[\hat{R}(\tau_u, \hat{\theta})]} = R \quad (D-31)$$

¹⁹ See Appendix I for a discussion of the conditions under which Eq. (D-30) is applicable.

and

$$\hat{R}(\tau_l, \hat{\theta}) - z_\alpha \sqrt{\text{var}[\hat{R}(\tau_l, \hat{\theta})]} = R \quad (\text{D-32})$$

where $\text{var}[\hat{R}(t, \hat{\theta})]$ is given by Eq. (D-29), and z_α is obtained from

$$1 - \alpha = \frac{1}{\sqrt{2\pi}} \int_{-z_\alpha}^{z_\alpha} e^{-\zeta^2/2} d\zeta \quad (\text{D-33})$$

Eqs. (D-31) and (D-32) give only approximate upper and lower limits for T in that they are obtained by treating $\hat{R}(t, \hat{\theta})$ as a normally distributed random variable, whereas it in fact follows a binomial distribution. The results are satisfactory, however, because the minimum acceptable sample sizes needed to infer the stochastic process are large enough to justify the use of the normal approximation to the binomial.

D.10 Method S2 Pros and Cons

D.10.1 Pros

1. Method S2 adjusts intervals to meet specified reliability targets.
2. Once installed on a computer, Method S2 is inexpensive to operate.
3. Reliability modeling in Method S2 involves the use of a variety of models. This ensures a considerable improvement in interval accuracy relative to methods S1 and A1 - A3.
4. Method S2 is compatible with the statistical identification of dogs, gems and other outliers.

D.10.2 Cons

1. Method S2 is expensive to design and implement. However, due to the accuracy of intervals obtained from its operation, design and development costs may be recovered quickly.
2. Method S2 requires a large size inventory to be cost-effective.

Appendix E Method S3 - Renewal Time Method

E.1 Developing the Likelihood Function

Method S3 is a refinement to Method S2 that has been developed to accommodate the variety of renewal policies encountered in practice. The method is developed in this Appendix from a hypothetical calibration history for an MTE attribute or item of equipment. The series is as follows:

Renewal			1		2			3	4	5	6		7	8	9	
As-found	I	I	A	I	F	I	I	A	A	F	A	I	F	F	A	I
Calibration	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16

In the above table, the as-found events are

- I - in-tolerance, no adjustment made
- A - in-tolerance, adjustment made
- F - out-of-tolerance, adjustment made

Let t_i be the time elapsed to the i th calibration. From the table, the “renewal times” are seen to be

$$\begin{aligned}
 \tau_1 &= t_3 - t_0 \\
 \tau_2 &= t_5 - t_3 \\
 \tau_3 &= t_8 - t_5 \\
 \tau_4 &= t_9 - t_8 \\
 \tau_5 &= t_{10} - t_9 \\
 \tau_6 &= t_{11} - t_{10} \\
 \tau_7 &= t_{13} - t_{11} \\
 \tau_8 &= t_{14} - t_{13} \\
 \tau_9 &= t_{15} - t_{14} \\
 \tau_{10} &= t_{16} - t_{15}
 \end{aligned}$$

Note that the zero time t_0 is included for formal reasons, and that a pseudo-“renewal” is forced at time t_{16} .

The likelihood function is given by

$$\begin{aligned}
 L &= R(t_1 - t_0) R(t_2 - t_0 | t_1 - t_0) R(t_3 - t_0 | t_2 - t_0) R(t_4 - t_3) [1 - R(t_5 - t_3 | t_4 - t_3)] \\
 &\quad \times R(t_6 - t_5) R(t_7 - t_5 | t_6 - t_5) R(t_8 - t_5 | t_7 - t_5) R(t_9 - t_8) \\
 &\quad \times [1 - R(t_{10} - t_9)] R(t_{11} - t_{10}) R(t_{12} - t_{11}) \\
 &\quad \times [1 - R(t_{13} - t_{11} | t_{12} - t_{11})] [1 - R(t_{14} - t_{13})] R(t_{15} - t_{14}) R(t_{16} - t_{15})
 \end{aligned}$$

In this expression the function $R(t_a - t_c | t_b - t_c)$ refers to the probability that the item is in-tolerance after a time $t_a - t_c$, given that it was in-tolerance after an interval of time $t_b - t_c$.

Because, if $t_a > t_b$,

$$R(t_a - t_c | t_b - t_c) = \frac{R(t_a - t_c, t_b - t_c)}{R(t_b - t_c)} = \frac{R(t_a - t_c)}{R(t_b - t_c)}, t_a > t_b$$

we can write

$$\begin{aligned} L = & R(t_3 - t_0) [R(t_4 - t_3) - R(t_5 - t_3)] R(t_8 - t_5) R(t_9 - t_8) \\ & \times [1 - R(t_{10} - t_9)] R(t_{11} - t_{10}) [R(t_{12} - t_{11}) - R(t_{13} - t_{11})] \\ & \times [R(t_{13} - t_{13}) - R(t_{14} - t_{13})] R(t_{15} - t_{14}) R(t_{16} - t_{15}) \end{aligned}$$

From the renewal times shown earlier, we can restate the above expression as

$$\begin{aligned} L = & R(\tau_1) [R(\tau_2 - I_2) - R(\tau_2)] R(\tau_3) R(\tau_4) \\ & \times [R(\tau_5 - I_5) - R(\tau_5)] R(\tau_6) [R(\tau_7 - I_7) - R(\tau_7)] \\ & \times [R(\tau_8 - I_8) - R(\tau_8)] R(\tau_9) R(\tau_{10}) \end{aligned}$$

where I_j is the calibration interval immediately preceding the j th renewal. Note that

$$\tau_5 - I_5 = t_{10} - t_9 - (t_{10} - t_9) = 0.$$

In keeping with the assumptions of other MLE methods, we assume that $R(0) = 1$. Hence,

$$R(\tau_5 - I_5) = R(0) = 1.$$

We now define the functions

$$r(\tau_j) \equiv R(\tau_j - I_j) \tag{E-1}$$

and rewrite the likelihood function as

$$\begin{aligned} L = & R(\tau_1) [r(\tau_2) - R(\tau_2)] R(\tau_3) R(\tau_4) \\ & \times [r(\tau_5) - R(\tau_5)] R(\tau_6) [r(\tau_7) - R(\tau_7)] \\ & \times [r(\tau_8) - R(\tau_8)] R(\tau_9) R(\tau_{10}). \end{aligned}$$

E.2 Generalizing the Likelihood Function

To extend the above to a computer algorithm, it will be helpful to define the function

$$x_j = \begin{cases} 1, & \text{if the } j\text{th renewal is for an in-tolerance item} \\ 0, & \text{otherwise.} \end{cases} \quad (\text{E-2})$$

With this, we can write the likelihood function as

$$L = \prod_{j=1}^{10} R(\tau_j)^{x_j} [r(\tau_j) - R(\tau_j)]^{1-x_j}. \quad (\text{E-3})$$

We now define the functions

$$\rho_j = R(\tau_j) / r(\tau_j) \quad (\text{E-4})$$

and rewrite the likelihood function as

$$\begin{aligned} L &= \prod_{j=1}^{10} r_j^{x_j} \rho_j^{x_j} r_j^{1-x_j} (1-\rho_j)^{1-x_j} \\ &= \prod_{j=1}^{10} r_j \rho_j^{x_j} (1-\rho_j)^{1-x_j} \end{aligned} \quad (\text{E-5})$$

where

$$r_j = r(\tau_j).$$

E.3 The Total Likelihood Function

In the above, we considered a single item. If there is a population or inventory of items to be dealt with, we need to add an additional subscript. For the i th item in an inventory of N items, Eq. (E-5) becomes

$$L_i = \prod_{j=1}^{n_i} r_{ij} \rho_{ij}^{x_{ij}} (1-\rho_{ij})^{1-x_{ij}} \quad (\text{E-6})$$

where n_i is the number of calibrations for the i th item. The total likelihood function is obtained as the product of the likelihood functions for each item

$$L = \prod_{i=1}^N \prod_{j=1}^{n_i} r_{ij} \rho_{ij}^{x_{ij}} (1-\rho_{ij})^{1-x_{ij}}. \quad (\text{E-7})$$

Taking the log of this function gives

$$\ln L = \sum_{i=1}^N \sum_{j=1}^{n_i} \left\{ x_{ij} \ln \rho_{ij} + (1-x_{ij}) \ln (1-\rho_{ij}) + \ln r_{ij} \right\}. \quad (\text{E-8})$$

The functions r_{ij} and ρ_{ij} are functions of the renewal times τ_{ij} and the calibration intervals I_{ij} . These functions are characterized by parameters that determine the functional relationships. The parameters are solved for by maximizing the likelihood function. We do this by setting the partial derivative of $\ln L$ equal to zero for each parameter. Letting $\hat{\theta}$ represent the parameter vector, we have

$$\begin{aligned} \frac{\partial}{\partial \hat{\theta}_v} \ln L &= \sum_{i=1}^N \sum_{j=1}^{n_i} \left\{ \frac{x_{ij}}{\rho_{ij}} \left(\frac{\partial \rho_{ij}}{\partial \hat{\theta}_v} \right) - \frac{(1-x_{ij})}{1-\rho_{ij}} \left(\frac{\partial \rho_{ij}}{\partial \hat{\theta}_v} \right) + \frac{1}{r_{ij}} \left(\frac{\partial r_{ij}}{\partial \hat{\theta}_v} \right) \right\} \\ &= \sum_{i=1}^N \sum_{j=1}^{n_i} \left\{ \frac{x_{ij}(1-\rho_{ij}) - (1-x_{ij})\rho_{ij}}{\rho_{ij}(1-\rho_{ij})} \left(\frac{\partial \rho_{ij}}{\partial \hat{\theta}_v} \right) + \frac{1}{r_{ij}} \left(\frac{\partial r_{ij}}{\partial \hat{\theta}_v} \right) \right\} \\ &= \sum_{i=1}^N \sum_{j=1}^{n_i} \frac{x_{ij} - \rho_{ij}}{\rho_{ij}(1-\rho_{ij})} \left(\frac{\partial \rho_{ij}}{\partial \hat{\theta}_v} \right) + \sum_{i=1}^N \sum_{j=1}^{n_i} \frac{1}{r_{ij}} \left(\frac{\partial r_{ij}}{\partial \hat{\theta}_v} \right) \\ &= 0, \quad v=1, 2, L, m. \end{aligned} \quad (E-9)$$

E.4 Grouping by Renewal Time

We now submerge the identity of the N items and visualize the functions ρ_{ij} and r_{ij} as the j th renewal function and j th failure indicator, respectively, in the i th renewal time sample. Then the maximizing equations can be written as

$$\sum_{i=1}^k \sum_{j=1}^{n_i} \frac{x_{ij} - \rho_{ij}}{\rho_{ij}(1-\rho_{ij})} \left(\frac{\partial \rho_{ij}}{\partial \theta_v} \right) + \sum_{i=1}^k \sum_{j=1}^{n_i} \frac{1}{r_{ij}} \left(\frac{\partial r_{ij}}{\partial \theta_v} \right) = 0, \quad v=1, 2, L, m \quad (E-10)$$

where k is the number of renewal time samples and n_i is now the number of observations within the i th renewal time sample. Equation (E-10) is the **general renewal time equation**. Eq. (E-10) applies to the renew-always, renew-if-failed and renew-as-needed cases.

E.5 Consistent Interval Cases

If $I_{ij} \cong I_i$ for all calibrations within a renewal time sample,²⁰ then $\tau_{ij} \rightarrow \tau_i$, $R(\tau_{ij}) \rightarrow R(\tau_i)$, and $r_{ij} \rightarrow R(\tau_i - I_i)$. Eq. (E-10) can then be written

$$\sum_{i=1}^k \frac{g_i - n_i \rho_i}{\rho_i(1-\rho_i)} \left(\frac{\partial \rho_i}{\partial \theta_i} \right) + \sum_{i=1}^k \frac{n_i}{r_i} \left(\frac{\partial r_i}{\partial \theta_v} \right) = 0, \quad v=1, 2, L, m, \quad (E-11)$$

where g_i is the number observed in-tolerance in the i th renewal time sample. We now define an “observed reliability”

²⁰ If the intervals I_{ij} , $j = 1, 2, \dots, n_i$ are not equal, it may be acceptable to set $I_i = (1/n_i) \sum_{j=1}^{n_i} I_{ij}$.

$$\hat{R}_i^0 = g_i / n_i \quad (\text{E-12})$$

for the i th renewal time. With this quantity, Eq. (D-11) becomes

$$\sum_{i=1}^k W_i (\hat{R}_i^0 - \rho_i) \left(\frac{\partial \rho_i}{\partial \theta_v} \right) + \sum_{i=1}^k w_i (1 - r_i) \left(\frac{\partial r_i}{\partial \theta_v} \right) = 0, \quad v = 1, 2, \dots, m \quad (\text{E-13})$$

where

$$W_i = \frac{n_i}{\rho_i (1 - \rho_i)}, \quad (\text{E-14})$$

and

$$w_i = \frac{n_i}{r_i (1 - r_i)} \quad (\text{E-15})$$

Equation (E-13) is the **consistent interval renewal time equation**.

E.6 Limiting Renewal Cases

E.6.1 Renew-Always

If the renew-always policy is adhered to, then $\tau_i = I_i$, $r_i \rightarrow 1$, and $\rho_i \rightarrow R_i$. The second term in Eq. (E-13) then becomes zero, and we have

$$\sum_{i=1}^k W_i (\hat{R}_i^0 - R_i) \left(\frac{\partial R_i}{\partial \theta_v} \right) = 0, \quad v = 1, 2, \dots, m, \quad (\text{E-16})$$

where

$$W_i = \frac{n_i}{R_i (1 - R_i)} \quad (\text{E-17})$$

A comparison of these expressions with Eq. (D-6) in Appendix D shows that the renew-always case can be derived as a special case of the general renewal time equations.

E.6.2 Renew-If-Failed

If renewals are performed only in the case of observed out-of-tolerances, then Eqs (E-2) and (E-3) yield

$$L = \prod_{i=1}^X [r(\tau_i) - R(\tau_i)],$$

where X is the number of observed out-of-tolerances. Differentiating the log of this expression with respect to the m components of the parameter vector $\hat{\theta}$ gives

$$\sum_{i=1}^X \frac{1}{r_i - R_i} \left(\frac{\partial r_i}{\partial \theta_v} - \frac{\partial R_i}{\partial \theta_v} \right) = 0, \quad v = 1, 2, \dots, m \quad (\text{E-18})$$

E.7 Example: Simple Exponential Model

E.7.1 General Case

The simple exponential model is

$$R(t) = e^{-\lambda t}. \quad (\text{E-19})$$

Substituting Eq. (E-19) in Eqs. (E-1) and (E-4) gives

$$r_{ij} = R_{ij} e^{\lambda I_{ij}}, \quad (\text{E-20})$$

and

$$\rho_{ij} = e^{-\lambda I_{ij}},$$

and after a little algebra, Eq. (E-9) becomes

$$\sum_{i=1}^k \sum_{j=1}^{n_i} I_{ij} \frac{1 - x_{ij}}{1 - e^{-\lambda I_{ij}}} - \tau = 0 \quad (\text{E-21})$$

where

$$\tau = \sum_{i=1}^k \sum_{j=1}^{n_i} \tau_{ij} \quad (\text{E-22})$$

Eq. (E-21) can be solved for λ by use of a Newton-Raphson or equivalent method.

E.7.2 Renew-Always Case

In the renew-always case, renewals occur at every calibration. Thus, we can group the terms in Eq. (E-21) by resubmission time, and the variable x_{ij} becomes

$$x_{ij} = \begin{cases} 1, & \text{If the } j\text{th observation of the } i\text{th resubmission time is in-tolerance} \\ 0, & \text{otherwise} \end{cases}$$

With this definition, Eq. (E-20) can be written

$$\sum_{i=1}^k I_i \frac{n_i - g_i}{1 - e^{-\lambda I_i}} - \tau = 0 \quad (\text{E-23})$$

where g_i and n_i are defined as before, and

$$\tau = \sum_{i=1}^k n_i \tau_i \quad (\text{E-24})$$

E.7.3 Renew-If-Failed Case

Substituting Eq. (E-19) and (E-20) in Eq. (E-18) gives

$$\sum_{i=1}^X \frac{I_i}{1 - e^{-\lambda I_i}} - \tau = 0 \quad (\text{E-25})$$

where the subscript j ranges over the X observed out-of-tolerances and

$$\tau = \sum_{i=1}^X \tau_i.$$

In Eq. (E-25), the variable I_i is the interval during which the i th observed out-of-tolerance occurred.

E.8 Method S3 Pros and Cons

E.8.1 Pros

1. Method S3 adjusts intervals to meet specified reliability targets.
2. Once installed on a computer, Method S3 is inexpensive to operate.
3. Reliability modeling in Method S3 involves the use of a variety of models. This ensures a considerable improvement in interval accuracy relative to methods S1 and A1 - A3.
4. Method S3 permits the statistical identification of dogs, gems and other outliers.
5. Method S3 accommodates all renewal policies (see Appendix G).

E.8.2 Cons

1. Method S3 is expensive to design and implement. However, due to the accuracy of intervals obtained from its operation, design and development costs may be recovered quickly.
2. Method S3 requires a large size inventory to be cost-effective.

Appendix F Adjusting Borrowed Intervals

F.1 Introduction

The reliability targets for a requiring organization and an external authority providing a calibration interval may not be the same. If this is the case, then the borrowed interval will need to be adjusted to be consistent with the requiring organization's target.

If the reliability model and parameters for a borrowed interval are known, it is possible to make this adjustment mathematically. Note, however, that this adjustment does not compensate for variations between organizations in specifications, use, stress, calibration methods, and other factors mentioned in Chapters 2 and 4.

F.2 General Case

If the reliability model from the external authority is $\hat{R}(t, \hat{\theta})$ and the reliability target for the requiring organization is R^* , then the required interval is obtained by solving for I_r from

$$\hat{R}(I_r, \hat{\theta}) = R^*.$$

As an example, the Weibull model is given by

$$R(t, \hat{\theta}) = e^{-(\lambda t)^\beta}.$$

The interval for the requiring organization is, accordingly, given by

$$I_r = \frac{(-\ln R^*)^{1/\beta}}{\lambda}.$$

Similar expressions can be obtained for the other reliability models described in this RP. A general treatment is given in Appendix I.

F.3 Exponential Model Case

In cases where the borrowed interval was computed using a simple exponential model, all that need be known are the external authority's reliability target and assigned interval. If these quantities are r^* and I_e , respectively, then the failure rate parameter λ can be obtained from

$$\lambda = -\frac{1}{I_e} \ln r^*.$$

RP-1: Establishment and Adjustment of Calibration Intervals

If the reliability target for the requiring organization is R^* , the appropriate interval is calculated as

$$\begin{aligned} I_r &= -\frac{1}{\lambda} \ln R^* \\ &= I_e \frac{\ln R^*}{\ln r^*}. \end{aligned}$$

Appendix G Renewal Policies

G.1 Introduction

This Appendix examines technical and management issues related to equipment renewal policies. This examination does not provide a definitive argument for one renewal policy over another but, instead, points toward deciding on an interval-analysis methodology. This disclaimer notwithstanding, makers of renewal policies might benefit from reading the following.

G.2 Decision Variables

G.2.1 Analytical Considerations

Comparing Eq. (I-2) with Eq. (I-9) in Appendix I suggests that, from the standpoint of solving for and assigning calibration intervals, the renew-always policy is to be preferred over the renew-if-failed and renew-as-needed policies. Moreover, if the renew-always policy is adopted, then Method S2 can be implemented without modification. This greatly reduces system development effort relative to that of S3 and enhances system applicability relative to Method S1. Method S2 is the simplest predictive method that takes into account the facts that (1) failure times are unknown in interval analysis, and (2) a variety of uncertainty growth mechanisms govern the process by which attribute transition from an in-tolerance state to an out-of-tolerance state.

G.2.2 Maintenance / Cost Considerations

While choosing a renewal policy on the basis of ease of interval-analysis and simplicity of interval assignment has some merit, it should be recognized that basing a renewal policy on these considerations alone would constitute having “the tail wag the dog.” Considerations of equipment stability, application, cost of adjustment, and so on should normally outweigh considerations of analytical convenience. Ideally, analysts should adjust to the requirements of the operating environment — not the other way around.

In past years, several articles have been written on the subject of whether to renew or not renew. Although many of these are neither rigorously developed nor completely objective, some have emerged that offer insights into the consequences of adopting one policy over another. To summarize, the relevant factors to consider are:

1. Does attribute adjustment disturb the equilibrium of an attribute, thereby hastening the occurrence of an out-of-tolerance condition?
2. Do attribute adjustments stress functioning components, thereby shortening the life of the MTE?
3. During calibration, the mechanism is established to optimize or “center-spec” attributes. The technician is there, the equipment is set up, the references are in-place. If it is desired to have attributes performing at their nominal values, is this not the best time to adjust?
4. By placing attribute values as far from the tolerance limits as possible, does adjustment to nominal extend the time required for re-calibration?
5. Do random effects dominate attribute value changes to the extent that adjustment is merely a futile attempt to control random fluctuations?
6. Do systematic effects dominate attribute value changes to the extent that adjustment is beneficial?
7. Is attribute drift information available that would lead us to believe that not adjusting to nominal would, in certain instances, actually extend the period required for re-calibration?

8. Is attribute adjustment prohibitively expensive?
9. If adjustment to nominal is *not* done at every calibration, are equipment users being short-changed?
10. What renewal practice is likely to be followed by calibrating personnel, irrespective of policy?
11. Which renewal policy is most consistent with a cost-effective interval-analysis methodology?

Except for item 11, the answer to each of these questions appears to be context sensitive. In other words, what may be optimal for one MTE would be suboptimal for another. In deciding on which policy to implement, then, it would be useful to have guidelines that address each of the eleven items above in such a way that the best policy can be found for a given MTE, within a given context.

G.2.3 Cost Guidelines

Viewed from a cost-management perspective, it may at first be thought that the “renew-if-failed” practice should be universally accepted. On paper, it would appear that leaving in-tolerance attributes alone is cheaper and less intrusive than adjusting them. This policy is especially attractive for MTE whose attribute value changes are randomly spontaneous, thereby rendering adjustment futile, or for MTE whose attributes tend to go out-of-tolerance more quickly if disturbed by adjustment. In these cases, the renew-if-failed practice may well be advisable.

In the vast majority of cases, however, it appears that systematic drift and response to external stress are the predominant mechanisms for transitioning an attribute from an in-tolerance to an out-of-tolerance condition. In these cases, a “renew always” practice is usually more cost effective than a renew-if-failed or even a renew-as-needed practice. This is because equipment renewal ordinarily extends the period required for out-of-tolerances to occur. In other words, the renew-always policy typically extends calibration intervals.

The deciding factors in evaluating whether to adjust or not on the grounds of cost accounting alone are those that balance the tradeoff between cost reductions due to extended calibration intervals and the cost penalties incurred by adjustment. These factors are items 1-8 and item 11 above. From the observations made in the preceding paragraph, it would appear that, from a cost standpoint, positive responses to items 3, 4, 6, and 11 favor a renew-always policy. On the other hand, positive responses to items 2, 5, 7 and 8 would tend to support a renew-if-failed policy.

It appears unlikely that any kind of general statement can be made that argues in favor of renew-always over renew-if-failed, or vice versa, on a cost-control basis alone. Unless a requiring organization is prepared to analyze the tradeoffs inherent in each policy on a case-by-case basis, it might be prudent to declare a tie with respect to cost factors and proceed to other considerations.

G.2.4 Random vs. Systematic Guidelines

In 1991, an analytical model was presented that examined the benefit of adjusting attributes during calibration, [BW91]. The model, which will be referred to here as the **Weiss model**, argues that a non-adjustment policy is preferable. In doing so, the model assumes that the mean of values of an attribute under study remains constant between successive calibrations, with all changes in value due to random effects. Other models based on this assumption also support a non-adjustment policy [KC94, KC95].

However, if a systematic mean value change mechanism, such as monotonic drift, is introduced into the model, the result can be quite different. For discussion purposes, modifications of the model that provide for systematic change mechanisms will be referred to as **Weiss-Castrup** models (unpublished).

By experimenting with different combinations of values for drift rate and extent of attribute fluctuation in a Weiss-Castrup model, it becomes apparent that the decision to adjust or not adjust depends on whether changes in attribute values are predominantly random or systematic. In addition to being supported by rigorous analysis, this result is intuitively appealing.

From the standpoint of random vs. systematic effects, it would appear that the central question is whether random fluctuations or systematic drift is the dominant attribute change mechanism. There are at least two cost-effective approaches that strive to answer this question.

G.2.4.1 Approach 1: Attribute Tolerance Evaluation

Suppose that changes in attribute values are due entirely to random fluctuations, as in the Weiss model. There are two possible outcomes:

1. If random fluctuations are contained within attribute tolerance limits, not only is adjustment during calibration not called for, but calibration itself is not beneficial.
2. If random fluctuations tend to cross tolerance limits, then the tolerance limits are too tight. Unless fluctuations occur with a periodicity that is at least on the order of the periodicity of the calibration interval, then attribute adjustment is futile – as is periodic calibration.

There is a simple, yet indirect, way to determine whether outcome 1 or 2 exists. The procedure requires the ability to classify as-found calibration results in terms of degree of out-of-tolerance and involves conducting statistical interval analysis, as described in Appendix D, with two reliability “models” added to the list of models in that Appendix. These models are the **no-fail model** and the **reject model**. The no-fail model is selected if, despite a number of calibration results sufficient for interval analysis, no out-of-tolerances have been recorded. The reject model is chosen if, after statistical analysis, all of the ten models described in Appendix D are rejected.

If the no-fail model is selected, we conclude that outcome 1 applies. In this case, periodic calibration is not required. We make this decision, however, only after experimenting with interval extensions out to the expected life span of the MTE in question.

If the reject model is selected for an MTE, we conclude that outcome 2 applies. In this case, we soften the out-of-tolerance criterion for the MTE and conduct a re-analysis with the new criterion. For instance, suppose that calibration history records contain as-found codes that indicate whether an as-found result was in-tolerance, within 1.0 to 1.5 times spec, 1.5 to 2.0 times spec, and so on. Suppose further, that we soften the failure criterion of the interval-analysis system to consider failures to be out-of-tolerances that exceed 1.5 times the tolerance limits. If a re-analysis using the new criterion results in the selection of a model other than the reject model, then we conclude that the MTE tolerance limits were originally too tight.

Incidentally, the procedure of softening failure criteria, followed by interval re-analysis, is useful for finding realistic tolerance limits for MTE that cannot meet desired reliability targets.

G.2.4.2 Approach 2: Attribute Response Modeling

In Appendix D, under the topic **Renewal Policy and the Drift Model**, a method was outlined for evaluating whether random or systematic effects dominate in changes in attribute value taking place during uncertainty growth. The method involves fitting the drift model to calibration-history time-series data and evaluating the ratio of the slope of the drift process to the standard deviation of this slope.

G.2.5 Quality Assurance Guidelines

Item 9 above addresses the issue of regarding a calibrated MTE as the “product” provided by the calibrating organization to the MTE user. If the calibrated MTE is returned to the user without its attributes being adjusted to nominal, is the calibrating organization putting out a flawed product? This may be the interpretation of many equipment users. If so, then it could be a factor in deciding on a renewal policy.

G.2.6 Interval Methodology Guidelines

Guidelines in this section address item 11. From the remarks under Analytical Considerations at the beginning of this Appendix, it would appear that “renew-always” is the optimal policy from an analytical standpoint. This is because the “renew-always” policy compliments Method S2 and Method S2 is the most cost-effective of the multi-model predictive methods. If, however, Method S1 is selected as the preferred method, then the renew-if-failed model is optimal. With Method S3, it does not matter which policy is in effect. All this method needs to know is whether a renewal took place during calibration.

G.2.7 Systemic Disturbance Guidelines

G.2.7.1 Hardware Corrections

If physical adjustments tend to stress functioning components or disturb equipment equilibrium (see items 1 and 2) then the renew-if-failed policy gets a point in its favor.

G.2.7.2 Software Corrections

If renewals consist of software corrections, e.g., bias offsets, then the renew-always policy is recommended.

G.2.8 Policy Adherence Considerations

Regardless of what the renewal policy might be, the actual practice may be inspired more by on-the-spot conditions and by the experience and personal preferences of calibrating technicians and/or supervisors. In other words, the policy may be renew-if-failed, while the practice is renew-always, or vice versa.

By nature, highly skilled calibrating technicians are “concerned citizens.” Many consider leaving an attribute in anything but a nominal state to be an irresponsible act. To tell such an individual that, despite an opportunity, a method and an abiding motive to make an optimizing adjustment, he or she should do otherwise seldom works. Several informal surveys conducted in calibrating organizations with a renew-if-failed policy find that technicians are employing a renew-always practice instead. In one such example, management stated with absolute certainty that the renew-if-failed policy was being adhered to. This was known to be, because exhaustive “tiger team” audits had just been conducted in this and other areas. However, a quick informal trip to one of this organization's cal labs and some brief discussions with calibrating technicians showed that the renew-always practice was actually in effect, at least at that organization.

For interval-analysis purposes, the important point to consider in evaluating the practice vs. policy issue is not so much whether to implement one policy over another, but rather whether to *assume* one policy over another.

G.3 Renewal Policy Selection

As stated under Maintenance / Cost Considerations above, for a renewal policy to be optimal, it should be applied on a case-by-case basis. If so, then the interval-analysis system needs (1) to be able to determine when renewal has taken place, and (2) employ a methodology that can accommodate whatever renewal actions have occurred. If a mixture of practices is anticipated, Method S3 is an obvious choice.

Applying renewal policies on a case-by-case basis requires that each model number of equipment undergo engineering pre-analysis that takes into account items 1, 2, 5, 6 and 7. In addition, cost analyses would be required regarding items 3, 4, 8 and 11; and management decisions would have to be made concerning items 3, 9 and 10.

Case-by-cases analyses of this sort are expected to be beyond the capability of most requiring organizations. Consequently, it would appear that some guidance is needed to assist in arriving at the optimal renewal policy at the organizational level.

From a practical standpoint, it seems that the optimal renewal policy for most organizations is renewal-always. The reasoning behind this assertion is as follows:

G.3.1 Point 1 - Quality Assurance

Item 3 is a major consideration regarding renewal policy. This claim should be taken in the context of Policy Adherence Considerations above. In this context, the instincts of the calibrating technicians are correct from a quality assurance standpoint. If in doubt, implement the most conservative policy. If the answers to items 1, 2, 5, 6 and 7 are unavailable, then the answer to item 4 will almost always be positive. The answers to items 3 and 9 follow immediately.

G.3.2 Point 2 - Majority Rule

Except for requiring organizations that specialize in restricted measurement technologies, the equipment inventories of most requiring organizations tend to be dominated by MTE designed to operate at nominal values, with adequate compensation for physical adjustment stress. Adjustment to nominal is likely to extend rather than shorten the life of such equipment. Adjustment to nominal is also likely to extend the time that equipment can remain in use in an in-tolerance state. In the absence of precise drift or other information to the contrary, the answer to item 4 is ordinarily positive. In addition, many MTE attribute adjustments are not physical but, rather, of the correction factor variety, where attribute corrections are made in the form of “soft” adjustments instead of physical tweaks.

With regard to item 8, attribute adjustments are usually designed to be fairly straightforward. In the past, the reverse was often the case. Anyone who has worked with MTE technology from the '50s and '60s will recall removing chassis and other impediments to get at trim pots or other adjustable components. Today, however, such gymnastics are rarely required. If so, then the offending MTE emerges as an exception rather than the rule. If it is desired, then, to forego adjustment on the grounds that adjustments are too expensive to make, it would appear that such a decision should be made on an “exception” basis rather than as a general policy.

G.3.3 Point 3 - Public Relations

As stated earlier, metrology's "product" consists primarily of calibrated MTE. If some sort of optimization is not performed during calibration, then the product will be perceived as being superfluous. MTE users already disgruntled over having to give up their equipment for periodic calibration are not likely to be enamored of an adjustment policy that gives them a return on investment only if attributes are found out-of-tolerance. While arguments that adjustments are intrusive or futile may be made on a case-by-case basis, it is difficult to see how any using community would accept them as generally valid axioms.

G.3.4 Point 4 - A Logical Predicament

If we can convince ourselves that adjustment of in-tolerance attributes should not be made, how then to convince ourselves that adjustment of out-of-tolerance attributes is somehow beneficial? For instance, if we conclude that attribute fluctuations are random, what is the point of adjusting attributes at all? What is special about attribute values that cross over a completely arbitrary line called a tolerance limit? Does traversing this line transform them into variables that can be controlled systematically? Obviously not.

In cases where the decision to adjust or not is based on economic considerations or on the grounds that adjustment shortens equipment lifetimes and/or calibration intervals, we face a similar dilemma. Do we assert that adjusting an attribute that is 1 % outside of spec is cost-effective, while adjusting an attribute that is 1 % within spec is not? Where do we draw the line? Some organizations employ the renew-as-needed policy, setting adjustment limits at some point inside attribute tolerance limits. If adjustment decisions are made on the basis of economics, however, then it would seem likely that adjustment limits should often be set *outside* tolerance limits. Such a practice would encourage adjustments only when absolutely necessary. Determining where to put such adjustment limits would, in each case, require a fairly sophisticated analysis of user needs vs. adjustment costs and impact on equipment longevity. To do this as a general practice seems extravagant.

G.3.5 Point 5 - Analytical Convenience

It was acknowledged under Interval Methodology Guidelines above that analytical convenience should not alone be the basis for an adjustment policy. However, analytical convenience *is* a factor. If implementation of methods such as S3 is beyond the capability of the requiring organization, then analytical convenience may suddenly become analytical *validity*.

To expand on this point, if it is desired to optimize intervals as discussed in Chapter 4, then the best methods are S2 and S3. Of these, Method S2 is by far the most tractable from an interval-analysis system development standpoint. Implementation of Method S3 requires a level of analytical sophistication that can embrace advanced statistics, probability theory and numerical analysis methodologies. As research continues in the field on interval analysis, Method S3 will become more approachable. At present, however, it must be considered an extremely tough nut to crack.

If Method S2 is the best method that can be reasonably implemented, then, because the method is ideally suited to the renew-always policy, analytical convenience argues in favor of renew-always.^{21, 22}

G.4 Analytical Policy Selection

If Method S3 can be implemented, then the analysis system is capable of dealing with whatever renewal practice is in place. In this regard, renewal *practice* should be contrasted with renewal *policy*. To be optimal, the analytical system needs to respond to what is actually occurring during calibration, not what is *supposed* to occur during calibration. If any doubt exists as to this issue, then the observations under Policy Adherence Considerations argue in favor of assuming a renew-always practice, regardless of whatever renewal policy is in effect.

This brings up an interesting conclusion. Even if method S3 could be implemented, would intervals emerging from the analysis system be valid? Suppose that the requiring organization provides an indicator in its calibration history database that flags whether adjustment took place or not. If a record indicates that no adjustments have been made, should we accept this at face value? If the policy is renew-if-failed, for instance, it would be unlikely to find a record showing that an in-tolerance MTE was adjusted, although this may have been the case. When confronted with questionable adjustment indicators, the appropriate analytical course is sometimes unclear. This course is even more obscure when adjustment indicators are unavailable.

At this point, it would appear that assuming a renew-always practice should serve as a reasonable default position. This position could be modified if strong evidence for other practices could be established.

G.5 Maintaining Condition Received Information

Whatever renewal policy is implemented, it cannot be overly stressed that adjustments should not be made before all relevant Condition Received information has been recorded (see Chapter 8). A procedure that reflects this recommendation can be found on the Internet in the HP Metrology Forum [HP95]. The essential points are

- An item received for calibration to a specification first undergoes a complete and thorough performance test.
- All test results are recorded. No adjustments are made at this stage.
- The results, with failed attributes highlighted (if relevant), are labeled ('pass' or 'fail').
- If any attributes were non-compliant, corrective adjustments are made.
- The full performance test is then carried out again, with all results recorded.

The only modification to this procedure suggested here is that, if adjustments do not negatively impact the stability of the MTE, then it may be cost-effective to optimize (adjust) in-tolerance attributes as well as out-of-tolerance ones following the recording of test results. As pointed out earlier, although this practice

²¹ Certain modifications to Method S2 can be made that more or less adapt it to the renew-if-failed and renew-as-needed policies. For amplification on these methods, contact the Calibration Interval Committee Chairman.

²² Arguments to the contrary may be found in various reports and papers written prior to the early 1980s. At the time of their writing, methods for analyzing type III censored data were not widely known, and Method S1 was the method in-place. As indicated in Chapter 6, Method S1 works best if the renew-if-failed policy is in effect.

incurs an additional adjustment cost it may lead to a net cost saving by extending the MTE calibration interval.

G.6 Summary

At present, no inexpensive systematic tools exist for deciding on the optimal renewal policy for a given MTE. While it can be argued that one policy over another should be implemented on an organizational level, there is a paucity of rigorously demonstrable tests that lead to a clear-cut decision as to what that policy should be. The implementation of reliability models, such as the drift model, that yield information on the relative contributions of random and systematic effects, seems to be a step in the right direction. The development of other tools is in the future. In the meantime, in the absence of solid evidence to the contrary, it may be most prudent for the interval-analysis system to assume a renew-always practice, regardless of which renewal policy is in effect.

Appendix H System Evaluation

H.1 Introduction

Once an interval-analysis system is in operation, it may be helpful to periodically test whether the intervals generated by the system lead to actual measurement reliabilities that are consistent with reliability targets. Indeed, quality standards or other documents may recommend or mandate validation of the interval-analysis system [Z540.3, IL07]. This appendix discusses an approach for such tests. In brief, the approach involves the following:

1. Compare observed in-tolerance percentages against reliability targets for each interval generated by the interval-analysis system.
2. Evaluate computed intervals and engineering overrides separately.
3. Focus only on calibration results for calibrations with resubmission times close to the computed intervals. This requires developing a window of time around each computed interval that serves as a resubmission time filter.
4. Perform a statistical test for each computed interval. Indicate whether intervals pass or fail the test.
5. Summarize and evaluate the test results.

It should be noted that recommendations 1, 3, and 4 above are inherent parts of Method A3, as is the adjustment of any intervals failing the test.

H.2 Developing a Sampling Window

Obviously, intervals that differ appreciably from those recommended by the interval-analysis system are not relevant to system evaluation. However, it is unreasonable to suppose that usable samples sizes will be obtained if only resubmission times are included that exactly match computed intervals. To obtain samples that are both relevant and sufficient for analysis requires the implementation of sampling windows.

A sampling window for a computed interval consists of a lower and upper limit around the interval that captures sufficient data for evaluation. At first glance, it would seem reasonable to set the width of each sampling window equal to a percentage (e.g., $\pm 10\%$) of the interval. Other assumptions come to mind. For one, it might be assumed that MTE resubmission times would, on average, be longer than assigned intervals. These and other assumptions were examined in an informal study performed in the late '70s and reported in 1988 [HC88].

H.2.1 Case Studies

The study examined only cases where intervals were assigned by the interval-analysis system. A principal objective was to isolate routine calibrations, performed as part of normal equipment recall, from calibrations that were due to some other requirement. It was reasoned that, for routine calibrations, most resubmission times would be close to the assigned intervals. It was assumed, however, that some lag time would normally be observed due to times required for shipping and handling and to the reluctance of users to surrender equipment for calibration. For this reason, the study did not assume that resubmission time mode values would be equal to assigned intervals. To capture representative calibrations, the study did the following:

1. Determined mode resubmission time values for each MTE with an assigned interval equal to the interval computed by the analysis systems.
2. Computed $\pm$ one sigma (68 % confidence) limits around each mode value.

H.2.2 Study Results

The results of the study were somewhat unexpected. They are the following

1. Mode values tended to be equal to assigned intervals. Evidently, as many users were eager to have their MTE calibrated in a timely manner as were reluctant to part with their equipment.
2. Sampling windows for intervals less than around twelve weeks tended to be approximately 25 % of the interval value. For instance, a ten-week interval tended to have approximately 68 % of resubmission times fall within ± 2.5 weeks.
3. Sampling windows for intervals greater than twelve weeks showed a strong tendency to be fixed – the overwhelmingly predominant value being ± 4 weeks (rounded off).

H.2.3 Sampling Window Recommendations

The results of the above study may not apply to all requiring organizations. To amplify on this point, the study was performed at a large aerospace facility where recall cycles were regularly enforced and where data validity, consistency and completeness were assured. These conditions may not be met, for example, within Department of Defense organizations where calibrations are performed at locations spread around the world, at several levels of sophistication, with varying degrees of data communication integrity, and with varying degrees of participation in calibration data management programs.

The conditions of the study may also not apply to organizations with less regular enforcement of recall schedules, where $\pm$ one-sigma limits may be looser than four weeks, or to organizations with small inventories that may be controlled to limits tighter than four weeks.

For these reasons, it is recommended that studies similar to the one outlined here be performed by each requiring organization, where feasible.

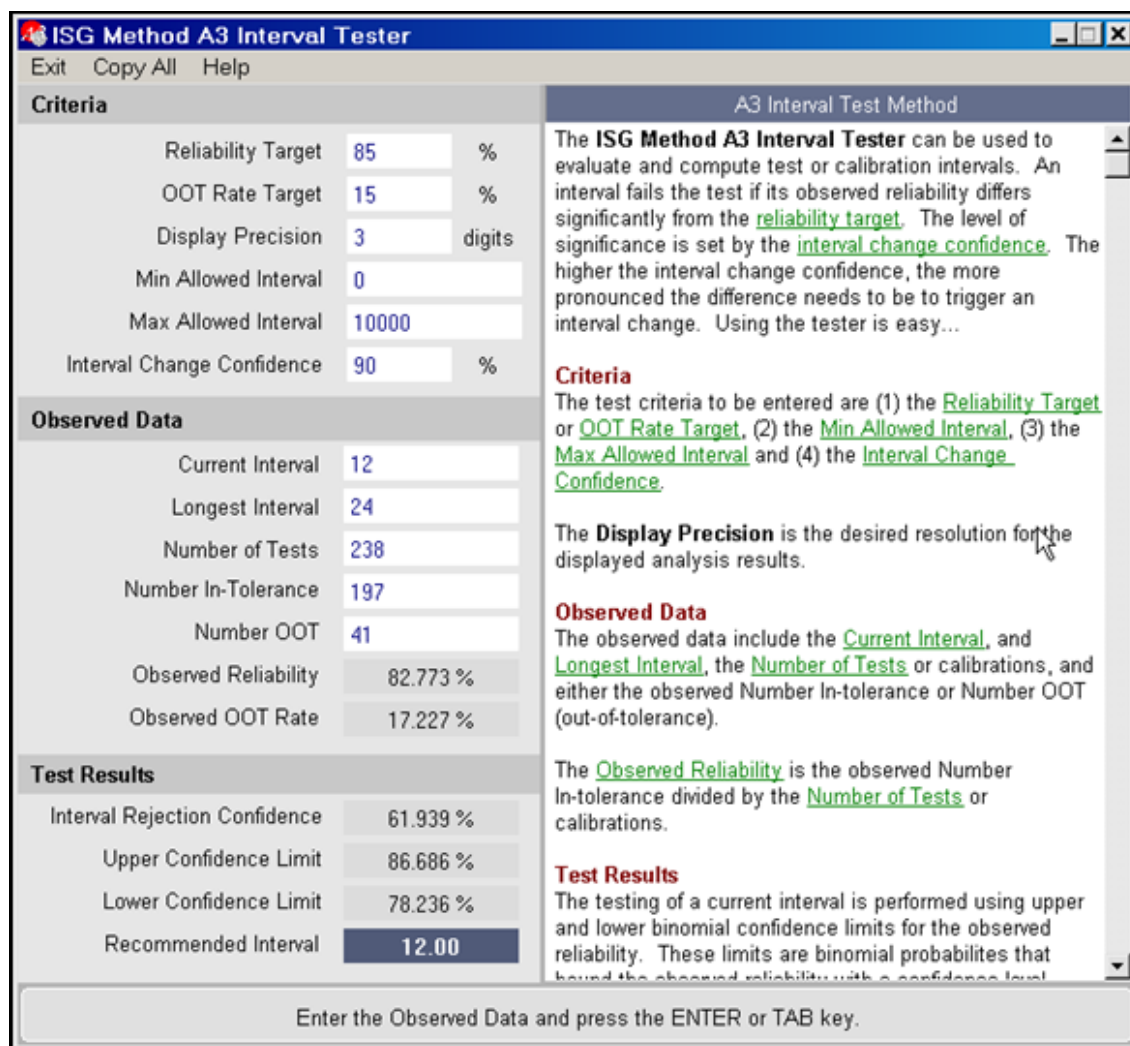
H.3 System Evaluation Guidelines

Once a resubmission window study has been performed, observed calibrations should be compared against reliability targets and tested for compliance.

H.3.1 Test Method

The recommended method computes upper and lower binomial confidence limits around observed measurement reliabilities. If the reliability target falls within the confidence limits for a given interval, then the system passes the test for that interval. If the reliability target falls outside the confidence limits, the system fails the test for that interval.

The computation of binomial confidence limits is described in most upper division statistics textbooks (see, for example, Ref. PH62, pp. 239-240). A small Windows PC application is freely available [IE08] that illustrates the computation. An example of its use is shown in Figure H-1.



ISG Method A3 Interval Tester

Exit Copy All Help

Criteria		A3 Interval Test Method
Reliability Target	85 %	The ISG Method A3 Interval Tester can be used to evaluate and compute test or calibration intervals. An interval fails the test if its observed reliability differs significantly from the <u>reliability target</u>. The level of significance is set by the <u>interval change confidence</u>. The higher the interval change confidence, the more pronounced the difference needs to be to trigger an interval change. Using the tester is easy... Criteria The test criteria to be entered are (1) the <u>Reliability Target</u> or <u>OOT Rate Target</u>, (2) the <u>Min Allowed Interval</u>, (3) the <u>Max Allowed Interval</u> and (4) the <u>Interval Change Confidence</u>. The Display Precision is the desired resolution for the displayed analysis results. Observed Data The observed data include the <u>Current Interval</u>, and <u>Longest Interval</u>, the <u>Number of Tests</u> or calibrations, and either the observed Number In-tolerance or Number OOT (out-of-tolerance). The <u>Observed Reliability</u> is the observed Number In-tolerance divided by the <u>Number of Tests</u> or calibrations. Test Results The testing of a current interval is performed using upper and lower binomial confidence limits for the observed reliability. These limits are binomial probabilities that bound the observed reliability with a confidence level
OOT Rate Target	15 %	
Display Precision	3 digits	
Min Allowed Interval	0	
Max Allowed Interval	10000	
Interval Change Confidence	90 %	
Observed Data		
Current Interval	12	
Longest Interval	24	
Number of Tests	238	
Number In-Tolerance	197	
Number OOT	41	
Observed Reliability	82.773 %	
Observed OOT Rate	17.227 %	
Test Results		
Interval Rejection Confidence	61.939 %	
Upper Confidence Limit	86.686 %	
Lower Confidence Limit	78.236 %	
Recommended Interval	12.00	

Enter the Observed Data and press the ENTER or TAB key.

Figure H-1. Binomial Confidence Test Example

H.3.2 Evaluation Reports

The results of system testing should be reported at the individual test level, with some summary information provided. A typical test report is shown below.

Table H-1. System Evaluation Test Results:

Interval Evaluations
Test Confidence Level = 0.90

Mfr / Model	Number Calibrated	Number In-tolerance	Observed Reliability	Reliability Target	Rejection Confidence	Test Result
Abc / 1234	15	12	0.800	0.85	0.208	PASS
Xyz / 3241	28	26	0.929	.90	0.081	PASS

Alpha / 2211	19	12	0.632	.85	0.967	FAIL
--------------	----	----	-------	-----	-------	------

Overall Results

Overall Observed Reliability: 0.835
 Number Mfr/Models Tested: 622
 Number Failed: 91
 Percent Passed: 85.4 %

H.3.3 System Evaluation

There is no clear guideline for how many interval test failures make a failed system. The choice of what number or percentage to use is largely a matter of system criticality and management taste. About the only general statement that can be made here is that system test results are relative. For example, of the results of two alternative methods of interval-analysis are available, the test results can be compared to pronounce one method better or worse than the other. If such comparisons are not available, then test results can be compared against what would be achieved if intervals were set randomly. In this case a better than 50 % pass rate may be acceptable. Such a conjecture should be supported by simulation.

Appendix I Solving for Calibration Intervals

I.1 Introduction

This appendix refines expressions found in the body of this RP for computing calibration interval in terms of inverse reliability functions and reliability targets. The basic equations are extended to cover renew-if-failed and renew-as-needed policies.

I.2 Special Cases

In this RP, expressions are found that set intervals to be commensurate with reliability targets. For the most part, these expressions take the form

$$R(T) = R^* \quad (\text{I-1})$$

from whence

$$T = R^{-1}(R^*). \quad (\text{I-2})$$

In these equations, T represents the calibration interval, $R(T)$ represents the measurement reliability at the end of the interval, R^* is the reliability target, and R^{-1} is the inverse of the reliability function.

I.3 General Cases

Strictly speaking, Eq. (I-2) is only approximate, except in cases where $R(T)$ is an exponential model or where the renew-always policy is in effect. If conditions are otherwise, a modification of Eq. (I-2) is needed. The first step in developing this modification is to define a variable T_n as

$$T_n = t_1 + t_2 + \dots + t_n, \quad (\text{I-3})$$

where

$$T_i = \text{ith interval since the last renewal, } i = 1, 2, \dots, n. \quad (\text{I-4})$$

If an item of MTE has gone three successive intervals without renewal, for instance, then $T_3 = t_1 + t_2 + t_3$. If the end-of-period reliability target is R^* , then, after n successive intervals without renewal, we have

$$R(T_{n+1}|T_n = R^*) \quad (\text{I-5})$$

where the notation $R(T_{n+1}|T_n)$ designates the conditional probability for an in-tolerance at time T_{n+1} , given that the MTE was in-tolerance at time T_n . From basic probability theory, the conditional probability in Eq. (I-5) can be written

$$R(T_{n+1}|T_n) = R(T_{n+1}, T_n) / R(T_n).$$

But $R(T_{n+1}, T_n)$ is just $R(T_{n+1})$, and Eq. (I-5) can be written

$$R(T_{n+1}) = R(T_n)R^*. \quad (\text{I-6})$$

By induction, Eq. (I-6) can be expressed as

$$R(T_{n+1}) = R(T_1)(R^*)^n,$$

and, because $R(T_1) = R^*$, we get

$$R(T_{n+1}) = (R^*)^{n+1} \quad (\text{I-7})$$

To solve for the interval we use Eq. (I-7) to obtain

$$T_{n+1} = R^{-1} \left[(R^*)^{n+1} \right] \quad (\text{I-8})$$

Eq. (I-8) contains the solution to the interval t_{n+1} . Because, by Eq. (I-3), $t_{n+1} = T_{n+1} - T_n$, we have

$$t_{n+1} = R^{-1} \left[(R^*)^{n+1} \right] - R^{-1} \left[(R^*)^n \right] \quad (\text{I-9})$$

Note that, if the renew-always policy is in effect, then $n = 0$, and, because $R^{-1}(1) = 0$, Eq. (I-9) reduces to Eq. (I-2).

I.4 Inverse Reliability Functions

Below are inverse functions $R^{-1}(x)$ for a few of the reliability models described in Appendix D. Inverse functions for other models are similarly determined. For some models, such as the modified gamma model, numerical methods are required to compute inverses.

Exponential Model:

$$R(t) = e^{-\lambda t}, \text{ and } t = -\frac{1}{\lambda} \ln [R(t)], \text{ so that}$$

$$R^{-1}(x) = -\frac{1}{\lambda} \ln x.$$

Weibull Model:

$$R(t) = e^{-(\lambda t)^\beta}, \text{ and } R^{(-1)}(x) = \frac{1}{\lambda} (-\ln x)^{1/\beta}.$$

Warranty Model:

$$R(t) = \frac{1}{1 + e^{\lambda(t-\tau)}}, \text{ and } R^{-1}(x) = \tau + \frac{1}{\lambda} \ln\left(\frac{1-x}{x}\right)$$

Mixed Exponential Model:

$$R(t) = \frac{1}{(1 + \lambda t)^\beta}, \text{ and } R^{-1}(x) = \frac{1}{\lambda} \left[(1/x)^{1/\beta} - 1 \right].$$

I.5 Adjustment Intervals

In Eq. (I-9), we seek an interval that corresponds to specified in-tolerance probability, R^* . The in-tolerance probability, of course, refers to the probability that MTE attributes will be found within their tolerance limits. We can use the same equation to estimate an interval corresponding to the probability that MTE attributes will be found within their *adjustment* limits.

In doing this, we replace the reliability target R^* with a renewal probability target r^* , defined as follows:

r^* - The probability that MTE attributes are within specified adjustment limits

Using r^* in place of R^* in Eq. (I-9) yields a calibration interval that is optimal with respect to considerations of renewal rather than reliability.

History

1989 Revision

Organization	Member or Contributor
Boeing Aerospace Company	Mr. Mack Van Wyck
Claremont McKenna College	Dr. John A. Ferling
Diversified Data Systems, Inc.	Mr. Donald Wyatt
General Dynamics	Mr. Anthony Adams
General Electric Company	Mr. Frank M. Butz, Mr. Alex Macarevich, Mr. Gerry Riesenber
Hayes Technology	Mr. Jerry L. Hayes
Integrated Sciences Group	Dr. Howard Castrup (Committee Chair)
John Fluke Manufacturing Company	Mr. Frank Capell, Mr. Ray Kletke, Mr. Joseph Martins
McDonnell Aircraft Company	Mr. James L. Ryan
Navy Metrology Engineering Center	Mr. John C. Larsen
Rockwell International Corporation	Mr. Rolf B.F. Schumacher
Solar Energy Research Institute	Mr. Robert Hansen

1996 Revision

Organization	Member or Contributor
A&MCA	Mr. Steven De Cenzo
Boeing Aerospace Company	Mr. Mack Van Wyck
Boeing Aerospace and Electronics Corporation	Mr. Patrick J. Snyder
Boeing Commercial Equipment	Mr. Derek Porter
Claremont McKenna College	Dr. John A. Ferling
Diversified Data Systems, Inc.	Mr. Donald Wyatt
Eastman Kodak Company	Mr. John Wehrmeyer
General Dynamics	Mr. Anthony Adams
General Electric Company	Mr. Frank Butz, Mr. Gerry Riesenber
Glaxo Pharmaceuticals	Mr. Ken Hoglund
Hewlett Packard Company	Mr. Dave Abell
Hughes Missile Systems Company	Mr. William Quigley
Integrated Sciences Group	Dr. Howard Castrup (Committee Chair)
Marine Instruments	Mr. John Miche
Naval Surface Warfare Center	Mr. Bruce Marshall
Naval Warfare Assessment Department	Mr. John C. Larsen
Texas Instruments	Mr. Dan Fory
Textron Lycoming	Mr. Joseph Balcher

AFTERWORD

This Recommended Practice has been prepared by the NCSL International toward a goal of uniformity in the field of instrumentation and measurement. To be of real value, this document should not be static, but should be subject to periodic review.

Permission to Reproduce

If you wish to take material from NCSLI technical publications (figures, extracts, abstracts, even full-text) and incorporate them into new works, you will need to obtain copyright permission. Please send a request to NCSL International requesting permission to reproduce/republish material. Permission to make fair use of the material contained in this publication provided that the following conditions are met:

- 1) The use is limited and noncommercial in nature, for teaching purposes.
- 2) The NCSLI copyright notice appears at the beginning of the publication.
- 3) The words “NCSL International Technical Publication” appears on each page reproduced.
- 4) The following disclaimer is incorporated and understood by all persons or organizations reproducing the publication:

Reproduction of any material in this publication is permitted only with the written permission of NCSL International. Requests for such permission should be addressed to NCSL International

Permission to Translate

Permission to translate part or all of this < pub class and title> is granted provided that the following conditions are met:

- 1) The NCSLI copyright notice appears at the beginning of the translation.
- 2) The words “Translated by [translator's name]” appear on each page translated.
- 3) The following disclaimer is included and understood by all persons or organizations translating this publication. If the translation is copyrighted, the translation must carry a copyright notice for both the translation and for the publication from which it is translated.

Disclaimer

The materials and information contained herein are provided and promulgated as an industry aid and guide, and are based on standards, formulae, and techniques recognized by NCSL International. The materials are prepared without reference to any specific federal, state, or local laws or regulations, and NCSL International does not warrant or guarantee any specific result when relied upon. The materials provide a guide or recommended practices and are not all-inclusive.

From time-to-time commercial equipment, instruments, or materials are identified in technical publications to foster understanding. Such identification does not imply recommendation or endorsement by the NCSL International, nor does it imply that the materials or equipment identified are necessarily the best available for the purpose.